

# 分析化学

## *Analytical Chemistry*

甘峰

2026 年 11 月 8 日

“如果觉得有用，请我喝杯茶吧。
咖啡也行。”
甘峰 的赞赏码

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# Preface

Before retirement, I wrote two analytical chemistry textbooks: *Fundamentals of Analytical Chemistry* (ISBN: 978-7-5025-9676-7) and *Analytical Chemistry* (ISBN: 978-7-03-063519-8). When writing the former, I lacked experience; when writing the latter, I lacked time. The first was a case of not knowing how to write, and the second a case of not having time to write. Taken together, the two barely qualify as having been written.

Yet a traditional printed textbook is frozen the moment it goes to press. Even when the author discovers errors or hits upon a better way of explaining something, he can only wait for the next revision, and that cycle often stretches over years or even more than a decade. Writing a book is, in the end, an exercise in regret.

After retirement, my time is largely my own, and I have naturally lost my best excuse, that of being too busy. So an idea took hold: why not write, at a comfortable pace, a textbook that truly satisfies me? That is how this book came about.

This book is released in electronic form precisely to break free from the limitations of print. It can be revised, supplemented, and refined at any time, so the reader always receives the latest version. In this sense, it is not a finished book but a book that keeps growing; or rather, a book that can never be fully finished.

Writing a book is no easy task; it weighs on the mind and consumes no small amount of coffee. If you find this book helpful, especially if it spares you from failing a course, consider buying me a coffee ! Your support is what keeps me going.

This book is not without its shortcomings. For any errors or infelicities in the text, I would sincerely appreciate your corrections. My usual email addresses are:

cesgf@mail.sysu.edu.cn and ganfeng1964@163.com. I usually reply as promptly as I can; if I do not, I am most likely off somewhere pretending to survey historic sites, as the cover suggests.

GAN, Feng

Guangzhou, 2026 年 11 月 8 日

# Chapter 4 Acid-Base Titration

**Chapter Navigation**

Acid-base titration is one of the most widely used methods in volumetric analysis. This chapter starts from the Brønsted-Lowry proton theory and builds up the distribution fraction, the central tool of the subject (Sections 1 and 2); it then uses that tool to calculate the hydrogen ion concentration of various acid-base systems (Section 3) and goes on to introduce the concept and the quantitative description of buffer solutions (Section 4). With these foundations in place, we begin with the simplest case, the titration of a strong acid with a strong base, and gradually construct de Levie's general equation for acid-base titration (Section 5). The general equation not only unifies every type of acid-base titration but also reveals the independent behavior of each component during the titration. Section 6 discusses how the endpoint is determined and how the titration error is evaluated, and Section 7 uses five representative applications drawn from national standards to show how acid-base titration is practiced.

Acid-base titration is a volumetric method built on acid-base reactions. It determines the content of an analyte by accurately measuring the volume of a standard solution that reacts exactly and completely with that analyte. As one of the most important techniques in chemical analysis, acid-base titration has developed over several hundred years, from the discovery of the earliest acid-base indicators to the widespread use of modern automatic titrators, and many scientists have made seminal contributions to the field.

In the mid-seventeenth century the English chemist Robert Boyle found by

chance, in 1663, that hydrochloric acid splashed onto violet petals turned them from purple to red. This observation prompted him to test systematically how various acids and bases affect violets and other plant pigments: acids redden the violets, whereas bases may turn them blue or give other colors. He then studied the color changes of dozens of plant extracts and found that litmus infusion is red in acid and blue in base. Because this color change was reproducible, litmus became the earliest acid-base indicator used systematically in the history of chemistry. He also soaked slips of paper in litmus infusion to make "litmus paper," which was convenient for everyday use and was the forerunner of modern pH paper. This work provided a basis for the qualitative analysis of acids and bases at the time and supplied the essential idea for judging the endpoint in the acid-base titrations that followed.

In the eighteenth century the Swedish chemist Carl Wilhelm Scheele was the first to study systematically the properties of organic and inorganic acids. This period also produced the early definition "an acid is a substance that reacts with a base to form a salt," which gave the classification of acids and bases its first criterion. In 1729 the French chemist Claude Joseph Geoffroy added potassium carbonate to acetic acid in small portions and took the cessation of gas evolution as the endpoint, thereby linking an acid-base reaction with quantitative analysis for the first time. In 1767 the English chemist William Lewis recorded an important advance in *Experiments and Observations upon American Potash*: while titrating potassium carbonate with hydrochloric acid, he used litmus paper to locate the endpoint and weighed the standard solution consumed, thus carrying out the first quantitative acid-base analysis. This result marks the transition of titration from qualitative observation to quantitative measurement and is an important forerunner of modern titrimetric analysis. In 1791 the French chemist

Henri Descroizilles further devised a graduated “alkalimeter” that metered the liquid drop by drop so that the volume could be measured accurately; the device is regarded as the forerunner of the modern burette and laid the foundation for the later development of volumetric apparatus.

At the beginning of the nineteenth century Joseph Louis Gay-Lussac brought titration to industrial application. While determining the available chlorine of bleaching powder he used an indigo solution to indicate the endpoint, which solved the difficulty of judging the endpoint in the traditional process. More importantly, he improved the graduated burette and replaced the earlier weighing method, raising the precision of volume measurement to the level of 0.1 mL. This invention took acid-base titration from the laboratory to the factory, where it became a core means of quality control. In 1891 Wilhelm Ostwald proposed the theory of indicator color change: an indicator is essentially an organic weak acid or weak base whose molecular and ionic forms differ in color, and the point at which the color changes depends on the hydrogen ion concentration. This theory was the first to explain the action of indicators from the standpoint of chemical equilibrium; although it did not fully reveal the quantitative relation between pH and color, it provided a theoretical foothold for accurate pH measurement.

In 1909 the Danish chemist Søren Peter Lauritz Sørensen, while studying the properties of protein solutions, proposed the concept of “pH,” defining acidity as the negative logarithm of the hydrogen ion concentration ($-\lg[H^+]$) and establishing a standardized scale from 0 to 14. This work turned the description of acid and base strength from a vague “strong or weak” into a quantifiable number and thoroughly changed the logic by which acidity is determined. He also invented the “formaldehyde titration,” in which formaldehyde reacts with the amino group of an amino acid and liberates hydrogen ions, so that the amino acid content can

be determined accurately; the application of titration was thereby extended from inorganic chemistry to biochemistry.

After the middle of the twentieth century, the penetration of electronics pushed titration into the age of automation: from the 1950s on, commercial potentiometric titrators became common and replaced visual observation of an indicator color change by continuous monitoring of the solution potential with electrodes; from the 1970s on, computer-controlled automatic titrating systems automated sample loading, titration, and data recording, and the efficiency of analysis rose greatly compared with manual operation. In addition, nonaqueous titration (for samples sparingly soluble in water) and mixed indicators (which sharpen the color change) extended titration to the analysis of oils, drugs, polymers, and other complex samples.

From Boyle's litmus paper to the modern intelligent titrating system, the history of acid-base titration is a miniature of chemical analysis itself, "from qualitative to quantitative, from experience to theory, from manual to intelligent." Today it remains a core method in environmental monitoring (water acidity), the food industry (acidity determination), and the pharmaceutical field (drug purity testing), and the scientific logic behind it, "observation, induction, theory, and application," still points the way to further innovation.

This chapter sets out the basic theory of acid-base titration and its applications. To that end it introduces the general equation established by Robert de Levie and uses it as a tool for treating all kinds of titrating systems in a single framework. De Levie's way of constructing the general equation is direct and concise; to help the reader follow where it comes from, this chapter proceeds from the simple to the complex: it first builds the titration equation for a monoprotic acid (or base), then the equation for a single polyprotic acid, and finally the

equation for a mixture of acids. This stepwise construction brings out the pattern followed by the titration equation when different types of analyte are titrated and leads finally to the general equation. The value of the general equation is that a mathematical analysis makes the dynamic change in the concentration of every species during the titration visible, which deepens our understanding of the titration process and of the logic underlying the design of existing acid-base titration methods.

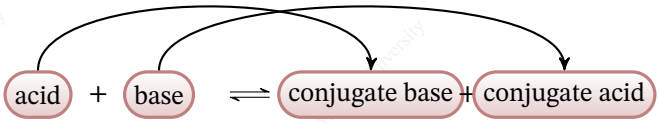
To achieve these goals the chapter concentrates on three themes. (1) The nature of the Brønsted-Lowry theory is reexamined. Taking "proton transfer" as its core, the theory describes the behavior of acids and bases in a unified way and reveals that the two are essentially interconvertible. A deeper understanding of the chemistry implied by proton transfer deepens our understanding of acid-base behavior and lifts the traditional view to a higher level. (2) The concept of the distribution fraction is understood accurately and its mathematical expression mastered. As a dimensionless parameter that expresses the proportion of a given species in an acid-base system, the distribution fraction has a clear physical meaning and a simple mathematical form; it is the essential basis for building mathematical models of acid-base equilibria. (3) The independence of the components of a polyprotic acid or an acid mixture during titration is fully recognized. Although several species coexist in such a system, the distribution of each changes independently with pH, so that each acid behaves independently during the titration. Grasping this property helps in understanding and handling the titration of complex acid-base systems.

## 4.1 The Brønsted-Lowry Theory and Dissociation Equilibria

The concepts of acid and base were first put forward by Svante August Arrhenius on the basis of his theory of electrolytic dissociation. That theory states that an acid is a substance that dissociates in aqueous solution to give hydrogen ions and the corresponding anion, and that a base dissociates to give hydroxide ions and the corresponding cation; Arrhenius was also the first to distinguish quantitatively between strong and weak acids by the degree of dissociation. As the scope of chemical research widened, however, the limitations of his theory became apparent. It cannot explain reasonably the basicity of many substances that contain no hydroxide ion and do not dissociate to give one, such as ammonia. Such substances usually acquire their basicity by reacting with other molecules, water among them, and producing hydroxide ions indirectly. A number of new theories of acids and bases were subsequently proposed, and after repeated testing and refinement the Brønsted-Lowry proton theory gained wide acceptance and became one of the mainstream theories by which acid-base properties are explained in chemistry today.

### 4.1.1 The Brønsted-Lowry Theory of Acids and Bases

In 1923 Thomas Martin Lowry and Johannes Nicolaus Brønsted independently proposed the proton theory of acids and bases, now called the Brønsted-Lowry model. According to this model, an acid is a proton donor and a base is a proton acceptor. An acid, after donating a proton, becomes its conjugate base, and a base, after accepting a proton, becomes its conjugate acid. The relation is summarized in the scheme below.




The proton theory has two distinguishing features. (1) The proton is the focus of the theory and proton transfer its key step, which breaks completely with the dependence on the hydroxide ion in the Arrhenius theory. (2) Acid and base are inseparably conjugate. When an acid donates a proton, the remainder becomes its conjugate base; correspondingly, when a base accepts a proton, the species formed is its conjugate acid. Acetic acid ($CH_3COOH$), for example, gives the acetate ion ($CH_3COO^-$) after donating a proton, and acetic acid and acetate form a conjugate acid-base pair.

More importantly, within this framework acid and base are a dialectical unity. We therefore need only describe proton transfer from the standpoint of the acid or from that of the base. We can go one step further: it is no longer necessary to decide whether a given substance is an acid or a base. All we need is to characterize quantitatively its ability to give up a proton. This strength of the proton theory will show itself more clearly when the general equation for acid-base titration is constructed later. Such a change of viewpoint may not come easily, but once it is grasped the reader's understanding rises to a higher level and the material that follows becomes much easier to follow.

**Stop and Think**

(1) Write the conjugate base and the conjugate acid of $HSO_4^-$. (2) Water acts as an acid in one reaction and as a base in another; what are the products of the two reactions? (3) Can you find a substance that can act only as an acid and never as a base? Can you find one that can act only as a base and never as an acid?

**Key Concept**

**The unified description in terms of $K_a$:** In the Brønsted-Lowry theory acids and bases are characterized uniformly by $K_a$. For any substance a single number, the $K_a$ of its conjugate acid, suffices to describe its acid-base behavior in aqueous solution. A large $K_a$ means a strong acid (and a very weak conjugate base), a small $K_a$ a weak acid (and a stronger conjugate base). This framework underlies the calculation of distribution fractions, the construction of titration equations, and the derivation of the $F$ function that follow.

The acid-base systems discussed in this book are mainly those in aqueous solution. In such systems the acid-base dissociation equilibrium between the solute and water is the central feature and occurs widely in chemical processes. A few simple examples are given below.

$$\underset{\text{acid}}{H_2O(l)} + \underset{\text{base}}{NH_3(aq)} \rightleftharpoons \underset{\text{conjugate base}}{OH^-(aq)} + \underset{\text{conjugate acid}}{NH_4^+(aq)}$$

$$\underset{\text{acid}}{HAc(aq)} + \underset{\text{base}}{H_2O(l)} \rightleftharpoons \underset{\text{conjugate base}}{Ac^-(aq)} + \underset{\text{conjugate acid}}{H_3O^+(aq)}$$

$$\underset{\text{acid}}{H_2O(l)} + \underset{\text{base}}{C_2H_5NH_2(aq)} \rightleftharpoons \underset{\text{conjugate base}}{OH^-(aq)} + \underset{\text{conjugate acid}}{C_2H_5NH_3^+(aq)}$$

Somewhat more complex acids such as $H_2CO_3$ and $H_3PO_4$ dissociate in water step by step. For $H_2CO_3$, for example,

$$\underset{\text{acid}}{H_2CO_3(aq)} + \underset{\text{base}}{H_2O(l)} \rightleftharpoons \underset{\text{conjugate base}}{HCO_3^-(aq)} + \underset{\text{conjugate acid}}{H_3O^+(aq)}$$

$$\underset{\text{acid}}{HCO_3^-(aq)} + \underset{\text{base}}{H_2O(l)} \rightleftharpoons \underset{\text{conjugate base}}{CO_3^{2-}(aq)} + \underset{\text{conjugate acid}}{H_3O^+(aq)}$$

Within the framework of conjugate pairs, water itself, the substance most familiar to us, shows a remarkable property: it is both an acid and a base. The property follows from the dissociation equilibrium of water itself, written as

$$\underset{\text{acid}}{H_2O(l)} + \underset{\text{base}}{H_2O(l)} \rightleftharpoons \underset{\text{conjugate base}}{OH^-(aq)} + \underset{\text{conjugate acid}}{H_3O^+(aq)}$$

Here $H_3O^+$ is the hydronium ion. The ionic radius of the hydrogen ion is so small that it cannot exist as a free species in aqueous solution; it is always present as the hydronium ion.

Another example is glycine, $NH_2CH_2COOH$, for which the following dissociation equilibria are established:

$$\underset{\text{acid}}{NH_2CH_2COOH} + \underset{\text{base}}{NH_2CH_2COOH} \rightleftharpoons \underset{\text{conjugate base}}{NH_2CH_2COO^-} + \underset{\text{conjugate acid}}{NH_3^+CH_2COOH}$$

### 4.1.2 Acid Dissociation Constants

The extent to which an acid dissociates in aqueous solution depends on the acid itself. At equilibrium the dissociation can be described by an equilibrium constant. Consider the acid dissociation equilibrium

$$HB(aq) + H_2O(l) \rightleftharpoons B^-(aq) + H_3O^+(aq)$$

The equilibrium concentrations of the ions in aqueous solution obey the relation

$$K^{\ominus}(HB) = \frac{\frac{[H_3O^+]}{c^{\ominus}} \times \frac{[B^-]}{c^{\ominus}}}{\frac{[HB]}{c^{\ominus}} \times \frac{[H_2O]}{c^{\ominus}}} \tag{4.1}$$

where $K^{\ominus}(HB)$ is the standard equilibrium constant, a dimensionless quantity; the square brackets denote the equilibrium concentration of the species enclosed, in mol/L; and $c^{\ominus}$ is the standard concentration, $c^{\ominus} = 1\,\text{mol/L}$.

Equation (4.1) contains the equilibrium concentration of water. In dilute acid solutions this concentration is essentially constant and roughly equal to 55.6 mol/L. It is therefore customary to merge the water concentration into the equilibrium constant and to define a parameter called the acid dissociation con-

stant, $K_a$ (the subscript a standing for acid), as follows:

$$K_a(\mathrm{HB}) = K^{\ominus}(\mathrm{HB}) \times \frac{[\mathrm{H_2O}]}{c^{\ominus}} = \frac{\frac{[\mathrm{H_3O^+}]}{c^{\ominus}} \times \frac{[\mathrm{B^-}]}{c^{\ominus}}}{\frac{[\mathrm{HB}]}{c^{\ominus}}} = \frac{\frac{[\mathrm{H^+}]}{c^{\ominus}} \times \frac{[\mathrm{B^-}]}{c^{\ominus}}}{\frac{[\mathrm{HB}]}{c^{\ominus}}} \tag{4.2}$$

where $[\mathrm{H^+}]$ has replaced $[\mathrm{H_3O^+}]$ for brevity.

According to equation (4.2), $K_a$ directly reflects the strength of the acid: the larger $K_a$, the stronger the tendency of the acid to transfer its proton to a water molecule (that is, to dissociate), and the stronger the acid. Note that $K_a$ depends markedly on the solvent, and the same acid has different dissociation constants in different solvents. Unless stated otherwise, what follows is restricted to aqueous systems.

In a like manner we may define the proton association constant of the conjugate base $\mathrm{B^-}$ of HB. The conjugate base $\mathrm{B^-}$ of the acid HB establishes the following equilibrium in aqueous solution:

$$\mathrm{B^-(aq) + H_2O(l) \rightleftharpoons HB(aq) + OH^-(aq)}$$

The proton association constant $K_b(\mathrm{B^-})$ is defined as

$$K_b(\mathrm{B^-}) = \frac{\frac{[\mathrm{HB}]}{c^{\ominus}} \times \frac{[\mathrm{OH^-}]}{c^{\ominus}}}{\frac{[\mathrm{B^-}]}{c^{\ominus}}} \tag{4.3}$$



In the traditional chemical literature $K_b$ is often called the base dissociation constant. Under the modern conjugate acid-base theory, however, a base is defined as a proton acceptor, and continuing to use the term "base dissociation constant" is logically inconsistent with that definition. This book therefore recommends renaming $K_b$ the "protonation constant of the base", a name that reflects its physical chemical nature more accurately.

Equations (4.2) and (4.3) give

$$\begin{aligned} K_a(\mathrm{HB}) \times K_b(\mathrm{B}^-) &= \frac{\frac{[\mathrm{H}^+]}{c^\ominus} \times \frac{[\mathrm{B}^-]}{c^\ominus}}{\frac{[\mathrm{HB}]}{c^\ominus}} \times \frac{\frac{[\mathrm{HB}]}{c^\ominus} \times \frac{[\mathrm{OH}^-]}{c^\ominus}}{\frac{[\mathrm{B}^-]}{c^\ominus}} \\ &= \frac{[\mathrm{H}^+]}{c^\ominus} \times \frac{[\mathrm{OH}^-]}{c^\ominus} \\ &= K_w = 1 \times 10^{-14} \end{aligned} \tag{4.4}$$

where $K_w$ is the ion-product constant for water.

This relation shows that $K_a$ and $K_b$ are strictly reciprocal: given either constant, the other follows from $K_w$. The relation reveals the essence of acid and base as a dialectical unity within the proton theory: they are opposed proton donor and acceptor, yet they are linked quantitatively through $K_w$.

We can go one step further: the value of $K_b$ is no longer needed, and $K_a$ alone suffices to characterize the ability of a substance to give up a proton. An acid has a large $K_a$; a base, a small one. For a strong base such as NaOH, which cannot give up a proton in water at all, $K_a(\mathrm{NaOH})$ may be given a value small enough, for example $K_a(\mathrm{NaOH}) = 10^{-16}$, that it has no detectable effect over the whole pH range of a titration. The benefit of this treatment will become apparent when distribution fractions are studied later.

In view of the above, and to respect historical usage, this book retains the $K_b$ value. To describe acid-base systems uniformly, however, the acid dissociation constant $K_a$ is used preferentially.

**Example 4.1** The reaction of $S^{2-}$ with water is given below. Find the dissociation constant $K_{a_2}$ of the conjugate acid of $S^{2-}$.

$$S^{2-}(aq) + H_2O(l) \rightleftharpoons HS^-(aq) + OH^-(aq), \quad K_{b_1} = 1.4$$

Solution: We are given

$$K_{b_1} = \frac{\frac{[HS^-]}{c^\ominus}\frac{[OH^-]}{c^\ominus}}{\frac{[S^{2-}]}{c^\ominus}}$$

Since the conjugate acid of $S^{2-}$ is $HS^-$, its dissociation reaction is

$$HS^-(aq) + H_2O(l) \rightleftharpoons S^{2-}(aq) + H_3O^+(aq)$$

Hence

$$K_{a_2} = \frac{\frac{[S^{2-}]}{c^\ominus}\frac{[H_3O^+]}{c^\ominus}}{\frac{[HS^-]}{c^\ominus}}$$

Because

$$K_{b_1} \times K_{a_2} = \frac{\frac{[HS^-]}{c^\ominus}\frac{[OH^-]}{c^\ominus}}{\frac{[S^{2-}]}{c^\ominus}} \times \frac{\frac{[S^{2-}]}{c^\ominus}\frac{[H_3O^+]}{c^\ominus}}{\frac{[HS^-]}{c^\ominus}} = K_w$$

we obtain

$$K_{a_2} = \frac{K_w}{K_{b_1}} = \frac{1 \times 10^{-14}}{1.4} = 7.1 \times 10^{-15}$$

A polyprotic acid dissociates in steps, and stepwise dissociation constants are needed to describe the process. For carbonic acid in aqueous solution, for example, the stepwise dissociations are

$$H_2CO_3(aq) + H_2O(l) \rightleftharpoons HCO_3^-(aq) + H_3O^+(aq)$$

$$HCO_3^-(aq) + H_2O(l) \rightleftharpoons CO_3^{2-}(aq) + H_3O^+(aq)$$

and the corresponding stepwise constants are

$$K_{a_1} = K_a(H_2CO_3) = \frac{\frac{[HCO_3^-]}{c^\ominus}\frac{[H^+]}{c^\ominus}}{\frac{[H_2CO_3]}{c^\ominus}} \tag{4.5}$$

and

$$K_{a_2} = K_a(HCO_3^-) = \frac{\frac{[CO_3^{2-}]}{c^\ominus}\frac{[H^+]}{c^\ominus}}{\frac{[HCO_3^-]}{c^\ominus}} \tag{4.6}$$

Here the subscripts of $K_{a_1}$ and $K_{a_2}$ denote the first and second dissociation constants of carbonic acid, whereas $K_a(H_2CO_3)$ and $K_a(HCO_3^-)$ express the same constants through the species concerned. For a polyprotic acid the subscript notation is relatively simple, and in this book it is used for stepwise constants whenever no confusion can arise.

## 4.2 The Distribution of Species in Acid-Base Solutions

A chemical system tends spontaneously toward equilibrium. When an acid or a base is mixed with water, interaction between the water molecules and the acid or base molecules sets up a proton transfer that continues until equilibrium is reached, and the solution then contains several acid and base species. The cases are treated separately below.

### 4.2.1 Distribution of Species in Monoprotic Acid Solutions

Take HB as an example. If part of it dissociates in aqueous solution to give $B^-$ while the rest remains as HB, both species are present at the same time. Let their equilibrium concentrations be [HB] and $[B^-]$; the sum of the two is the initial concentration of HB, $c$(HB), that is,

$$c(HB) = [HB] + [B^-] \tag{4.7}$$

Let $\delta$(HB) be the fraction of HB in the total concentration and $\delta(B^-)$ the fraction of $B^-$. These fractions are called the distribution fractions of the corre-

sponding species, and

$$\delta(\mathrm{HB}) = \frac{[\mathrm{HB}]}{c(\mathrm{HB})} = \frac{[\mathrm{HB}]}{[\mathrm{HB}] + [\mathrm{B}^-]} \tag{4.8}$$

$$\delta(\mathrm{B}^-) = \frac{[\mathrm{B}^-]}{c(\mathrm{HB})} = \frac{[\mathrm{B}^-]}{[\mathrm{HB}] + [\mathrm{B}^-]} \tag{4.9}$$

Since

$$K_\mathrm{a}(\mathrm{HB}) = \frac{\frac{[\mathrm{H}^+]}{c^\ominus}\frac{[\mathrm{B}^-]}{c^\ominus}}{\frac{[\mathrm{HB}]}{c^\ominus}} \tag{4.10}$$

equations (4.9) and (4.10) may be rearranged to give

$$\delta(\mathrm{HB}) = \frac{[\mathrm{H}^+]}{[\mathrm{H}^+] + K_\mathrm{a}(\mathrm{HB})c^\ominus} \tag{4.11}$$

$$\delta(\mathrm{B}^-) = \frac{K_\mathrm{a}(\mathrm{HB})c^\ominus}{[\mathrm{H}^+] + K_\mathrm{a}(\mathrm{HB})c^\ominus} \tag{4.12}$$

and the following relation holds:

$$\delta(\mathrm{HB}) + \delta(\mathrm{B}^-) = 1 \tag{4.13}$$

Equations (4.11) and (4.12) show that the distribution fractions depend only on the acid dissociation constant $K_\mathrm{a}$ and the hydrogen ion concentration, not on the total concentration. Plotting the distribution fractions of the species against pH gives the distribution fraction curves. Figure 4.1 shows the distribution fractions of $CH_3COOH$ and $CH_3COO^-$ in an acetic acid solution.

Four conclusions may be drawn from Figure 4.1: (1) $\delta(CH_3COOH)$ decreases as pH rises, while $\delta(CH_3COO^-)$ increases; (2) when $\delta(CH_3COOH) = \delta(CH_3COO^-) = 0.5$, acetic acid and acetate each account for half of the molecules in solution; (3) when $\mathrm{pH} < \mathrm{p}K_\mathrm{a}$ the predominant form is acetic acid; and (4) when $\mathrm{pH} > \mathrm{p}K_\mathrm{a}$ the predominant form is acetate.

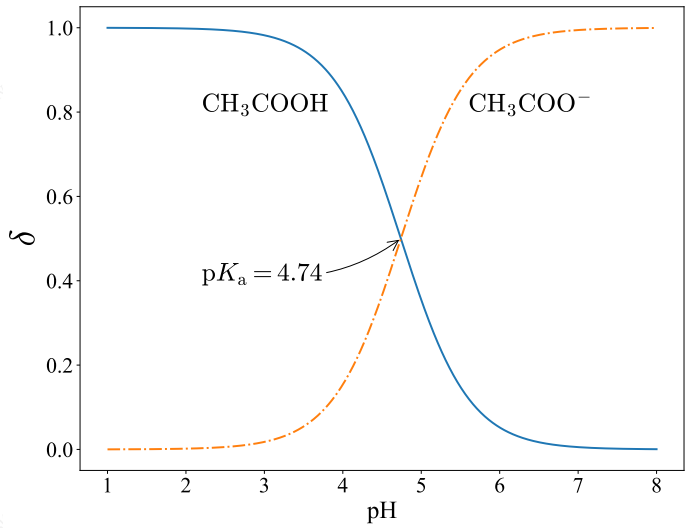



Figure 4.1: Distribution fraction curves of the two species in an acetic acid solution.

At the crossing point of the two curves the two species have the same concentration. From

$$K_\mathrm{a}(\mathrm{CH_3COOH}) = \frac{\frac{[\mathrm{H^+}]}{c^\ominus}\frac{[\mathrm{CH_3COO^-}]}{c^\ominus}}{\frac{[\mathrm{CH_3COOH}]}{c^\ominus}} \tag{4.14}$$

and $[\mathrm{CH_3COO^-}] = [\mathrm{CH_3COOH}]$ we obtain

$$K_\mathrm{a}(\mathrm{CH_3COOH}) = \frac{[\mathrm{H^+}]}{c^\ominus} \tag{4.15}$$

Taking the negative logarithm of both sides gives

$$-\lg K_\mathrm{a}(\mathrm{CH_3COOH}) = \mathrm{p}K_\mathrm{a}(\mathrm{CH_3COOH}) = \mathrm{pH} = -\lg \frac{[\mathrm{H^+}]}{c^\ominus} \tag{4.16}$$

The result tells us that the pH at the crossing point of the two curves is exactly the negative logarithm of the dissociation constant of acetic acid. In some cases this result can be used to estimate the dissociation constant of a weak acid approximately.

A strong monoprotic acid tends to dissociate completely in aqueous solution, so that only one species is important. Hydrochloric acid, for example, has $pK_a = -8.00$ and therefore $K_a = 10^{8.00}$. The distribution fractions of the two species in a dilute hydrochloric acid solution are

$$\delta(\mathrm{HCl}) = \frac{[\mathrm{H^+}]}{[\mathrm{H^+}] + 10^{8.00} c^\ominus} \approx 0 \tag{4.17}$$

$$\delta(\mathrm{Cl^-}) = \frac{10^{8.00} c^\ominus}{[\mathrm{H^+}] + 10^{8.00} c^\ominus} \approx 1 \tag{4.18}$$

These results show that hydrochloric acid is completely dissociated in water: no molecular species HCl remains, and the calculation agrees with the actual situation.

Textbooks differ in how they treat strong acids. Some state simply that they are "completely dissociated" and list no $K_a$ value; others give a large $K_a$ (the values in different sources are not always the same). The value $pK_a = -8.00$ for hydrochloric acid was used above to show that a strong acid is almost completely dissociated; to bring a strong acid into the unified treatment of distribution fractions and titration equations, it suffices to choose a value large enough, and the derivations that follow in this book take $K_a = 10^{10}$ as the limit of complete dissociation. The size of the value does not affect the calculated result; it serves only to ensure $\delta(\mathrm{Cl^-}) \approx 1$.

### 4.2.2 Distribution of Species in Monoprotic Base Solutions

The distribution fractions of the species of a weak monoprotic base in aqueous solution can be calculated directly with equations of the form of (4.11) and (4.12). For example, NaB establishes the following equilibrium in water:

$$\mathrm{B^-(aq) + H_2O(l) \rightleftharpoons HB(aq) + OH^-(aq)}$$

The species in solution are again $B^-$ and HB. The derivation of the distribution fractions of a monoprotic acid has already shown that the distribution fraction of a species depends only on the acid dissociation constant and on the pH of the solution. The distribution fractions of the species formed from NaB are therefore identical with those formed from HB, and equations (4.11) and (4.12) apply directly.

An example makes this clear. $NH_3$ establishes the following equilibrium in water:

$$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$

The species present are $NH_3$ and $NH_4^+$. Their distribution fractions depend only on the hydrogen ion concentration $[H^+]$ and on the acid dissociation constant of $NH_4^+$, not on how the species were formed. Hence

$$\delta(NH_3) = \frac{K_a(NH_4^+)c^\ominus}{[H^+] + K_a(NH_4^+)c^\ominus} \tag{4.19}$$

$$\delta(NH_4^+) = \frac{[H^+]}{[H^+] + K_a(NH_4^+)c^\ominus} \tag{4.20}$$

If any doubt remains about this procedure, the reader may turn to Exercise 4.6, which asks for the distribution fractions of the two species in an aqueous ammonia solution derived from the proton association constant. Using the proton association constant alone gives expressions different from (4.19) and (4.20); converted into acid dissociation constants, however, they agree with (4.19) and (4.20) exactly. These results confirm once more that, because acid and base form a dialectical unity, the behavior of an acid-base system can be described completely from the acid dissociation side alone.

For a compound such as the strong base NaOH, the same line of reasoning assigns a dissociation constant small enough, say $K_a = 10^{-16}$, so that it

fits into the unified framework of distribution fraction calculations.

### 4.2.3 Distribution of Species in Polyprotic Acid Solutions

The dissociation of a polyprotic acid is a little more complicated. Take oxalic acid, $H_2C_2O_4$ ($pK_{a_1} = 1.23$, $pK_{a_2} = 4.19$): its aqueous solution contains the three species $H_2C_2O_4$, $HC_2O_4^-$, and $C_2O_4^{2-}$, and the total concentration is

$$c(H_2C_2O_4) = [H_2C_2O_4] + [HC_2O_4^-] + [C_2O_4^{2-}] \tag{4.21}$$

The distribution fractions of the three species are

$$\delta(H_2C_2O_4) = \frac{[H_2C_2O_4]}{[H_2C_2O_4] + [HC_2O_4^-] + [C_2O_4^{2-}]} \tag{4.22}$$

$$\delta(HC_2O_4^-) = \frac{[HC_2O_4^-]}{[H_2C_2O_4] + [HC_2O_4^-] + [C_2O_4^{2-}]} \tag{4.23}$$

$$\delta(C_2O_4^{2-}) = \frac{[C_2O_4^{2-}]}{[H_2C_2O_4] + [HC_2O_4^-] + [C_2O_4^{2-}]} \tag{4.24}$$

With the relations

$$K_{a_1} = \frac{\frac{[H^+]}{c^\ominus}\frac{[HC_2O_4^-]}{c^\ominus}}{\frac{[H_2C_2O_4]}{c^\ominus}} \tag{4.25}$$

$$K_{a_2} = \frac{\frac{[H^+]}{c^\ominus}\frac{[C_2O_4^{2-}]}{c^\ominus}}{\frac{[HC_2O_4^-]}{c^\ominus}} \tag{4.26}$$

equations (4.22), (4.23), and (4.24) may be rearranged into the forms

$$\delta(H_2C_2O_4) = \frac{[H^+]^2}{[H^+]^2 + K_{a_1}c^\ominus[H^+] + K_{a_1}K_{a_2}(c^\ominus)^2} \tag{4.27}$$

$$\delta(HC_2O_4^-) = \frac{K_{a_1}c^\ominus[H^+]}{[H^+]^2 + K_{a_1}c^\ominus[H^+] + K_{a_1}K_{a_2}(c^\ominus)^2} \tag{4.28}$$

$$\delta(C_2O_4^{2-}) = \frac{K_{a_1}K_{a_2}(c^\ominus)^2}{[H^+]^2 + K_{a_1}c^\ominus[H^+] + K_{a_1}K_{a_2}(c^\ominus)^2} \tag{4.29}$$

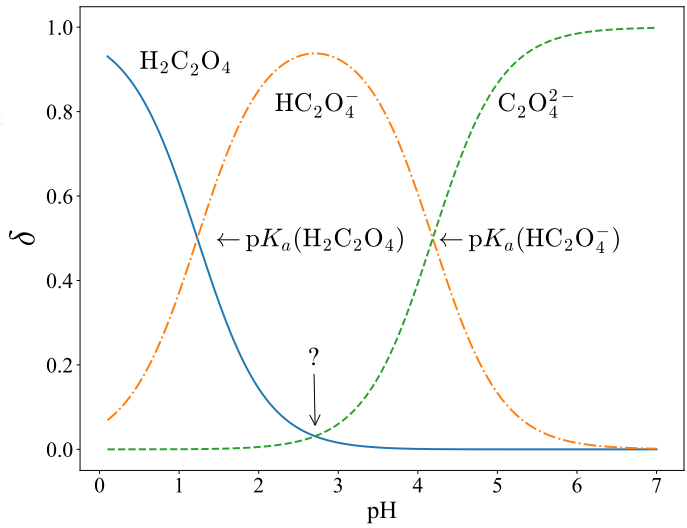



Figure 4.2: Distribution fraction curves of the species in an oxalic acid solution.

Figure 4.2 shows the distribution fraction curves of the species of oxalic acid. When pH $<$ $pK_{a_1}$ the predominant species is $H_2C_2O_4$; when pH $>$ $pK_{a_2}$ it is $C_2O_4^{2-}$; and when $pK_{a_2}$ $>$ pH $>$ $pK_{a_1}$ it is $HC_2O_4^-$. Within this interval there is a second crossing point, where $[H_2C_2O_4] = [C_2O_4^{2-}]$; a short calculation gives the pH at this point as

$$\mathrm{pH} = \frac{pK_{a_1} + pK_{a_2}}{2} = \frac{1.23 + 4.19}{2} = 2.71 \tag{4.30}$$

At this crossing point $\delta(HC_2O_4^-)$ reaches its maximum, while the distribution fractions of $H_2C_2O_4$ and $C_2O_4^{2-}$ are only 0.031.

For an $n$-protic acid $H_nB$, the distribution fractions of the species take the general form

$$\delta(H_pB) = \frac{[H^+]^p \prod_{j=0}^{n-p}(K_{a_j} c^{\ominus})}{\sum_{i=0}^{n}\left([H^+]^i \prod_{j=0}^{n-i}(K_{a_j} c^{\ominus})\right)} \tag{4.31}$$

where $K_{a_0} \equiv 1$ and $\prod$ denotes a continued product.

With equation (4.31) the distribution fractions of the species in an aqueous

solution of $H_3PO_4$ may be written as

$$\delta(H_3PO_4) = \frac{[H^+]^3}{[H^+]^3 + K_{a_1}c^\ominus[H^+]^2 + K_{a_1}K_{a_2}(c^\ominus)^2[H^+] + K_{a_1}K_{a_2}K_{a_3}(c^\ominus)^3} \tag{4.32}$$

$$\delta(H_2PO_4^-) = \frac{K_{a_1}c^\ominus[H^+]^2}{[H^+]^3 + K_{a_1}c^\ominus[H^+]^2 + K_{a_1}K_{a_2}(c^\ominus)^2[H^+] + K_{a_1}K_{a_2}K_{a_3}(c^\ominus)^3} \tag{4.33}$$

$$\delta(HPO_4^{2-}) = \frac{K_{a_1}K_{a_2}(c^\ominus)^2[H^+]}{[H^+]^3 + K_{a_1}c^\ominus[H^+]^2 + K_{a_1}K_{a_2}(c^\ominus)^2[H^+] + K_{a_1}K_{a_2}K_{a_3}(c^\ominus)^3} \tag{4.34}$$

$$\delta(PO_4^{3-}) = \frac{K_{a_1}K_{a_2}K_{a_3}(c^\ominus)^3}{[H^+]^3 + K_{a_1}c^\ominus[H^+]^2 + K_{a_1}K_{a_2}(c^\ominus)^2[H^+] + K_{a_1}K_{a_2}K_{a_3}(c^\ominus)^3} \tag{4.35}$$

where $K_{a_1}$, $K_{a_2}$, and $K_{a_3}$ are the three dissociation constants of phosphoric acid.

Equation (4.31) does look hard to remember, but it need not be memorized. The expressions for the distribution fractions follow a very clear pattern, which becomes apparent on looking at those for carbonic and phosphoric acids; it may be summarized as follows.

1. For an $n$-protic parent acid there are $n + 1$ species in solution and therefore $n + 1$ distribution fractions. The denominator of each fraction contains $n + 1$ terms.
2. In the denominator the highest power of $[H^+]$ is $n$ (for $H_3PO_4$, the third power).
3. Moving from one term to the next, the exponent of $[H^+]$ falls by one until it reaches zero (the constant term), while the stepwise dissociation constants $K_{a_1}$, $K_{a_2}$, and so on take the place of the corresponding powers of $[H^+]$.
4. The numerator is simply a cyclic permutation of the terms of the denominator.

Once this pattern is grasped, the distribution fractions of any polyprotic acid can be written down with ease. It applies as well to acid salts, for which the distribution fractions of the species in solution may be written directly by the same rule. The interested reader may try it.

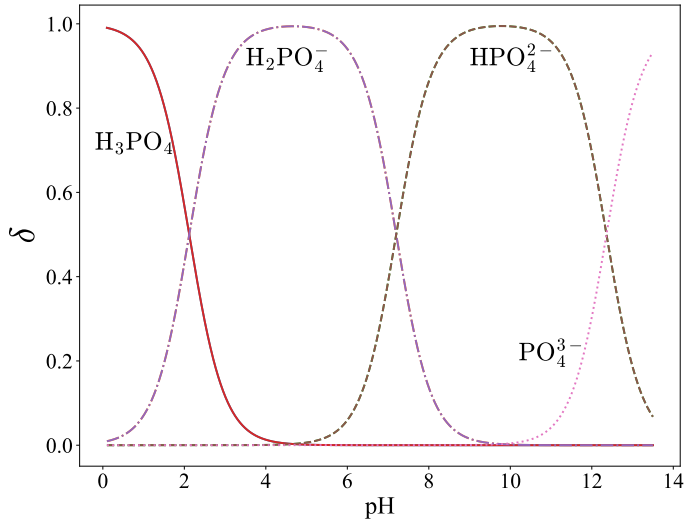



Figure 4.3: Distribution fraction curves of the species in a phosphoric acid solution.

Figure 4.3 shows the distribution fraction curves of the species of phosphoric acid. The reader is invited to analyze this system from the pattern of the curves; see Exercise 4.4. Note that the intersections of the curves are directly related to the acid dissociation constants, and that familiarity with this relation helps in following the changes in each species.

Calculating distribution fractions by hand is tedious. The author has written an Octave program (`abt_delta.m`) for the purpose, listed in Appendix B, together with a demonstration program `demo_ac_abt.m` that shows how the parameters are set and the program is run.

**Key Concept**

**The two core properties of the distribution fraction: (1)** it depends only on $K_a$ and $[H^+]$, not on the analytical concentration, so that whether one starts from HB or from NaB, the proportions of the species at the same pH are the same; (2) at $pH = pK_a$, $\delta(HB) = \delta(B^-) = 0.5$, so the crossing point locates $pK_a$ directly.

## 4.3 Calculating the Hydrogen Ion Concentration of Acid-Base Solutions

The expressions for the distribution fractions show that the distribution fraction of each species in an acid (or base) solution depends on the hydrogen ion concentration of the solution. Once that concentration is fixed, the distribution fractions and hence the equilibrium concentrations of all species follow. To describe the concentration of every component of an acid-base solution quantitatively, therefore, the hydrogen ion concentration must first be calculated accurately. The problem reduces to setting up a mathematical equation for the hydrogen ion concentration, and this section treats the various cases separately.

### 4.3.1 Hydrogen Ion Concentration in Solutions of a Strong Monoprotic Acid (Base)

Consider an aqueous solution of hydrochloric acid of total concentration $c(HCl)$, in which the two components HCl and $H_2O$ establish the two dissociation equilibria

$$HCl(aq) \rightleftharpoons H^+(aq) + Cl^-(aq)$$

$$H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$$

In this solution the hydrogen ion has two sources, the dissociation of hy-

drochloric acid and the self-dissociation of water. Because hydrogen ions from the two sources are chemically identical, their origin cannot be distinguished. More importantly, the dissociation of hydrochloric acid strongly suppresses the self-dissociation of water (the common-ion effect), while the self-dissociation in turn has a slight influence on the dissociation of the acid. This mutual constraint means that the exact hydrogen ion concentration cannot be obtained simply from the initial concentration; an equation describing the equilibrium system must be set up and solved.

Hydrochloric acid is completely dissociated in water, so

$$c(\mathrm{HCl}) = [\mathrm{Cl}^-] \tag{4.36}$$

From the charge balance,

$$[\mathrm{H}^+] = [\mathrm{OH}^-] + [\mathrm{Cl}^-] \tag{4.37}$$

and therefore

$$[\mathrm{H}^+] = [\mathrm{OH}^-] + c(\mathrm{HCl}) \tag{4.38}$$

Since $K_\mathrm{w} = \frac{[\mathrm{OH}^-]}{c^\ominus}\frac{[\mathrm{H}^+]}{c^\ominus}$, that is, the hydroxide concentration is conjugate to the hydrogen ion concentration, equation (4.38) is in essence an equation for the equilibrium concentration of the hydrogen ion and is also called the proton balance equation. Rearranged, it gives the quadratic

$$[\mathrm{H}^+]^2 - c(\mathrm{HCl})[\mathrm{H}^+] - K_\mathrm{w}(c^\ominus)^2 = 0 \tag{4.39}$$

whose exact solution is

$$[\mathrm{H}^+] = \frac{c(\mathrm{HCl}) + \sqrt{c(\mathrm{HCl})^2 + 4K_\mathrm{w}(c^\ominus)^2}}{2} \tag{4.40}$$

Equation (4.40) shows that the controlling variable is the initial concentration of hydrochloric acid. When $c(\mathrm{HCl})$ is large enough compared with $K_\mathrm{w}$, the

hydrogen ion concentration may be calculated as

$$[\mathrm{H^+}] \approx c(\mathrm{HCl}) \tag{4.41}$$

As the concentration of the acid decreases, the hydrogen ions released by water become non-negligible and equation (4.41) must be used with care. Since the hydrogen ion concentration from water is about $1 \times 10^{-7}$ mol/L, however, the approximation still holds as long as the acid concentration is reasonably large. With $c(\mathrm{HCl}) = 1.000 \times 10^{-6}$ mol/L, for example, the exact equation gives $[\mathrm{H^+}] = 1.010 \times 10^{-6}$ mol/L, and the relative error introduced by the simplest equation is

$$\begin{aligned} \epsilon_r([\mathrm{H^+}]) &= \frac{[\mathrm{H^+}]_{\text{近似}} - [\mathrm{H^+}]_{\text{准确}}}{[\mathrm{H^+}]_{\text{准确}}} \times 100\% \\ &= \frac{1.000 \times 10^{-6} - 1.010 \times 10^{-6}}{1.010 \times 10^{-6}} \times 100\% \\ &= -1.0\% \end{aligned} \tag{4.42}$$

Even at such a low concentration of hydrochloric acid, the error caused by the approximation is negligible.

The hydrogen ion concentration of a solution of a strong monoprotic base is calculated in a similar way. For an aqueous solution of NaOH, the proton balance equation is

$$[\mathrm{OH^-}] = [\mathrm{H^+}] + c(\mathrm{NaOH}) \tag{4.43}$$

and again, when the concentration of sodium hydroxide is reasonably large, the approximation

$$[\mathrm{OH^-}] \approx c(\mathrm{NaOH}) \tag{4.44}$$

may be used.

### 4.3.2 Hydrogen Ion Concentration in Solutions of a Weak Monoprotic Acid (Base)

A weak acid differs from a strong acid in that it dissociates only partly in aqueous solution; the number of hydrogen ions it supplies to the solution is limited by its dissociation constant. For an aqueous solution of the weak monoprotic acid HB the equilibrium is

$$\mathrm{HB(aq)} \rightleftharpoons \mathrm{H^+(aq) + B^-(aq)}, \quad K_\mathrm{a} = \frac{\frac{[\mathrm{H^+}]}{c^\ominus}\frac{[\mathrm{B^-}]}{c^\ominus}}{\frac{[\mathrm{HB}]}{c^\ominus}}$$

and the proton balance equation is

$$[\mathrm{H^+}] = [\mathrm{OH^-}] + [\mathrm{B^-}] \tag{4.45}$$

that is,

$$[\mathrm{H^+}] = \frac{K_\mathrm{w}c^\ominus}{\frac{[\mathrm{H^+}]}{c^\ominus}} + c(\mathrm{HB}) \times \frac{K_\mathrm{a}c^\ominus}{[\mathrm{H^+}] + K_\mathrm{a}c^\ominus} \tag{4.46}$$

Expanding equation (4.46) gives

$$[\mathrm{H^+}]^3 + K_\mathrm{a}c^\ominus[\mathrm{H^+}]^2 - \left(K_\mathrm{a}c^\ominus c(\mathrm{HB}) + K_\mathrm{w}(c^\ominus)^2\right)[\mathrm{H^+}] - K_\mathrm{a}K_\mathrm{w}(c^\ominus)^3 = 0 \tag{4.47}$$

Equation (4.47) is the exact expression for the hydrogen ion concentration of a weak monoprotic acid. Its solution by hand is still laborious, and an Octave program that solves it will be given when the titration equations are discussed. Here we look for reasonable approximations. When the product of the acid concentration and the dissociation constant is large enough, the dissociation of water may be ignored, that is, the term $K_\mathrm{a}K_\mathrm{w}(c^\ominus)^3$ may be dropped. It is generally accepted that when $K_\mathrm{a}c^\ominus c(\mathrm{HB}) > 20K_\mathrm{w}(c^\ominus)^2$, the term $K_\mathrm{w}(c^\ominus)^2$ in the bracket may be ignored, and then

$$[\mathrm{H^+}]^2 + K_\mathrm{a}c^\ominus[\mathrm{H^+}] - K_\mathrm{a}c^\ominus c(\mathrm{HB}) = 0 \tag{4.48}$$

whose solution is

$$[\mathrm{H}^+] = \frac{-K_\mathrm{a}c^\ominus + \sqrt{(K_\mathrm{a}c^\ominus)^2 + 4K_\mathrm{a}c^\ominus c(\mathrm{HB})}}{2} \tag{4.49}$$

Equation (4.49) can be simplified further. When $c(\mathrm{HB}) > 500K_\mathrm{a}c^\ominus$, the free $K_\mathrm{a}$ terms may all be ignored, and

$$[\mathrm{H}^+] = \sqrt{K_\mathrm{a}c^\ominus c(\mathrm{HB})} \tag{4.50}$$

The hydrogen ion concentration of a solution of a weak monoprotic base is calculated in the same way; it suffices to replace $[\mathrm{H}^+]$ by $[\mathrm{OH}^-]$ and $K_\mathrm{a}$ by $K_\mathrm{b}$, because a base normally takes hydrogen ions from water and thereby produces $[\mathrm{OH}^-]$ indirectly, its behavior being in fact similar to that of an acid. The reader may try to summarize the results.

**Example 4.2** Calculate the hydrogen ion concentration of a 0.10 mol/L solution of HF. For HF, $K_\mathrm{a} = 6.6 \times 10^{-4}$.

Solution: Since

$$K_\mathrm{a}c(\mathrm{HF}) = 0.10\,\mathrm{mol/L} \times 6.6 \times 10^{-4} = 6.6 \times 10^{-5}\,\mathrm{mol/L} > 20K_\mathrm{w}c^\ominus$$

but

$$c(\mathrm{HF}) = 0.10\,\mathrm{mol/L} < 500K_\mathrm{a}c^\ominus = 0.33\,\mathrm{mol/L}$$

we have

$$\begin{aligned}[\mathrm{H}^+] &= \frac{-K_\mathrm{a}c^\ominus + \sqrt{(K_\mathrm{a}c^\ominus)^2 + 4K_\mathrm{a}c^\ominus c(\mathrm{HF})}}{2} \\ &= \frac{-6.6 \times 10^{-4}\,\mathrm{mol/L} + \sqrt{(6.6 \times 10^{-4}\,\mathrm{mol/L})^2 + 4 \times 0.10 \times 6.6 \times 10^{-4}(\mathrm{mol/L})^2}}{2} \\ &= 7.8 \times 10^{-3}\,\mathrm{mol/L}\end{aligned}$$

**Example 4.3** Calculate the hydrogen ion concentration of a 0.10 mol/L solution of acetic acid. For acetic acid, $K_a = 1.8 \times 10^{-5}$.

Solution: Since

$$c(CH_3COOH)K_a = 1.8 \times 10^{-6}\ \text{mol/L} > 20K_w c^{\ominus}$$

and

$$c(CH_3COOH) > 500 \times K_a c^{\ominus} = 9.0 \times 10^{-3}\ \text{mol/L}$$

the approximation may be used:

$$\begin{aligned}[H^+] &= \sqrt{K_a c(CH_3COOH)c^{\ominus}} \\ &= \sqrt{0.10 \times 1.8 \times 10^{-5}(\text{mol/L})^2} \\ &= 1.3 \times 10^{-3}\ \text{mol/L}\end{aligned}$$

### 4.3.3 Hydrogen Ion Concentration in Solutions of a Polyprotic Acid (Base)

The stepwise dissociation equilibria in a solution of a polyprotic acid make the calculation of the hydrogen ion concentration more involved. For $H_2B$, for example, the two dissociation equilibria are

$$H_2B \rightleftharpoons H^+ + HB^-, \quad K_{a_1} = \frac{\frac{[H^+]}{c^{\ominus}} \times \frac{[HB^-]}{c^{\ominus}}}{\frac{[H_2B]}{c^{\ominus}}}$$

$$HB^- \rightleftharpoons H^+ + B^{2-}, \quad K_{a_2} = \frac{\frac{[H^+]}{c^{\ominus}} \times \frac{[B^{2-}]}{c^{\ominus}}}{\frac{[HB^-]}{c^{\ominus}}}$$

where $K_{a_1}$ and $K_{a_2}$ are the first and second dissociation constants. The proton balance equation of the system is

$$[H^+] = [HB^-] + 2[B^{2-}] + [OH^-] \tag{4.51}$$

Substituting the equilibrium concentrations gives

$$[H^+] = \frac{c(H_2B)K_{a_1}c^\ominus[H^+]}{[H^+]^2 + K_{a_1}c^\ominus[H^+] + K_{a_1}K_{a_2}(c^\ominus)^2} + \ldots$$
$$\ldots + \frac{2c(H_2B)K_{a_1}K_{a_2}(c^\ominus)^2}{[H^+]^2 + K_{a_1}c^\ominus[H^+] + K_{a_1}K_{a_2}(c^\ominus)^2} + \frac{K_w c^\ominus}{\frac{[H^+]}{c^\ominus}} \tag{4.52}$$

which rearranges to

$$[H^+]^4 + K_{a_1}c^\ominus[H^+]^3 + \left(K_{a_1}K_{a_2}(c^\ominus)^2 - c(H_2B)K_{a_1}c^\ominus - K_w(c^\ominus)^2\right)[H^+]^2 - \ldots$$
$$\ldots - \left(2c(H_2B)K_{a_1}K_{a_2}(c^\ominus)^2 + K_w(c^\ominus)^3K_{a_1}\right)[H^+] - K_wK_{a_1}K_{a_2}(c^\ominus)^4 = 0 \tag{4.53}$$

This is a quartic equation, and solving it by hand is extremely difficult. Approximate solutions may still be sought in certain cases. If the acid is fairly concentrated and fairly strong, for example, $K_{a_1}$ is large and the $K_w$ terms may be ignored, so that equation (4.53) reduces to

$$[H^+]^3 + K_{a_1}c^\ominus[H^+]^2 + \left(K_{a_1}K_{a_2}(c^\ominus)^2 - c(H_2B)K_{a_1}c^\ominus\right)[H^+] - \ldots$$
$$\ldots - 2c(H_2B)K_{a_1}K_{a_2}(c^\ominus)^2 = 0 \tag{4.54}$$

If the first dissociation is complete, $K_{a_1} = 10^{10}$; dividing both sides by $K_{a_1}$ and rearranging gives

$$[H^+]^2 + \left(K_{a_2}c^\ominus - c(H_2B)\right)[H^+] - 2c(H_2B)K_{a_2}c^\ominus = 0 \tag{4.55}$$

This is a quadratic equation and can be solved by hand. Simplification for special cases is feasible, but the result is approximate in any case. More accu-

rate results may be obtained with the Kapok program written by the author. Figures 4.4 and 4.5 show the parameter file and the calculated titration curve produced by the program. For a 0.1000 mol/L solution of $H_3PO_4$ the initial pH is about 1.62.

Note that the initial pH reported by the Kapok program is the value obtained when the volume ratio of titrant to sample, $\eta = V_t/V_0$, is small enough; it is not the value at $\eta = 0$.

### 4.3.4 Hydrogen Ion Concentration in Solutions of Amphoteric Substances

An amphoteric substance is one that can both donate and accept a proton. Common examples include acid salts (such as $NaHCO_3$, $NaH_2PO_4$, and $Na_2HPO_4$) and salts of a weak acid with a weak base (such as $NH_4Ac$). Such a substance behaves in aqueous solution in a way intermediate between an acid and a base, and its hydrogen ion concentration cannot be obtained simply by applying the formula for a weak acid or a weak base; it must be derived systematically from the proton balance equation.

#### Amphoteric Substances of the NaHB Type

Take the acid salt NaHB as an example. It forms from the diprotic acid $H_2B$ and NaOH in a 1:1 molar ratio. In aqueous solution $HB^-$ may dissociate as an acid and donate a proton,

$$HB^-(aq) + H_2O(l) \rightleftharpoons B^{2-}(aq) + H_3O^+(aq)$$

or accept a proton as a base:

$$HB^-(aq) + H_2O(l) \rightleftharpoons H_2B(aq) + OH^-(aq)$$

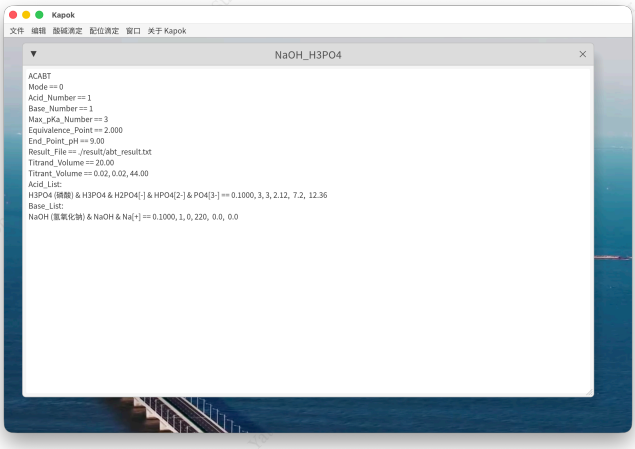



Figure 4.4: Parameter file for the titration of $H_3PO_4$ with NaOH.

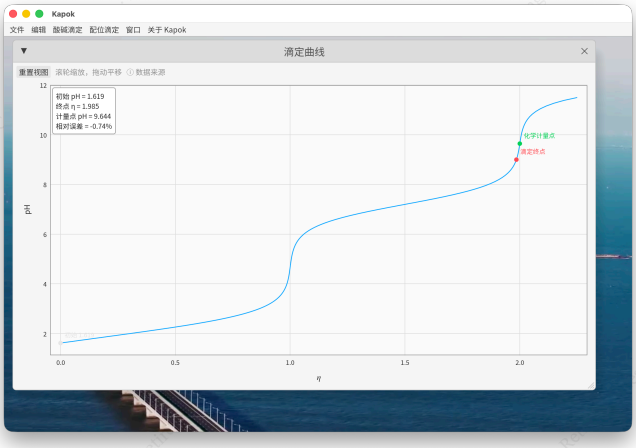



Figure 4.5: Titration curve and related parameters for the titration of $H_3PO_4$ with NaOH.

Let the analytical concentration of NaHB be $c$(NaHB). The species present are $H_2B$, $HB^-$, $B^{2-}$, $H^+$, $OH^-$, and $Na^+$. The mass balances are

$$[Na^+] = c(\text{NaHB}) \tag{4.56}$$

$$[H_2B] + [HB^-] + [B^{2-}] = c(\text{NaHB}) \tag{4.57}$$

The charge balance is

$$[Na^+] + [H^+] = [HB^-] + 2[B^{2-}] + [OH^-] \tag{4.58}$$

Substituting (4.56) into (4.58) and eliminating $[HB^-]$ with the aid of (4.57) gives the proton balance equation

$$[H^+] + [H_2B] = [B^{2-}] + [OH^-] \tag{4.59}$$

This relation may be understood directly in terms of proton reference levels. With $HB^-$ and $H_2O$ as the references, $H_2B$ and $H_3O^+$ (that is, $H^+$) are the products that have gained a proton, while $B^{2-}$ and $OH^-$ are the products that have lost one, and the two groups are equal in number.

Expressing the equilibrium concentrations of the species through distribution fractions,

$$[H_2B] = \delta(H_2B) \times c(\text{NaHB}), \quad [B^{2-}] = \delta(B^{2-}) \times c(\text{NaHB}) \tag{4.60}$$

Substituting into (4.59) gives

$$[H^+] + \delta(H_2B)c(\text{NaHB}) = \delta(B^{2-})c(\text{NaHB}) + \frac{K_w c^\ominus}{[H^+]/c^\ominus} \tag{4.61}$$

where the distribution fractions of the diprotic acid are

$$\delta(H_2B) = \frac{[H^+]^2}{[H^+]^2 + K_{a_1}c^\ominus[H^+] + K_{a_1}K_{a_2}(c^\ominus)^2}$$

$$\delta(\mathrm{B}^{2-}) = \frac{K_{\mathrm{a}_1}K_{\mathrm{a}_2}(c^{\ominus})^2}{[\mathrm{H}^+]^2 + K_{\mathrm{a}_1}c^{\ominus}[\mathrm{H}^+] + K_{\mathrm{a}_1}K_{\mathrm{a}_2}(c^{\ominus})^2}$$

Equation (4.61) is the exact expression for the hydrogen ion concentration of a NaHB solution. Substituting the two distribution fractions and rearranging gives a quartic equation in $[\mathrm{H}^+]$:

$$\begin{aligned}[\mathrm{H}^+]^4 + \left(K_{\mathrm{a}_1}c^{\ominus} + c(\mathrm{NaHB})\right)[\mathrm{H}^+]^3 + \left(K_{\mathrm{a}_1}K_{\mathrm{a}_2} - K_{\mathrm{w}}\right)(c^{\ominus})^2[\mathrm{H}^+]^2 \\ - \left(c(\mathrm{NaHB})K_{\mathrm{a}_1}K_{\mathrm{a}_2} + K_{\mathrm{w}}K_{\mathrm{a}_1}c^{\ominus}\right)(c^{\ominus})^2[\mathrm{H}^+] - K_{\mathrm{w}}K_{\mathrm{a}_1}K_{\mathrm{a}_2}(c^{\ominus})^4 = 0\end{aligned} \tag{4.62}$$

Solving the quartic by hand is inconvenient, and the step is not carried out here; numerical solutions may be obtained with the Kapok program. Solutions of amphoteric substances share a common feature, however: as long as the analytical concentration is not very small, the species that predominates in solution is the amphoteric substance itself, so that

$$[\mathrm{HB}^-] \approx c(\mathrm{NaHB})$$

This approximation is the source of every simplification in this section and is independent of the two dissociation constant expressions themselves. Treating $[\mathrm{HB}^-]$ for the moment as unknown, the definitions of $K_{\mathrm{a}_1}$ and $K_{\mathrm{a}_2}$ give the concentrations of the other two species directly:

$$[\mathrm{H_2B}] = \frac{[\mathrm{H}^+][\mathrm{HB}^-]}{K_{\mathrm{a}_1}c^{\ominus}}, \qquad [\mathrm{B}^{2-}] = \frac{K_{\mathrm{a}_2}c^{\ominus}[\mathrm{HB}^-]}{[\mathrm{H}^+]} \tag{4.63}$$

This step merely rewrites the definitions and introduces no new approximation. Substituting the two expressions into the proton balance equation (4.59),

$$[\mathrm{H}^+] + \frac{[\mathrm{H}^+][\mathrm{HB}^-]}{K_{\mathrm{a}_1}c^{\ominus}} = \frac{K_{\mathrm{a}_2}c^{\ominus}[\mathrm{HB}^-]}{[\mathrm{H}^+]} + \frac{K_{\mathrm{w}}c^{\ominus}}{[\mathrm{H}^+]/c^{\ominus}}$$

Multiplying both sides by $[\mathrm{H}^+]$ and collecting the terms in $[\mathrm{HB}^-]$,

$$[\mathrm{H}^+]^2\left(1 + \frac{[\mathrm{HB}^-]}{K_{\mathrm{a}_1}c^{\ominus}}\right) = K_{\mathrm{a}_2}c^{\ominus}[\mathrm{HB}^-] + K_{\mathrm{w}}(c^{\ominus})^2$$

Substituting $[HB^-] \approx c(\mathrm{NaHB})$ then gives the intermediate expression for the hydrogen ion concentration of an amphoteric solution:

$$[H^+] = c^\ominus \sqrt{\frac{K_{a_1}\left(K_{a_2}\dfrac{c(\mathrm{NaHB})}{c^\ominus} + K_w\right)}{K_{a_1} + \dfrac{c(\mathrm{NaHB})}{c^\ominus}}} \tag{4.64}$$

The intermediate expression contains the conclusions within it: if $c(\mathrm{NaHB}) \gg K_{a_1}c^\ominus$, the $K_{a_1}$ in the denominator may be dropped; if $K_{a_2}c(\mathrm{NaHB}) \gg K_w c^\ominus$, the $K_w$ in the numerator may be dropped. When both conditions hold at once, the hydrogen ion concentration is fixed by the two dissociation constants alone:

$$[H^+] \approx \sqrt{K_{a_1}K_{a_2}}\, c^\ominus \tag{4.65}$$

$$\mathrm{pH} \approx \frac{pK_{a_1} + pK_{a_2}}{2} \tag{4.66}$$

Equations (4.65) and (4.66) are the simplest expressions for the pH of a NaHB-type amphoteric solution. They lead to a neat conclusion: the pH of such a solution is approximately independent of the analytical concentration $c(\mathrm{NaHB})$ and depends only on the geometric mean of two adjacent dissociation constants. In chemical terms, $HB^-$ dissociates as an acid and as a base at the same time, and the two tendencies restrain each other. When the amount of $H_2B$ formed equals the amount of $B^{2-}$ formed, the system is at equilibrium and the pH is exactly halfway between $pK_{a_1}$ and $pK_{a_2}$, which agrees with the conclusion drawn at the crossing point $\delta(H_2B) = \delta(B^{2-})$ of the distribution fraction curves (see equation (4.30)).

It should be pointed out that the intermediate expression (4.64), accurate as it is compared with the simplest expression, still rests on $[HB^-] \approx c(\mathrm{NaHB})$. When $c(\mathrm{NaHB})$ is comparable to or smaller than $K_{a_1}c^\ominus$, the approximation fails and both expressions deviate noticeably from the exact value; one must then return to the

proton balance equation (4.59) or obtain a numerical solution with the Kapok program.

### Amphoteric Substances of the $NaH_2PO_4$ Type

For an acid salt of a triprotic acid such as $NaH_2PO_4$, there are two directions of acid dissociation and one of base dissociation. With $H_2PO_4^-$ and $H_2O$ as the proton reference levels, the proton balance equation is

$$[H^+] + [H_3PO_4] = [HPO_4^{2-}] + 2[PO_4^{3-}] + [OH^-] \tag{4.67}$$

There are four terms in the equation. At ordinary analytical concentrations both $[H^+]$ and $[OH^-]$ may be ignored. $PO_4^{3-}$ is the product of the third dissociation, and the expression for $K_{a_3}$ gives

$$[PO_4^{3-}] = K_{a_3} c^{\ominus} [HPO_4^{2-}]/[H^+]$$

which in an acid salt solution is more than seven orders of magnitude smaller than $[HPO_4^{2-}]$, so that the term $2[PO_4^{3-}]$ may also be dropped. Since the acid and base dissociations of $H_2PO_4^-$ are both slight, $[H_2PO_4^-] \approx c(NaH_2PO_4)$. Equation (4.67) then reduces to

$$[H_3PO_4] \approx [HPO_4^{2-}] \tag{4.68}$$

The remaining steps are exactly parallel to those of Section 3.4.1. The concentrations of the two species concerned follow from the definitions of $K_{a_1}$ and $K_{a_2}$:

$$[H_3PO_4] = \frac{[H^+][H_2PO_4^-]}{K_{a_1} c^{\ominus}}, \qquad [HPO_4^{2-}] = \frac{K_{a_2} c^{\ominus} [H_2PO_4^-]}{[H^+]}$$

Substituting these two expressions into (4.68) and approximating $[H_2PO_4^-]$ by $c(NaH_2PO_4)$ gives an equation of exactly the same form as the intermediate expression (4.64), with $H_3PO_4$'s two adjacent constants in place of $K_{a_1}$ and $K_{a_2}$.

At ordinary analytical concentrations the two simplifying conditions, $c \gg K_{a_1} c^{\ominus}$ and $K_{a_2} c \gg K_w c^{\ominus}$, both hold and the intermediate expression reduces to

$$[H^+] \approx \sqrt{K_{a_1} K_{a_2}}\, c^{\ominus} \tag{4.69}$$

$$pH \approx \frac{pK_{a_1} + pK_{a_2}}{2} \tag{4.70}$$

For an amphoteric substance of the $Na_2HPO_4$ type, with $HPO_4^{2-}$ and $H_2O$ as the proton reference levels, the proton balance equation has the same form as (4.67) with every subscript shifted by one. Following the same steps, $K_{a_2}$ and $K_{a_3}$ replace $K_{a_1}$ and $K_{a_2}$ in the intermediate expression (4.64), the two conditions become $c \gg K_{a_2} c^{\ominus}$ and $K_{a_3} c \gg K_w c^{\ominus}$, and hence

$$[H^+] \approx \sqrt{K_{a_2} K_{a_3}}\, c^{\ominus} \tag{4.71}$$

$$pH \approx \frac{pK_{a_2} + pK_{a_3}}{2} \tag{4.72}$$

**Amphoteric Substances of the $NH_4Ac$ Type**

A salt of a weak acid with a weak base is also an amphoteric substance. In a solution of $NH_4Ac$, for example, $NH_4^+$ donates a proton as an acid while $Ac^-$ accepts one as a base. With $NH_4^+$, $Ac^-$, and $H_2O$ as the proton reference levels, the proton balance equation is

$$[H^+] + [HAc] = [NH_3] + [OH^-] \tag{4.73}$$

When the analytical concentration $c(NH_4Ac)$ is not too small, both $[H^+]$ and $[OH^-]$ may be ignored and (4.73) reduces at once to $[HAc] \approx [NH_3]$. Writing the distribution fractions of the two species and setting them equal,

$$\frac{c(NH_4Ac)[H^+]}{[H^+] + K_a(HAc)c^{\ominus}} \approx \frac{c(NH_4Ac)K_a(NH_4^+)c^{\ominus}}{[H^+] + K_a(NH_4^+)c^{\ominus}}$$

Cancelling $c(\mathrm{NH_4Ac})$ on both sides and cross-multiplying,

$$[\mathrm{H^+}]\left([\mathrm{H^+}] + K_\mathrm{a}(\mathrm{NH_4^+})c^\ominus\right) = K_\mathrm{a}(\mathrm{NH_4^+})c^\ominus\left([\mathrm{H^+}] + K_\mathrm{a}(\mathrm{HAc})c^\ominus\right)$$

Expanding both sides, the terms containing $K_\mathrm{a}(\mathrm{NH_4^+})c^\ominus[\mathrm{H^+}]$ cancel exactly and only

$$[\mathrm{H^+}]^2 = K_\mathrm{a}(\mathrm{HAc})K_\mathrm{a}(\mathrm{NH_4^+})(c^\ominus)^2$$

remains. This is exactly the result of the intermediate expression (4.64) when the two simplifying conditions hold, except that $K_1 = K_\mathrm{a}(\mathrm{HAc})$ and $K_2 = K_\mathrm{a}(\mathrm{NH_4^+})$ now come from two different acids: for a salt of a weak acid with a weak base, "two adjacent dissociation constants" is only a convenient phrase. Thus

$$[\mathrm{H^+}] \approx \sqrt{K_\mathrm{a}(\mathrm{NH_4^+})K_\mathrm{a}(\mathrm{HAc})}\, c^\ominus \tag{4.74}$$

$$\mathrm{pH} \approx \frac{\mathrm{p}K_\mathrm{a}(\mathrm{NH_4^+}) + \mathrm{p}K_\mathrm{a}(\mathrm{HAc})}{2} \tag{4.75}$$

Collecting the results for the three systems gives a unified rule for amphoteric solutions. Let $K_1$ be the dissociation constant of the acid formed when the amphoteric substance accepts a proton (the larger of the two), and $K_2$ the dissociation constant of the amphoteric substance itself acting as an acid (the smaller). Then, provided that $c \gg K_1 c^\ominus$ and $K_2 c \gg K_\mathrm{w} c^\ominus$, the hydrogen ion concentration of the solution is fixed by the geometric mean of the two constants and the pH by the arithmetic mean of their negative logarithms. Acid salts take two adjacent constants; salts of a weak acid with a weak base take the constants of the cationic acid and of the molecular acid, as for $\mathrm{NH_4Ac}$, where $K_\mathrm{a}(\mathrm{HAc})$ and $K_\mathrm{a}(\mathrm{NH_4^+})$ are used. The chemical meaning of this result is that the two tendencies of an amphoteric substance, to gain and to lose a proton, restrain each other, and the system reaches equilibrium when the product that has gained a proton equals the product that has lost one; this is precisely the crossing point of the distribution fraction curves (see equation (4.30)).

The simplest expression and the intermediate expression hold only under the two conditions above, and those conditions are not always met. When the analytical concentration is too low, $[HB^-] \approx c(\text{NaHB})$ no longer holds and the $K_1$ in the denominator cannot be dropped; when the second dissociation is so weak that $K_2 c$ is comparable to or smaller than $K_w c^\ominus$, the self-dissociation of water cannot be ignored and the $K_w$ in the numerator cannot be dropped. In either case one should turn to the corresponding intermediate expression (such as (4.64)) or solve the proton balance equation directly. How far apart the two dissociation constants are decides whether $HB^-$ can be the predominant species in solution and whether the titration can be carried out in steps; it is a different question from whether the simplest expression is valid.

**Example 4.4** Calculate the pH of a 0.10 mol/L solution of $NaHCO_3$. For $H_2CO_3$, $pK_{a_1} = 6.38$ and $pK_{a_2} = 10.25$.

Solution: $NaHCO_3$ is an amphoteric substance of the NaHA type, and the simplest expression (4.66) may be considered. Check the two conditions first: $c(\text{NaHCO}_3) = 0.10\,\text{mol/L}$, far larger than $K_{a_1} c^\ominus = 4.2 \times 10^{-7}\,\text{mol/L}$; and $K_{a_2} c = 5.6 \times 10^{-12}$, which is 560 times $K_w c^\ominus = 1.0 \times 10^{-14}$ and satisfies $K_{a_2} c \gg K_w c^\ominus$. Both conditions hold, so the simplest expression (4.66) applies:

$$\begin{aligned} \text{pH} &\approx \frac{pK_{a_1} + pK_{a_2}}{2} \\ &= \frac{6.38 + 10.25}{2} = 8.32 \end{aligned}$$

As a check, the intermediate expression (4.64) gives pH = 8.3146, only 0.0004 below the 8.3150 of the simplest expression, so the latter is entirely adequate here.

**Example 4.5** Calculate the pH of a 0.050 mol/L solution of $NaH_2PO_4$. For $H_3PO_4$, $pK_{a_1} = 2.12$ and $pK_{a_2} = 7.20$.

Solution: With $H_2PO_4^-$ and $H_2O$ as the reference levels, $NaH_2PO_4$ is an amphoteric substance of the NaHA type. Here $c = 0.050\,\text{mol/L}$ while $K_{a_1}c^\ominus = 7.6 \times 10^{-3}\,\text{mol/L}$, only about 7 times smaller, so the $K_{a_1}$ in the denominator cannot be dropped completely and the intermediate expression (4.64) is used directly:

$$\begin{aligned}[H^+] &= c^\ominus \sqrt{\frac{K_{a_1}\left(K_{a_2}\dfrac{c}{c^\ominus} + K_w\right)}{K_{a_1} + \dfrac{c}{c^\ominus}}} \\ &= \sqrt{\frac{7.6 \times 10^{-3} \times (6.3 \times 10^{-8} \times 0.050 + 1.0 \times 10^{-14})}{7.6 \times 10^{-3} + 0.050}}\,\text{mol/L} \\ &= 2.0 \times 10^{-5}\,\text{mol/L}\end{aligned}$$

The pH is therefore 4.69. The simplest expression (4.70), which takes the arithmetic mean of the two $pK_a$ values, gives 4.66, 0.03 too low. For an acid salt such as $NaH_2PO_4$, whose first dissociation is fairly strong, the intermediate expression should be preferred.

## 4.4 Acid-Base Buffer Solutions

An acid-base buffer solution is a solution that resists the addition of small amounts of strong acid or strong base, or moderate dilution, and keeps its pH relatively constant. Such systems usually consist of a weak acid and its conjugate base (or a weak base and its conjugate acid) in fairly high concentration; familiar examples are $HAc - NaAc$, $NH_4Cl - NH_3$, and $NaH_2PO_4 - Na_2HPO_4$. The central function of a buffer is to hold the pH of the solution within a certain range, so as

to meet the demands for pH stability in chemical reactions, biological processes, and analytical measurements.

### 4.4.1 Calculating the pH of an Acid-Base Buffer Solution

A buffer solution made of HB and NaB contains the equilibrium

$$\mathrm{HB(aq)} \rightleftharpoons \mathrm{H^+(aq) + B^-(aq)}$$

and the hydrogen ion concentration of the system is

$$\frac{[\mathrm{H^+}]}{c^\ominus} = K_\mathrm{a} \times \frac{\frac{[\mathrm{HB}]}{c^\ominus}}{\frac{[\mathrm{B^-}]}{c^\ominus}} \tag{4.76}$$

The mass balance and the charge balance of the system are

$$[\mathrm{HB}] + [\mathrm{B^-}] = c(\mathrm{HB}) + c(\mathrm{NaB}) \tag{4.77}$$

and

$$[\mathrm{H^+}] + [\mathrm{Na^+}] = [\mathrm{OH^-}] + [\mathrm{B^-}] \tag{4.78}$$

Solving (4.77) and (4.78) gives

$$[\mathrm{HB}] = c(\mathrm{HB}) + [\mathrm{OH^-}] - [\mathrm{H^+}] \tag{4.79}$$

and

$$[\mathrm{B^-}] = c(\mathrm{NaB}) + [\mathrm{H^+}] - [\mathrm{OH^-}] \tag{4.80}$$

Substituting (4.79) and (4.80) into (4.76) gives

$$\frac{[\mathrm{H^+}]}{c^\ominus} = K_\mathrm{a} \times \frac{c(\mathrm{HB}) + [\mathrm{OH^-}] - [\mathrm{H^+}]}{c(\mathrm{NaB}) + [\mathrm{H^+}] - [\mathrm{OH^-}]} \tag{4.81}$$

Equation (4.81) is the exact expression for the pH of a buffer solution and is also called the Henderson-Hasselbalch equation. Since $c$(HB) and $c$(NaB) are usually large, equation (4.81) may be simplified to

$$\frac{[\mathrm{H^+}]}{c^\ominus} \approx K_\mathrm{a} \times \frac{c(\mathrm{HB})}{c(\mathrm{NaB})} \tag{4.82}$$

or

$$\mathrm{pH} \approx \mathrm{p}K_\mathrm{a} + \lg \frac{c(\mathrm{NaB})}{c(\mathrm{HB})} \tag{4.83}$$

Equation (4.83) shows that the pH of a buffer is controlled on the one hand by the dissociation constant of the weak acid and on the other by the ratio of the concentrations of the acid and its conjugate base. If the buffer is diluted with a suitable amount of distilled water, the ratio of $c$(HB) to $c$(NaB) is unchanged, so the pH does not change. When the amount of acid or base added from outside is not too large, the concentration ratio changes but little and the pH changes little as well. This is why a buffer resists mild external disturbance.

**Example 4.6** A buffer solution is made of $NH_4Cl$ and $NH_3$, with analytical concentrations of $NH_3$ and $NH_4Cl$ equal to 0.10 mol/L and 0.10 mol/L, respectively. Calculate its pH. For $NH_3$, $K_\mathrm{b} = 1.8 \times 10^{-5}$.

Solution: The dissociation constant of $NH_4^+$ is

$$K_\mathrm{a}(NH_4^+) = \frac{K_\mathrm{w}}{K_\mathrm{b}(NH_3)} = \frac{1.0 \times 10^{-14}}{1.8 \times 10^{-5}} = 5.5 \times 10^{-10}$$

According to equation (4.83),

$$\begin{aligned} \mathrm{pH} &= \mathrm{p}K_\mathrm{a}(NH_4^+) - \lg \frac{c(NH_4^+)}{c(NH_3)} \\ &= 9.26 - \lg \frac{0.10}{0.10} \\ &= 9.26 \end{aligned}$$

The interested reader may vary $c(NH_4^+)$ and $c(NH_3)$ and repeat the calculation; the pH of this buffer is found to vary only in the neighborhood of $\mathrm{p}K_\mathrm{a}(NH_4^+)$. To construct a buffer system, therefore, one starts from the acid dissociation constant and selects a suitable weak acid and its conjugate base.

### 4.4.2 Buffer Index and Buffer Range

The example above shows that the base pH of a buffer is fixed by the p$K_a$ of its weak acid, and that the concentration ratio of acid to conjugate base can adjust it only within limits: the pH shifts by one unit only when the ratio changes by a factor of ten, so over the range of ratios used in practice the effect of the ratio on pH is limited.

To describe buffering power numerically, the buffer index is defined as

$$\beta = \frac{\mathrm{d}c_b}{\mathrm{dpH}} = -\frac{\mathrm{d}c_a}{\mathrm{dpH}} \tag{4.84}$$

where $\beta$ is the buffer index and $\mathrm{d}c_b$ and $\mathrm{d}c_a$ are the changes in concentration of added base or acid.

To examine how the buffer index depends on concentration, add NaOH to the buffer above; the effect of the strong base may be measured by the change in $c$(NaB). From equation (4.80),

$$c(\mathrm{NaB}) = [\mathrm{B}^-] - [\mathrm{H}^+] + [\mathrm{OH}^-] \tag{4.85}$$

Differentiating with respect to pH,

$$\begin{aligned}\frac{\mathrm{d}c(\mathrm{NaB})}{\mathrm{dpH}} &= \frac{\mathrm{d}[\mathrm{B}^-]}{\mathrm{dpH}} + \frac{\mathrm{d}[\mathrm{OH}^-]}{\mathrm{dpH}} - \frac{\mathrm{d}[\mathrm{H}^+]}{\mathrm{dpH}} \\ &= \beta([\mathrm{B}^-]) + \beta([\mathrm{OH}^-]) + \beta([\mathrm{H}^+])\end{aligned} \tag{4.86}$$

The last two terms are relatively small and are neglected, so that

$$\beta = \frac{\mathrm{d}c(\mathrm{NaB})}{\mathrm{dpH}} = \frac{\mathrm{d}[\mathrm{B}^-]}{\mathrm{dpH}} = \beta([\mathrm{B}^-]) \tag{4.87}$$

Since

$$[\mathrm{B}^-] = \frac{K_a(\mathrm{HB})c^\ominus\Big(c(\mathrm{HB}) + c(\mathrm{NaB})\Big)}{[\mathrm{H}^+] + K_a(\mathrm{HB})c^\ominus} \tag{4.88}$$

the buffer index is

$$\beta = \frac{\mathrm{d}[\mathrm{B}^-]}{\mathrm{dpH}} = 2.30 \times \frac{K_\mathrm{a}(\mathrm{HB})c^\ominus\Big(c(\mathrm{HB}) + c(\mathrm{NaB})\Big)[\mathrm{H}^+]}{\Big([\mathrm{H}^+] + K_\mathrm{a}(\mathrm{HB})c^\ominus\Big)^2} \tag{4.89}$$

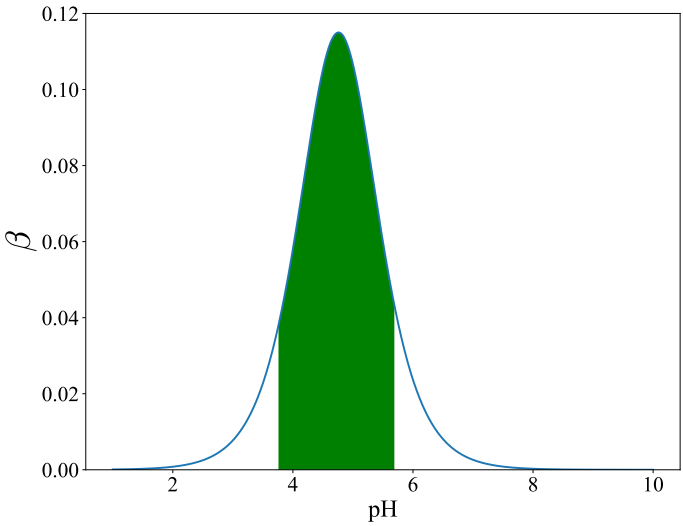



Figure 4.6: Buffer index curves of HAc and NaAc buffer solutions, both of concentration 0.10 mol/L.

Calculating the buffer index at different hydrogen ion concentrations gives the buffer index curve shown in Figure 4.6. The shaded region of the curve corresponds to $\mathrm{pH} = \mathrm{p}K_\mathrm{a} \pm 1$; this region is called the buffer range of the solution.

Differentiating (4.89) with respect to $[\mathrm{H}^+]$, setting the derivative to zero, and rearranging gives

$$[\mathrm{H}^+]_{\beta_\mathrm{max}} = K_\mathrm{a}(\mathrm{HB})\, c^\ominus \tag{4.90}$$

and substituting (4.90) into (4.89) gives

$$\beta_\mathrm{max} = 0.575 \times \Big(c(\mathrm{HB}) + c(\mathrm{NaB})\Big) \tag{4.91}$$

The maximum buffer index is thus proportional to the total concentration. A buffer of larger capacity therefore requires a sufficiently large total concentration.

The total amount of the buffer must of course be large enough as well; otherwise it cannot offset the acid (or base) added from outside.

### 4.4.3 Principles for Choosing a Buffer Solution

Many buffer solutions are used in analytical chemistry to control pH, and the choice depends on the circumstances. The principles are: (1) the buffer should not interfere with the measurement; (2) the $pK_a$ of the weak acid in the buffer should be close to the pH to be controlled; (3) the buffer index should be large enough; and (4) the buffer substances should be inexpensive and readily available, and should not cause pollution.

Table 4.1 lists several common buffer solutions.

Table 4.1: Common buffer solutions

| Buffer composition | Acid form | Base form | $pK_a$ |
|---|---|---|---|
| glycine–HCl | $^+NH_3CH_2COOH$ | $^+NH_3CH_2COO^-$ | $2.35(pK_{a_1})$ |
| HAc − NaAc | HAc | $Ac^-$ | 4.74 |
| hexamethylenetetramine–HCl | $(CH_2)_6N_4H^+$ | $(CH_2)_6N_4$ | 5.15 |
| triethanolamine–HCl | $^+HN(CH_2CH_2OH)_3$ | $N(CH_2CH_2OH)_3$ | 7.76 |
| $NaHCO_3 - Na_2CO_3$ | $HCO_3^-$ | $CO_3^{2-}$ | $10.25(pK_{a_2})$ |

**Stop and Think**

(1) To prepare a buffer of pH = 4.5, which systems in Table 4.1 are candidates, and on what basis? (2) If the experiment requires the largest possible buffer index, what else should be considered besides matching the $pK_a$? (3) The glycine–HCl system has $pK_a = 2.35$; by adjusting the ratio of acid to base, can it cover pH = 1.5 ∼ 3.0? Look for the answer in the Henderson-Hasselbalch equation.

## 4.5 Acid-Base Titration Equations

An acid-base titration equation is a mathematical equation that describes how the pH of a solution of unknown concentration changes as a titrant (an acid or a base) is added, that is, the relation between the pH and the amount of titrant added. Many researchers have studied titration equations, and the work of Robert de Levie is the most representative; he established the general equation of acid-base titration, which can in principle describe any arbitrarily complex system. This section begins with the simpler systems and gradually builds up the general equation.

### 4.5.1 Titrating a Strong Monoprotic Acid with a Strong Base

Taking the titration of hydrochloric acid with sodium hydroxide as an example, the reaction is

$$\mathrm{HCl(aq) + NaOH(aq) = NaCl(aq) + H_2O(l)}$$

Let the analytical concentration of hydrochloric acid at the start be $c(\mathrm{HCl})$ and its initial volume $V_0$. When the acid is titrated with sodium hydroxide of analytical concentration $c(\mathrm{NaOH})$ and the volume of base added is $V_t$, the mass balances are

$$\frac{c(\mathrm{HCl})V_0}{V_0 + V_t} = [\mathrm{Cl^-}] \tag{4.92}$$

$$\frac{c(\mathrm{NaOH})V_t}{V_0 + V_t} = [\mathrm{Na^+}] \tag{4.93}$$

The charged species present during the titration are $\mathrm{Na^+}$, $\mathrm{H^+}$, $\mathrm{OH^-}$, and $\mathrm{Cl^-}$, and their charge balance is

$$[\mathrm{Na^+}] + [\mathrm{H^+}] = [\mathrm{OH^-}] + [\mathrm{Cl^-}] \tag{4.94}$$

Substituting the terms of (4.92) and (4.93) into (4.94) gives

$$\frac{c(\mathrm{NaOH})V_\mathrm{t}}{V_0+V_\mathrm{t}}+[\mathrm{H^+}]=[\mathrm{OH^-}]+\frac{c(\mathrm{HCl})V_0}{V_0+V_\mathrm{t}} \tag{4.95}$$

Defining $\eta=\frac{V_\mathrm{t}}{V_0}$ and $\Delta=[\mathrm{H^+}]-[\mathrm{OH^-}]$, equation (4.95) rearranges to

$$\begin{aligned}\eta&=\frac{c(\mathrm{HCl})-([\mathrm{H^+}]-[\mathrm{OH^-}])}{c(\mathrm{NaOH})+([\mathrm{H^+}]-[\mathrm{OH^-}])}\\&=\frac{c(\mathrm{HCl})-\Delta}{c(\mathrm{NaOH})+\Delta}\end{aligned} \tag{4.96}$$

Equation (4.96) is the titration equation for hydrochloric acid titrated with sodium hydroxide. Since $[\mathrm{H^+}]$ and $[\mathrm{OH^-}]$ are conjugate to each other, (4.96) is in effect a quantitative relation between the volume ratio $\eta$ and the hydrogen ion concentration $[\mathrm{H^+}]$. Calculating the volume ratio at various hydrogen ion concentrations gives the relation between the two, that is, the titration curve shown in Figure 4.7. The ordinate of a titration curve is usually the negative logarithm of the concentration (pH), which displays the abrupt change in hydrogen ion concentration more clearly.

Figure 4.7 has two key positions, the stoichiometric point and the titration jump. The stoichiometric point is the position $(\eta, \mathrm{pH})$ on the titration curve at which the amount of sodium hydroxide added exactly equals the amount of hydrochloric acid in the solution. From the stoichiometry,

$$c(\mathrm{NaOH})\times V_\mathrm{sp}=c(\mathrm{HCl})\times V_0 \tag{4.97}$$

so that

$$\eta_\mathrm{sp}=\frac{V_\mathrm{sp}}{V_0}=\frac{c(\mathrm{HCl})}{c(\mathrm{NaOH})} \tag{4.98}$$

where the subscript sp stands for stoichiometric point. For titration at equal concentrations, that is $c(\mathrm{HCl})=c(\mathrm{NaOH})$, $\eta_\mathrm{sp}=1$. The stoichiometric point of the system shown in Figure 4.7 is (1.000, 7.00).

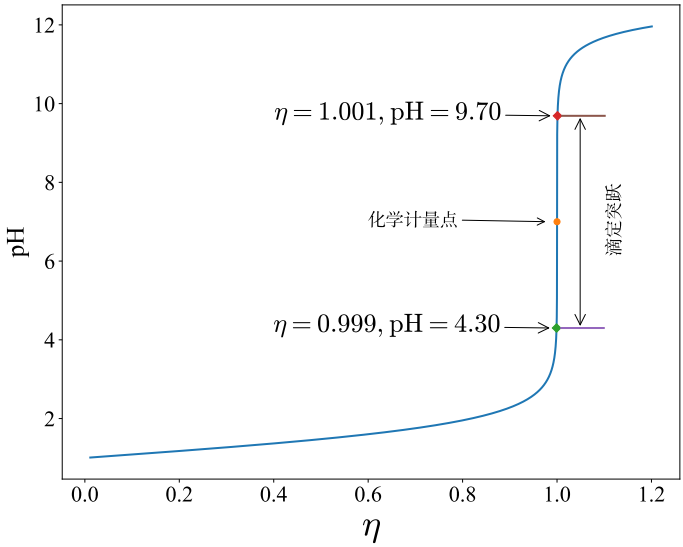



Figure 4.7: Titration curve for 0.1000 mol/L NaOH titrating HCl of the same concentration.

The titration jump is the stretch in which $\eta$ runs from 0.999 to 1.001, where the pH leaps from 4.30 to 9.70; the range $0.999 \sim 1.001$ corresponds to a degree of titration of $99.9\% \sim 100.1\%$.

Concentration is another factor that affects the titration jump of a strong acid with a strong base. For example, when 0.1 mol/L NaOH titrates 0.1 mol/L HCl, the jump is from pH 4.30 to 9.70, whereas when 0.01 mol/L NaOH titrates 0.01 mol/L HCl, the jump is only from pH 5.3 to 8.7. Figure 4.8 shows the titration curves of NaOH against HCl of the same concentration for the two cases; the higher concentration gives the wider jump (solid line) and the lower concentration the narrower jump (dash-dotted line).

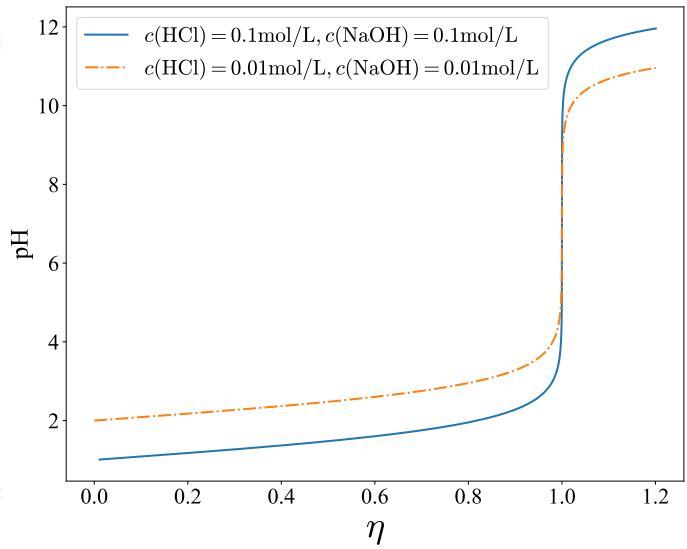



Figure 4.8: Effect of concentration on the titration jump.

### 4.5.2 Titrating a Weak Monoprotic Acid with a Strong Base

Taking the titration of a weak acid HB with sodium hydroxide as an example, the reaction is

$$\mathrm{HB(aq) + NaOH(aq) = NaB(aq) + H_2O(l)}$$

Let the analytical concentration of HB at the start be $c$(HB) and its initial volume $V_0$. When the acid is titrated with sodium hydroxide of analytical concentration $c$(NaOH) and the volume added is $V_\mathrm{t}$, the mass balances are

$$\frac{c(\mathrm{HB})V_0}{V_0 + V_\mathrm{t}} = [\mathrm{B^-}] + [\mathrm{HB}] \tag{4.99}$$

$$\frac{c(\mathrm{NaOH})V_\mathrm{t}}{V_0 + V_\mathrm{t}} = [\mathrm{Na^+}] \tag{4.100}$$

The charged species present during the titration are $\mathrm{Na^+}$, $\mathrm{H^+}$, $\mathrm{OH^-}$, and $\mathrm{B^-}$, and their charge balance is

$$[\mathrm{Na^+}] + [\mathrm{H^+}] = [\mathrm{B^-}] + [\mathrm{OH^-}] \tag{4.101}$$

The concentration of $B^-$ may be written as

$$[B^-] = \delta(B^-) \times \frac{c(HB)V_0}{V_0 + V_t} \tag{4.102}$$

Substituting the relevant terms of (4.100) and equation (4.102) into the charge balance and rearranging gives

$$\eta = \frac{\delta(B^-)c(HB) - \Delta}{c(NaOH) + \Delta} \tag{4.103}$$

Equation (4.103) is the titration equation for the titration of the weak acid HB with sodium hydroxide. Formally it closely resembles (4.96); the difference is that the analytical concentration of the weak acid carries a factor $\delta(B^-)$.

Figure 4.9 shows the titration curve of acetic acid with sodium hydroxide, together with the curve for hydrochloric acid drawn as a dashed line for comparison. The stretch before the jump is seen to be lifted markedly for the weak acid. The reason is that acetic acid is a weak acid and dissociates only partly, so that the hydrogen ion concentration it supplies is lower than its analytical concentration and the initial pH is higher than that calculated from the analytical concentration.

During the early part of the titration the added NaOH reacts with $CH_3COOH$ to form $CH_3COO^-$, and the acetate and acetic acid form a buffer, so the curve changes gently; this is characteristic of weak-acid titrations. At the stoichiometric point all the acetic acid has become sodium acetate, the solution is weakly basic, and the pH is usually no lower than 7. Comparison with the hydrochloric acid curve shows that the jump for acetic acid narrows to pH 7.74 ~ 9.70.

Figure 4.10 shows the titration curves of weak acids of different $K_a$ titrated with NaOH, the concentrations of both acid and base being 0.1000 mol/L. The smaller $K_a$ is, the narrower the jump; a small $K_a$ also makes the curve change gently rather than abruptly over the range 0.999 ~ 1.001, and this affects the accuracy of the titration.

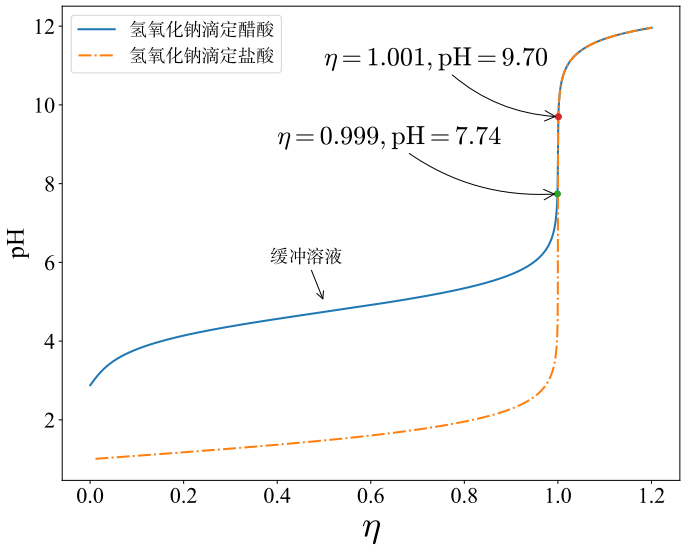



Figure 4.9: Titration curve for 0.1000 mol/L sodium hydroxide titrating acetic acid of the same concentration. $K_\mathrm{a} = 1.8 \times 10^{-5}$.

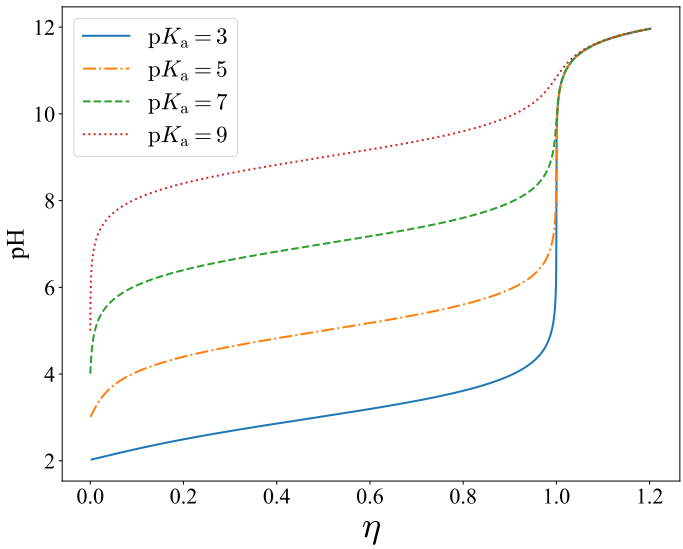



Figure 4.10: Titration curves of weak acids of different $\mathrm{p}K_\mathrm{a}$ titrated with NaOH at equal concentrations.

The positions of key points on the titration curve may also be calculated from the titration equation. At the starting point, for example, $\eta = 0.000$ and the titration equation becomes

$$\frac{\delta(\mathrm{B}^-)c(\mathrm{HB}) - \Delta}{c(\mathrm{NaOH}) + \Delta} = 0.000 \tag{4.104}$$

Expanded, this is simply the proton balance equation; the interested reader may verify it. For titration at equal concentrations the stoichiometric point corresponds to $\eta = 1.000$, and equation (4.103) gives

$$\frac{\delta(\mathrm{B}^-)c(\mathrm{HB}) - \Delta}{c(\mathrm{NaOH}) + \Delta} = 1.000 \tag{4.105}$$

This can be reduced to an equation in $[\mathrm{H}^+]$ whose solution gives the pH at the stoichiometric point. The boundaries of the jump are found in the same way.

Equation (4.103) is relatively simple in form and may be expanded into an analytical expression:

$$\begin{aligned}(\eta+1)[\mathrm{H}^+]^3 + \left(\eta c_1 + (\eta+1)K_\mathrm{a}c^\ominus\right)[\mathrm{H}^+]^2 + \ldots \\ \ldots + \left(\eta c_1 K_\mathrm{a}c^\ominus - (\eta+1)K_\mathrm{w}(c^\ominus)^2 - K_\mathrm{a}c^\ominus c_0\right)[\mathrm{H}^+] - (\eta+1)K_\mathrm{a}K_\mathrm{w}(c^\ominus)^3 = 0\end{aligned} \tag{4.106}$$

where $c_0$ denotes the initial concentration of HB, $c_1$ the initial concentration of sodium hydroxide, and $K_\mathrm{a}$ the dissociation constant of the weak acid.

Although (4.106) was constructed for a weak monoprotic acid titrated with sodium hydroxide, it applies as well to the special case of a strong monoprotic acid. Substituting $K_\mathrm{a} = 10^{10}$ and rearranging gives

$$(\eta+1)[\mathrm{H}^+]^2 + (\eta c_1 - c_0)[\mathrm{H}^+] - (\eta+1)K_\mathrm{w}(c^\ominus)^2 = 0 \tag{4.107}$$

Equation (4.107) is in fact the analytical form of the titration equation (4.96) for a strong acid titrated with sodium hydroxide; the reader may expand (4.96)

to verify it. Equation (4.106) is a cubic and in general has a real solution. Appendix B gives an Octave program, `pH_at_eta.m`, that computes the real solution of (4.106), and the program `demo_ac_abt.m` shows how it is used to calculate the stoichiometric point and the titration curve. Using Kapok to compute the curve is of course even more convenient.

### 4.5.3 Titrating a Polyprotic Acid with a Strong Base

Taking the titration of phosphoric acid with sodium hydroxide as an example, the reaction proceeds step by step:

$$\mathrm{NaOH(aq) + H_3PO_4(aq) = NaH_2PO_4(aq) + H_2O(l)}$$
$$\mathrm{NaOH(aq) + NaH_2PO_4(aq) = Na_2HPO_4(aq) + H_2O(l)}$$
$$\mathrm{NaOH(aq) + Na_2HPO_4(aq) = Na_3PO_4 + H_2O(l)}$$

Let the analytical concentration of phosphoric acid at the start be $c(\mathrm{H_3PO_4})$ and its initial volume $V_0$. When it is titrated with sodium hydroxide of analytical concentration $c(\mathrm{NaOH})$ and the volume added is $V_\mathrm{t}$, the mass balances are

$$\frac{c(\mathrm{H_3PO_4})V_0}{V_0 + V_\mathrm{t}} = [\mathrm{H_3PO_4}] + [\mathrm{H_2PO_4^-}] + [\mathrm{HPO_4^{2-}}] + [\mathrm{PO_4^{3-}}] \tag{4.108}$$
$$\frac{c(\mathrm{NaOH})V_\mathrm{t}}{V_0 + V_\mathrm{t}} = [\mathrm{Na^+}] \tag{4.109}$$

The charged species present during the titration are

$$\mathrm{H^+} \quad \mathrm{Na^+} \quad \mathrm{OH^-} \quad \mathrm{H_2PO_4^-} \quad \mathrm{HPO_4^{2-}} \quad \mathrm{PO_4^{3-}}$$

and their charge balance is

$$[\mathrm{Na^+}] + [\mathrm{H^+}] = [\mathrm{OH^-}] + [\mathrm{H_2PO_4^-}] + 2[\mathrm{HPO_4^{2-}}] + 3[\mathrm{PO_4^{3-}}] \tag{4.110}$$

Since the concentrations of the phosphate species are

$$[\mathrm{H_2PO_4^-}] = \delta(\mathrm{H_2PO_4^-}) \times \frac{c(\mathrm{H_3PO_4})V_0}{V_0 + V_\mathrm{t}} \tag{4.111}$$

$$[\mathrm{HPO_4^{2-}}] = \delta(\mathrm{HPO_4^{2-}}) \times \frac{c(\mathrm{H_3PO_4})V_0}{V_0 + V_\mathrm{t}} \tag{4.112}$$

$$[\mathrm{PO_4^{3-}}] = \delta(\mathrm{PO_4^{3-}}) \times \frac{c(\mathrm{H_3PO_4})V_0}{V_0 + V_\mathrm{t}} \tag{4.113}$$

substituting them into the charge balance gives

$$\eta = \frac{\Big(\delta(\mathrm{H_2PO_4^-}) + 2\delta(\mathrm{HPO_4^{2-}}) + 3\delta(\mathrm{PO_4^{3-}})\Big)c(\mathrm{H_3PO_4}) - \Delta}{c(\mathrm{NaOH}) + \Delta} \tag{4.114}$$

Equation (4.114) is the titration equation for phosphoric acid titrated with sodium hydroxide. Comparing it with the equations for the two cases above brings out both the similarity and the difference in form: each has a separate $\Delta$ term, and each has $c(\mathrm{NaOH}) + \Delta$ in the denominator, while the analytical concentration of the acid appears in the numerator as a factor, multiplied by a factor that differs from acid to acid.

Figure 4.11 shows the titration curve of phosphoric acid with sodium hydroxide. There are two jumps, at $\eta = 1$ and $\eta = 2$, but none at $\eta = 3$. The reason is that $\mathrm{p}K_{\mathrm{a}_3} = 12.36$ for phosphoric acid, which is an extremely weak acid, very close to $\mathrm{p}K_\mathrm{w}$ and behaving much like water itself.

To make a jump appear at the third stoichiometric point, the solvent must be changed. Ethanol, for example, has a self-dissociation constant $\mathrm{p}K_\mathrm{s}(\mathrm{CH_3CH_2OH}) = 19.1$, five units larger than $\mathrm{p}K_\mathrm{w} = 14.00$ for water; its tendency to self-dissociate is much weaker, and the third dissociation of phosphoric acid is thereby set further apart from the self-dissociation of the solvent: the two differ by 1.6 p$K$ units in water and by 6.7 in ethanol. Strictly speaking, the dissociation constants of phosphoric acid in ethanol differ from those in water; to illustrate the principle, the aqueous values of $K_\mathrm{a}$ are retained here and only $\mathrm{p}K_\mathrm{w}$ is replaced by $\mathrm{p}K_\mathrm{s}$. Replacing $\mathrm{p}K_\mathrm{w}$ by $\mathrm{p}K_\mathrm{s}(\mathrm{CH_3CH_2OH})$ and recalculating the titration curve gives Figure 4.12, in which a third jump indeed appears. Titration

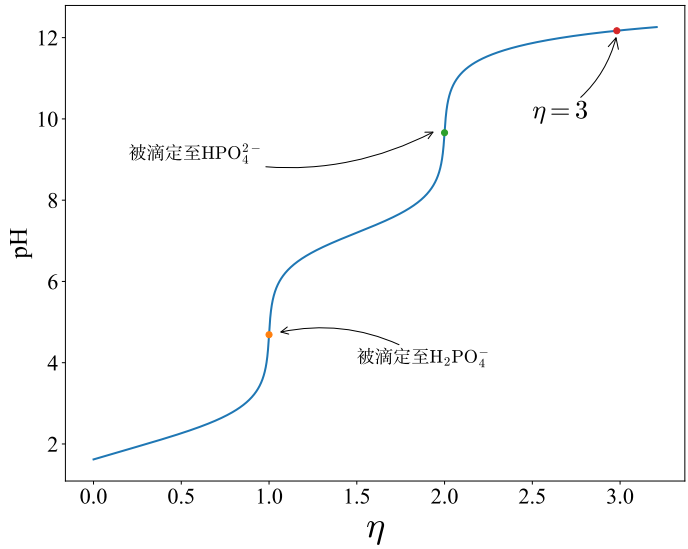



Figure 4.11: Titration curve of phosphoric acid in aqueous solution titrated with NaOH. All concentrations are 0.1000 mol/L.

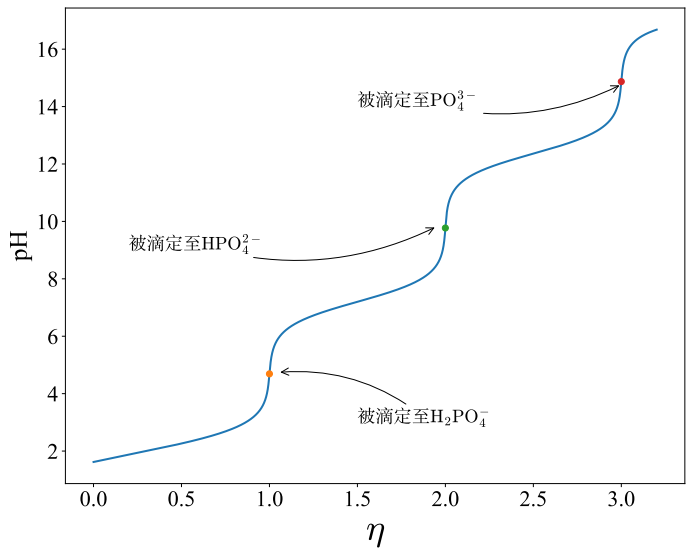



Figure 4.12: Titration curve of phosphoric acid in ethanol titrated with NaOH. All concentrations are 0.1000 mol/L.

in a nonaqueous solvent is called nonaqueous acid-base titration; space does not permit a fuller discussion here.

### 4.5.4 The General Equation of Acid-Base Titration

The examples above show that the titration equations for the three acids titrated with sodium hydroxide are very similar in form, so that a general equation of acid-base titration may be expected. Suppose that sodium hydroxide titrates a mixture of HCl, $CH_3COOH$, and $H_3PO_4$; the concentrations, initial volumes, and dissociation constants of the components are listed in Table 4.2.

Table 4.2: Concentrations and volume parameters of the components

| Acid or base | Initial concentration | Initial volume | Acid dissociation constant |
|---|---|---|---|
| NaOH | $c(\mathrm{NaOH})$ | 0 | |
| HCl | $c(\mathrm{HCl})$ | $V_0$ | |
| HAc | $c(\mathrm{CH_3COOH})$ | $V_0$ | $K_a$ |
| $H_3PO_4$ | $c(\mathrm{H_3PO_4})$ | $V_0$ | $K_{a_1}, K_{a_2}, K_{a_3}$ |

When the volume of sodium hydroxide consumed is $V_t$, the mass balances are

$$\frac{c(\mathrm{NaOH})V_t}{V_0+V_t} = [\mathrm{Na^+}] \tag{4.115}$$

$$\frac{c(\mathrm{HCl})V_0}{V_0+V_t} = [\mathrm{Cl^-}] \tag{4.116}$$

$$\frac{c(\mathrm{CH_3COOH})V_0}{V_0+V_t} = [\mathrm{CH_3COOH}] + [\mathrm{CH_3COO^-}] \tag{4.117}$$

$$\frac{c(\mathrm{H_3PO_4})V_0}{V_0+V_t} = [\mathrm{H_3PO_4}] + [\mathrm{H_2PO_4^-}] + [\mathrm{HPO_4^{2-}}] + [\mathrm{PO_4^{3-}}] \tag{4.118}$$

and the charge balance is

$$[\mathrm{H^+}] + [\mathrm{Na^+}] = [\mathrm{OH^-}] + [\mathrm{Cl^-}] + [\mathrm{CH_3COO^-}] + [\mathrm{H_2PO_4^-}] + 2[\mathrm{HPO_4^{2-}}] + 3[\mathrm{PO_4^{3-}}] \tag{4.119}$$

Substituting the terms of the mass balances into the corresponding terms of the charge balance and rearranging gives the titration equation

$$\eta = -\frac{F(\mathrm{HCl})c(\mathrm{HCl}) + F(\mathrm{CH_3COOH})c(\mathrm{CH_3COOH}) + F(\mathrm{H_3PO_4})c(\mathrm{H_3PO_4}) + \Delta}{F(\mathrm{NaOH})c(\mathrm{NaOH}) + \Delta} \tag{4.120}$$

where

- $F(\mathrm{NaOH}) = 1$;
- $F(\mathrm{HCl}) = -1$;
- $F(\mathrm{CH_3COOH}) = -\delta(\mathrm{CH_3COO^-})$;
- $F(\mathrm{H_3PO_4}) = -\delta(\mathrm{H_2PO_4^-}) - 2\delta(\mathrm{HPO_4^{2-}}) - 3\delta(\mathrm{PO_4^{3-}})$.

The $F$ in equation (4.120) is called the proton dissociation function (or proton association function); it is a quantity related to the form and behavior of an acid or base during the titration. For a component that tends to dissociate, the proton dissociation function is negative; for one that tends to accept a proton, it is positive. A strong acid loses its proton completely and is assigned $F = -1$, whereas a strong base gains a proton completely and is assigned $F = 1$.

The general form of the $F$ function is as follows[1]:

$$F = \frac{\sum_{j=0}^{p}(p-q-j)[\mathrm{H^+}]^{p-j}\prod_{i=0}^{j}K_{\mathrm{a},i}}{\sum_{j=0}^{p}[\mathrm{H^+}]^{p-j}\prod_{i=0}^{j}K_{\mathrm{a},i}} \tag{4.121}$$

where $K_{\mathrm{a},0} \equiv 1$; $p$ is the maximum number of protons the acid can dissociate, and $q$ the number of protons the species can still dissociate in its present state.

The derivation shows that the $F$ function is essentially expressed through

[1]The expression applies to weak acid (base) systems; for a strong base, $F = 1$ is taken directly by the convention of this book and the expression is not used. De Levie adopted a different auxiliary treatment in his book, and the question deserves further study.

distribution fractions and follows a regular pattern. A few examples will illustrate it:

- For $H_3PO_4$, which tends only to lose protons, one at a time,

$$F = -\delta(H_2PO_4^-) - 2\delta(HPO_4^{2-}) - 3\delta(PO_4^{3-}) \tag{4.122}$$

- For $NaH_2PO_4$ there are two possible directions: it may lose one proton to become $HPO_4^{2-}$, or two to become $PO_4^{3-}$, or it may gain one to become $H_3PO_4$. Hence

$$F = -\delta(HPO_4^{2-}) - 2\delta(PO_4^{3-}) + \delta(H_3PO_4) \tag{4.123}$$

- For $(NH_4)_2HPO_4$, the compound is best treated as two substances: the $NH_4^+$ tends to lose a proton and there are two moles of it, while the $HPO_4^{2-}$ may either lose or gain one (as in the preceding example). Hence

$$F = -\delta(PO_4^{3-}) + \delta(H_2PO_4^-) + 2\delta(H_3PO_4) - 2\delta(NH_3) \tag{4.124}$$

The linear summation of the terms in equation (4.120) reflects the relative independence of the components during an acid-base titration. Comparing (4.96), (4.103), (4.114), and (4.120), we see that the concentration terms of the first three equations add up to the concentration terms of the last.

In the examples above the acid is always titrated with a single sodium hydroxide solution, so that the denominator of (4.120) contains only one term. From the relative independence of the components, however, de Levie reasoned that for a mixture of bases titrating an acid, the denominator should be the arithmetic sum of the corresponding terms for the bases. The general equation of acid-base titration may therefore be written as

$$\eta = -\frac{\sum F_s c_s + \Delta}{\sum F_t c_t + \Delta} \tag{4.125}$$

where the subscripts t and s denote the titrant and the sample to be titrated.

Solving a complex system with the general equation by hand is of course impractical. The author has written the Kapok software package, which runs under Windows and includes the calculation of acid-base titration curves. Interested readers may request it by e-mail at ganfeng1964@163.com; the package contains instructions for its use, and Appendix **??** gives only a brief demonstration.

With Kapok the titration curve of the mixture of HCl, $CH_3COOH$, and $H_3PO_4$ titrated with NaOH can be calculated, as shown in Figure 4.13. Only one pronounced jump appears, at $\eta = 4$; this shows that sodium hydroxide titrates the mixture only as a whole and cannot determine the components one by one.

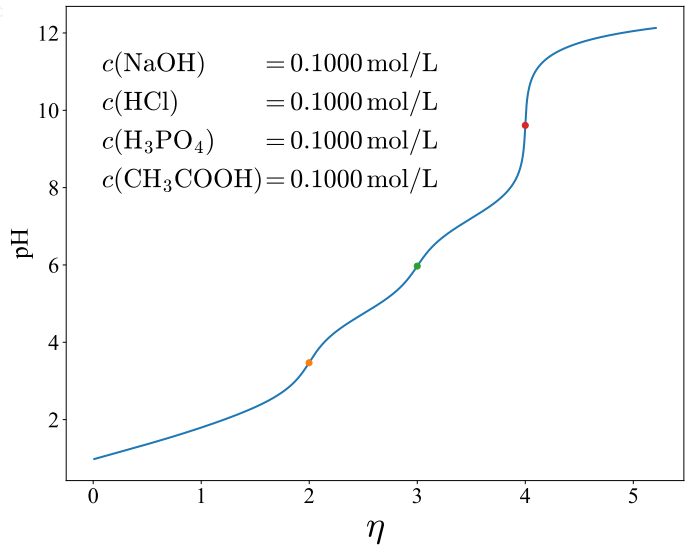



Figure 4.13: Titration curve of the acid mixture of hydrochloric, acetic, and phosphoric acids titrated with sodium hydroxide. Each acid is 0.1000 mol/L.

Figure 4.14 shows in detail how the principal species change during the titration of this mixture. Four facts stand out: (1) below $\eta < 1$ all the acids are titrated to some extent, and this mixed effect makes it impossible to see any single acid titrated completely at $\eta = 1$, although in theory hydrochloric acid should be completely titrated there; (2) at $\eta = 2$, $[H_3PO_4] \approx 0$, showing that $H_3PO_4$ is essentially

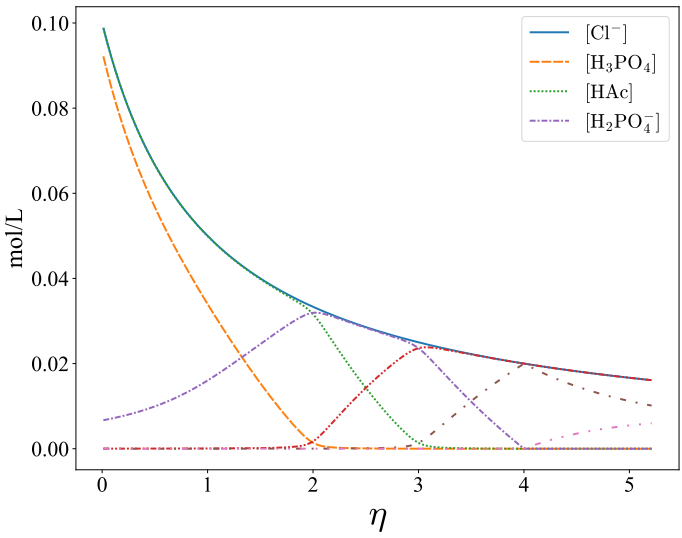



Figure 4.14: Concentration curves of the principal species during the titration of the acid mixture of hydrochloric, acetic, and phosphoric acids with sodium hydroxide. Each acid is 0.1000 mol/L.

titrated completely, which also brings $[H_2PO_4^-]$ to its maximum; (3) at $\eta = 3$ acetic acid is titrated completely, while $H_2PO_4^-$ is only incidentally titrated a little; and (4) at $\eta = 4$, $H_2PO_4^-$ is titrated to $HPO_4^{2-}$.

The establishment of the general equation is an important achievement of analytical chemistry. It brings the whole range of titration systems under one concise mathematical framework, and its value lies in revealing the additivity principle and the standardized treatment of multi-component acid-base titrations. Through the proton dissociation function (the $F$ function) it quantifies and linearly superposes the chemical behavior of components of different strength (strong or weak) and different proticity (monoprotic or polyprotic), thereby passing from single systems to complex mixtures. This deepens our understanding of the essence of titration (proton transfer and the distribution of species) and frees the calculation of titration curves from laborious hand derivation, providing

a theoretical basis for accurate prediction and simulation by computer.[2]

**Key Concept**

**How to use the general equation of acid-base titration:** $\eta = -\frac{\sum F_s c_s + \Delta}{\sum F_t c_t + \Delta}$. Proceed in three steps: (1) decide whether each component is an acid or a base and write its $F$ function (strong acid $F = -1$, strong base $F = +1$, weak acids as a combination of distribution fractions); (2) put the components of the sample into the numerator and those of the titrant into the denominator; (3) choose $[H^+]$, compute $\eta$, and plot the titration curve point by point. The titration behavior of any complex system reduces to these three steps.

## 4.6 Determining the Endpoint and the Titration Error

The classical way to determine the endpoint of an acid-base titration is to use an acid-base indicator, observing the abrupt color change of the indicator near the stoichiometric point. The method is simple and fast; an experienced analyst needs only about 5 minutes for a single titration. It has a subjective limitation, however: the accuracy of the endpoint depends strongly on how well the analyst perceives the color change, and different operators may differ in sensitivity and in the criterion they apply.

The method also faces a fundamental contradiction: an acid-base indicator is itself usually a weak acid or a weak base and therefore inevitably takes part in the reaction during the titration. This "both referee and player" character means that we must rely on the indicator to locate the endpoint and at the same time accept the extra disturbance it brings to the titration. This inherent defect is the theoretical difficulty that the classical indicator method cannot overcome, and it will be discussed specifically later.

[2]The authors have applied for a national invention patent on this basis, ZL 2022 1 1464305.9.

With the rapid development of analytical instrumentation, the potentiometric titrator has become standard equipment in the modern analytical laboratory. It monitors the change in pH during the titration in real time, constructs the complete titration curve, and locates the endpoint from the extremum of the first-derivative curve. This removes the subjective bias of the indicator method and markedly improves the reproducibility and accuracy of the results. The technique still faces challenges in practice: near the stoichiometric point a micro-titration mode must be used to obtain a precise derivative curve, which reduces the amplitude of the signal change and makes the measuring system more sensitive to random noise.

In the modern laboratory manual and instrumental titration coexist. Although the potentiometric titrator is clearly superior in accuracy and automation, a single titration usually takes longer than a manual one, a limitation that becomes conspicuous when many samples are to be analyzed. Manual titration therefore remains a core experimental skill that every analyst must master, and a thorough understanding of how acid-base indicators work is part of the foundation of analytical chemistry. For these reasons this section concentrates on the principles and the application of the acid-base indicator method.

### 4.6.1 Principles of Acid-Base Indicators

An acid-base indicator is an organic weak acid or weak base whose structure usually contains double bonds and benzene rings, so that the molecule forms a large conjugated $\pi$ electron system and often shows a definite color. When a hydrogen ion from the solution binds to the indicator, the original conjugated structure is altered and the color changes. This property makes the indicator a tracer of the hydrogen ion concentration of an acid-base solution.

Methyl orange, for example, is an organic weak base: its basic form has an azo structure and is yellow, while its acid form has a quinonoid structure and is red. Below pH $\leq$ 3.1 methyl orange exists mainly in the acid form and the solution is red; above pH $\geq$ 4.4 it exists mainly in the basic form and the solution is yellow. Between pH = 3.1 and 4.4 the two forms coexist, and the solution shows the mixture of red and yellow, that is, orange. Figure 4.15 shows the color change of methyl orange, and Table 4.3 lists several common acid-base indicators.

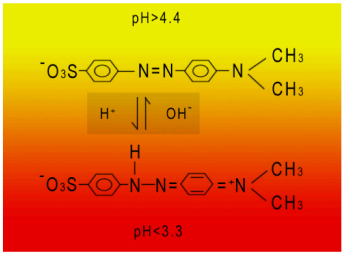



Figure 4.15: Color change of methyl orange.

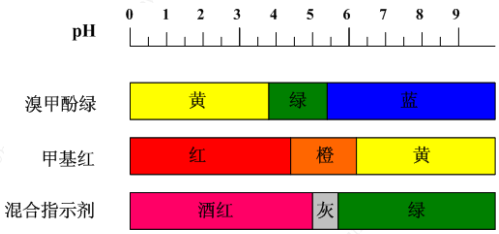



Figure 4.16: Color regions of a mixed indicator.

In practice the color change of an indicator must sometimes be confined to a very narrow range so that the endpoint is shown accurately; a single indicator then often cannot meet the requirement, and a mixed indicator is used. There are two kinds. One is a mixture of two or more indicators, whose colors are complementary so that the change becomes sharper. Bromocresol green, for example, is yellow in acid and blue in base with green as the intermediate color, while methyl red is red in acid and yellow in base with orange as the intermediate color, as shown in Figure 4.16.

When 0.1% bromocresol green is mixed with 0.2% methyl red in a ratio of 3:1, the green and orange intermediate colors are complementary and the mixture shows a very pale gray. At the same time the acid colors combine to a wine red and the base colors to green, a strong contrast. The mixed indicator therefore

changes sharply over a narrow range of intermediate color, which makes it easy to observe and confines the change to a very small interval of pH.

The other kind is a mixture of an indicator with an inert dye. The color of the dye does not change with pH; it serves only as a background and, through color complementarity, sharpens the change. Methyl orange mixed with sodium indigo disulfonate (blue), for example, is purple below pH $< 3.1$, gray at about pH $= 4.1$, and green above pH $> 4.4$, a very sharp change.

Table 4.3: Some common acid-base indicators

| Indicator | Transition range | Color | | $pK_a$ | Preparation |
|---|---|---|---|---|---|
| | | Acid | Base | | |
| Methyl orange | 3.1 ~ 4.4 | red | yellow | 3.4 | 1 g in 1 L of water |
| Methyl red | 4.4 ~ 6.2 | red | yellow | 5.2 | 0.1 g in 100 mL of 20% ethanol |
| Bromothymol blue | 6.0 ~ 7.6 | yellow | blue | 7.4 | 0.05 g in 100 mL of 20% ethanol |
| Phenol red | 6.7 ~ 8.4 | yellow | red | 8.0 | 0.1 g in 100 mL of 20% ethanol |
| Phenolphthalein | 8.2 ~ 10.0 | colorless | red | 9.1 | 0.1 g in 100 mL of 60% ethanol |
| Thymolphthalein | 9.4 ~ 10.6 | colorless | blue | 10.0 | 0.1 g in 100 mL of 90% ethanol |
| Alizarin yellow R | 10.1 ~ 12.1 | yellow | pale violet | 11.0 | 1 g in 1 L of water |
| Titan yellow | 12.0 ~ 13.0 | yellow | red | 12.5 | 1 g in 1 L of water |

### 4.6.2 Theoretical Color-Change Point and the Titration Error

Take the weak-acid indicator HIn as an example. Its dissociation equilibrium in aqueous solution is

$$\mathrm{HIn(aq)} \rightleftharpoons \mathrm{H^+(aq) + In^-(aq)}, \quad K_a = \frac{\frac{[\mathrm{H^+}]}{c^\ominus}\frac{[\mathrm{In^-}]}{c^\ominus}}{\frac{[\mathrm{HIn}]}{c^\ominus}}$$

From the expression for the dissociation constant,

$$\mathrm{pH} = \mathrm{p}K_a + \log\frac{[\mathrm{In^-}]}{[\mathrm{HIn}]} \tag{4.126}$$

The theoretical color-change point of the indicator is defined as the state in which $\frac{[\mathrm{In^-}]}{[\mathrm{HIn}]} = 1$, at which pH $= \mathrm{p}K_a$. Ideally the theoretical color-change point

would coincide exactly with the stoichiometric point of the system titrated, but this hardly ever happens. Because the two differ, a titration error arises, also called the endpoint error: the titrant has been added either in insufficient or in excess amount. Since the reaction follows a fixed stoichiometry, the endpoint error is calculated from the amount of substance, and is usually defined as

$$\epsilon_{\mathrm{ep}} = \frac{\text{终点时滴定剂过量（或不足）的物质的量}}{\text{待测物的物质的量}} \times 100\% \tag{4.127}$$

When NaOH of concentration $c(\mathrm{NaOH})$ titrates a volume $V_0$ of HCl of concentration $c(\mathrm{HCl})$, let the volume of NaOH consumed at the stoichiometric point be $V_{\mathrm{sp}}$ and that consumed at the endpoint be $V_{\mathrm{ep}}$. Equation (4.127) then becomes

$$\epsilon_{\mathrm{ep}} = \frac{c(\mathrm{NaOH})V_{\mathrm{ep}} - c(\mathrm{HCl})V_0}{c(\mathrm{HCl})V_0} \times 100\% \tag{4.128}$$

Rearranging (4.128),

$$\begin{aligned} \epsilon_{\mathrm{ep}} &= \frac{c(\mathrm{NaOH})V_{\mathrm{ep}} - c(\mathrm{HCl})V_0}{c(\mathrm{HCl})V_0} \times 100\% \\ &= \frac{\frac{V_{\mathrm{ep}}}{V_0} - \frac{c(\mathrm{HCl})}{c(\mathrm{NaOH})}}{\frac{c(\mathrm{HCl})}{c(\mathrm{NaOH})}} \times 100\% \end{aligned} \tag{4.129}$$

Since at the stoichiometric point

$$c(\mathrm{NaOH}) \times V_{\mathrm{sp}} = c(\mathrm{HCl}) \times V_0 \tag{4.130}$$

the volume ratio at the stoichiometric point is

$$\eta_{\mathrm{sp}} = \frac{V_{\mathrm{sp}}}{V_0} = \frac{c(\mathrm{HCl})}{c(\mathrm{NaOH})} \tag{4.131}$$

Substituting (4.131) into (4.129) gives the endpoint error

$$\epsilon_{\mathrm{ep}} = \frac{\eta_{\mathrm{ep}} - \eta_{\mathrm{sp}}}{\eta_{\mathrm{sp}}} \times 100\% \tag{4.132}$$

where $\eta_{ep} = \frac{V_{ep}}{V_0}$ is the volume ratio at the endpoint. Because the endpoint is judged by the theoretical color-change point of the indicator, this volume ratio may be calculated from the titration equation.

**Example 4.7** Calculate the endpoint error when 0.1000 mol/L NaOH standard solution titrates 0.1000 mol/L HCl with methyl orange as the indicator to pH = 4.00.

Solution: Since the titration is at equal concentrations, the volume ratio at the stoichiometric point is

$$\eta_{sp} = \frac{0.1000\,\text{mol/L}}{0.1000\,\text{mol/L}} = 1.000$$

The titration equation is

$$\eta = \frac{c(\text{HCl}) + [\text{OH}^-] - [\text{H}^+]}{c(\text{NaOH}) + [\text{H}^+] - [\text{OH}^-]}$$

At pH = 4.00,

$$\eta_{ep} = \frac{0.1000\,\text{mol/L} + 10^{-10.00}\,\text{mol/L} - 10^{-4.00}\,\text{mol/L}}{0.1000\,\text{mol/L} + 10^{-4.00}\,\text{mol/L} - 10^{-10.00}\,\text{mol/L}} = 0.998$$

so the endpoint error is

$$\epsilon_{ep} = \frac{0.998 - 1.000}{1.000} \times 100\% = -0.2\%$$

The titration of a weak acid with a strong base is similar; the key is again the volume ratio at the endpoint, as the next example shows.

**Example 4.8** Calculate the endpoint error when 0.1000 mol/L NaOH standard solution titrates 0.1000 mol/L $CH_3COOH$ to pH = 9.20.

Solution: The volume ratio at the stoichiometric point is $\eta_{sp} = 1.000$, and that at the endpoint is

$$\begin{aligned}\eta_{ep} &= \frac{\delta(CH_3COO^-)c(CH_3COOH) + [OH^-] - [H^+]}{c(NaOH) + [H^+] - [OH^-]} \\ &= \frac{\frac{K_a c^\ominus}{[H^+]+K_a c^\ominus} \times c(CH_3COOH) + [OH^-] - [H^+]}{c(NaOH) + [H^+] - [OH^-]} \\ &= \frac{\frac{1.8\times10^{-5}\,\text{mol/L}}{10^{-9.20}\,\text{mol/L}+1.8\times10^{-5}\,\text{mol/L}} \times 0.1000\,\text{mol/L} + \frac{10^{-14.00}}{10^{-9.20}}\,\text{mol/L} - 10^{-9.20}\,\text{mol/L}}{0.1000\,\text{mol/L} + 10^{-9.20}\,\text{mol/L} - \frac{10^{-14.00}}{10^{-9.20}}\,\text{mol/L}} \\ &= 1.000\end{aligned}$$

So the endpoint error is

$$\epsilon_{ep} = \frac{1.000 - 1.000}{1.000} \times 100\% = 0.0\%$$

Note: keeping a few more digits in $\eta_{ep}$ gives +0.03%.

One more example, a strong base titrating a mixture of acids, judges the feasibility of the titration from the endpoint error.

**Example 4.9** Can 0.02000 mol/L NaOH standard solution titrate selectively a mixed acid solution that is 0.02000 mol/L in HCl and 0.02000 mol/L in $CH_3COOH$?

Solution: Under the stated conditions there are two stoichiometric points, at $\eta = 1.000$ and $\eta = 2.000$. The titration curve is shown in Figure 4.17: only a small jump appears at $\eta = 1.000$ and a large one at $\eta = 2.000$. Whether the mixture can be titrated selectively must be decided by calculating the endpoint error at each stoichiometric point. Since the judgment of an indicator color change is uncertain by about ±0.30, the feasibility of

a stepwise titration may be judged by calculating the error over this range. The results are given in the tables below.

Error at the first stoichiometric point

| Indicator | Color-change point (pH) | Point ±0.30 (pH) | Relative error (%) |
|---|---|---|---|
| Methyl yellow | 3.30 | 3.00–3.60 | -7.8–4.2 |
| Methyl orange | 3.40 | 3.10–3.70 | -5.5–6.3 |
| Bromophenol blue | 4.10 | 3.80–4.40 | 8.6–31 |

Error at the second stoichiometric point

| Indicator | Color-change point (pH) | Point ±0.30 (pH) | Relative error (%) |
|---|---|---|---|
| Neutral red | 7.40 | 7.10–7.70 | -0.22–0.052 |
| Phenol red | 8.00 | 7.70–8.30 | -0.052–0.001 |
| Phenolphthalein | 9.10 | 8.80–9.40 | 0.043–0.19 |

The results show that the error at the first stoichiometric point is far too large for an accurate titration, whereas the second can be titrated accurately with phenolphthalein. A calculation shows that at the first stoichiometric point a titration error of ±0.1% corresponds to a pH range of only 3.38 ∼ 3.39, too narrow a jump to be judged with a chemical indicator.

The example shows how important the shape of the titration jump is for the feasibility of a quantitative determination. To make this clearer, the first jump of Figure 4.17 is redrawn in Figure 4.18. The two horizontal lines are drawn 0.30 pH units above and below the theoretical color-change point of methyl orange, $pK_a = 3.40$; the interval between them is 0.60 pH units, the range of acidity over which the color change may be judged by eye. The dotted line at $\eta = 1.000$ marks the first stoichiometric point, where pH = 3.38. The two horizontal lines meet the curve at $\eta = 0.945$ and $\eta = 1.063$; the lower line lies before the stoichiometric point and the upper one after it, so the whole range of color change straddles the stoichiometric point. The theoretical color-change point of methyl orange meets

the curve at $\eta = 1.004$, almost coinciding with the stoichiometric point, so the origin of the error is not the position of the color-change point but the width of the color-change interval. For a normal color change the titration must run from the lower triangle to the upper one on the curve, a change in volume ratio of $\Delta\eta = 0.118$.

What does this mean in practice? If the initial volume of the system is 20.00 mL, the change in volume ratio corresponds to a volume of NaOH of $20.00 \times 0.118 = 2.36$ mL. Since one drop from a burette is about 0.05 mL, no fewer than $2.36/0.05 \approx 47$ extra drops of NaOH would be needed to obtain the minimum color change. It is then easy to understand why the endpoint error at the first stoichiometric point is so large, and why a sufficiently large jump at the endpoint is essential.

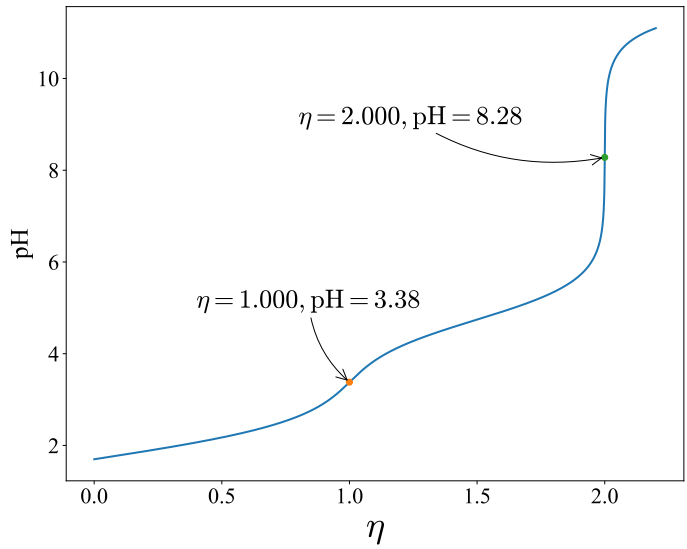



Figure 4.17: Titration curve of a mixture of hydrochloric and acetic acids, each 0.02000 mol/L, titrated with 0.02000 mol/L sodium hydroxide.

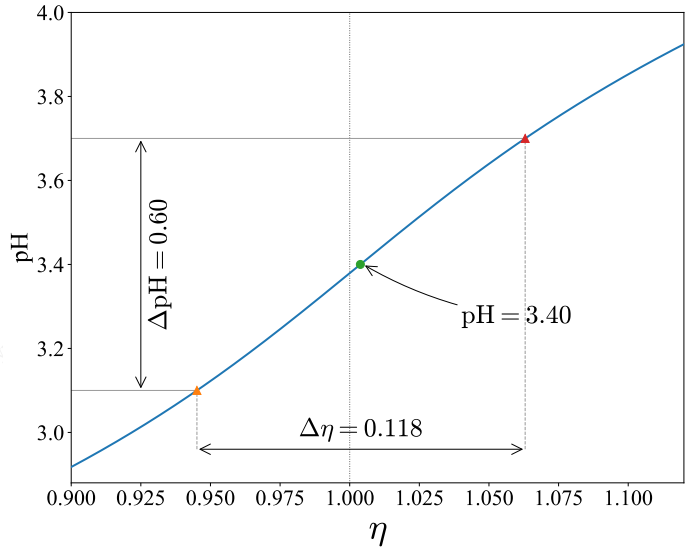



Figure 4.18: Enlarged view of the titration curve of a mixture of hydrochloric and acetic acids, each 0.02000 mol/L, titrated with 0.02000 mol/L sodium hydroxide.

### 4.6.3 The Influence of the Indicator on the Titration*

An indicator used in an acid-base titration is itself a weak acid (or weak base) and inevitably takes part in the reaction, which raises two questions: (1) how does the indicator influence the titration, and (2) how much indicator should actually be added?

Consider the simplest system, NaOH titrating HCl. With both at 0.1000 mol/L the volume ratio at the stoichiometric point is $\eta = 1.000$ and pH = 7.00. By the usual criteria bromothymol blue would be an ideal indicator, since its theoretical color-change point, $pK_a = 7.40$, lies very close to the stoichiometric point and its transition range, pH 6.00 ∼ 7.60, covers the jump of the system well.

The titration curves of the NaOH–HCl system were calculated for four cases and are shown in Figure 4.19. Figure 4.19(A) is a blank experiment without indi-

cator; its stoichiometric point lies at pH = 7.00, as expected for a strong base titrating a strong acid. When bromothymol blue is present at 1% of the concentration of the acid (Figure 4.19(B)), however, the stoichiometric point shifts markedly to pH = 5.20, clearly below neutrality. The presence of the indicator thus changes the position of the stoichiometric point.

Looking further at Figures 4.19(C) and (D), bromothymol blue shows a similar interference under different conditions. The shift of the stoichiometric point arises because the indicator, being a weak acid, takes part directly in the titration. In other words, bromothymol blue is no longer merely the "referee" that judges the endpoint but has become a "player" that affects the course of the reaction. This dual role inevitably causes a systematic error, and it is the theoretical defect inherent in the classical indicator method.

Table 4.4 lists the concentration of the basic form of bromothymol blue during the titration at various initial concentrations. When the initial concentration of bromothymol blue exceeds $1.00\times10^{-3}$ mol/L, the pH at $\eta = 1.001$ is $6.16 < 7.40$, so the acidity needed for a pronounced color change is not reached. To see the blue color one must titrate to a higher pH, which forces $\eta > 1.001$ and finally exceeds the permitted error. When the initial concentration is below the $10^{-4}$ mol/L level, the pH at $\eta = 1.001$ already exceeds 7.60 and a good color change is in principle possible; the concentration of $[In^-]$, however, is then $10^{-5}$ mol/L or less, and a color change at such a low concentration is hard for the eye to detect. This is another reason why the indicator method demands more of the analyst.

Only when the concentration of bromothymol blue is of the order of $10^{-4}$ mol/L, that is, about one thousandth of the concentration of the HCl to be titrated, can the change from yellow to blue be seen with ease while the error remains acceptable. In practice, however, the concentration of the analyte is not

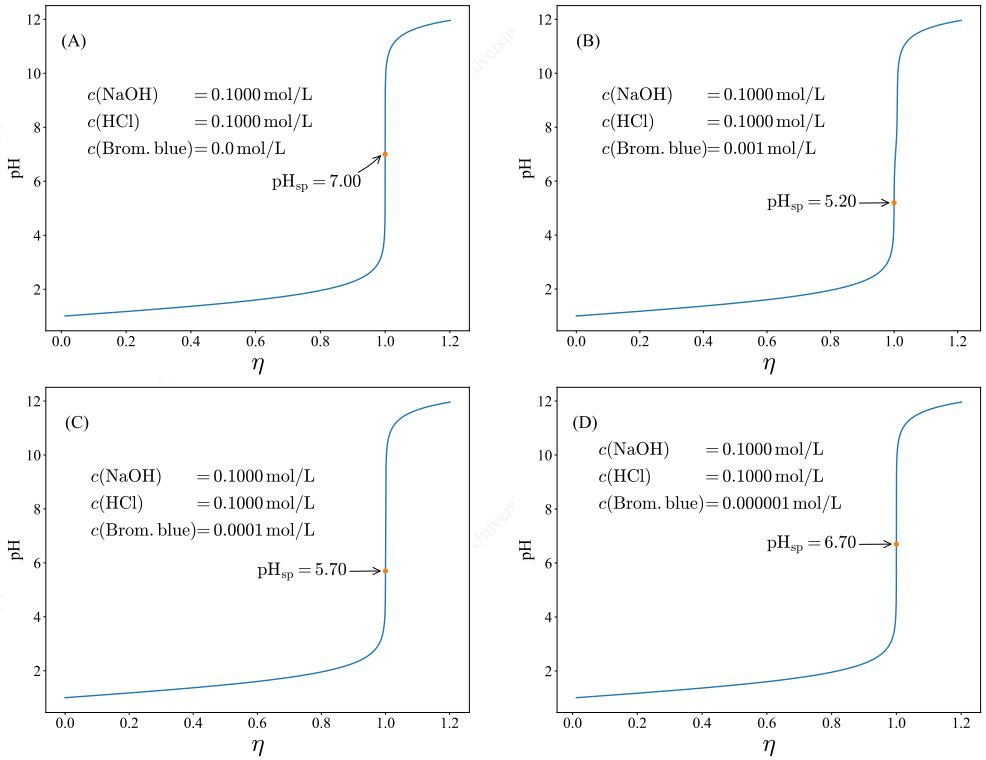



Figure 4.19: Titration curves of HCl with NaOH using bromothymol blue of different concentrations as indicator.

known in advance, so how can the optimal amount of indicator be added? Fortunately, its approximate value is known in routine analysis. For an acid produced by a factory, for example, the concentration of the final product is fixed by the process; the analyst only needs to determine a suitable amount of indicator by repeated trials, so that the determination of that product is not much affected by the indicator.

## 4.7 Applications

Acid-base titration is widely used in research and in production. The main components of many chemical products, such as caustic soda, soda ash, ammo-

Table 4.4: Concentration of the basic form of bromothymol blue during the titration

| Initial concentration (mol/L) | pH ($\eta = 0.999, 1.000, 1.001$) | $[In^-]$ ($\eta = 0.999, 1.000, 1.001$) |
|---|---|---|
| 0.100 | 4.04, 4.20, 4.37 | $4.35 \times 10^{-5}$, $6.29 \times 10^{-5}$, $9.29 \times 10^{-5}$ |
| $1.00 \times 10^{-3}$ | 4.30, 5.20, 6.16 | $7.92 \times 10^{-7}$, $6.22 \times 10^{-6}$, $5.15 \times 10^{-5}$ |
| $1.00 \times 10^{-4}$ | 4.31, 5.70, 8.39 | $8.10 \times 10^{-8}$, $1.91 \times 10^{-6}$, $4.75 \times 10^{-5}$ |
| $1.00 \times 10^{-5}$ | 4.31, 6.22, 9.66 | $8.10 \times 10^{-9}$, $5.81 \times 10^{-7}$, $4.98 \times 10^{-6}$ |
| $1.00 \times 10^{-6}$ | 4.31, 6.70, 9.70 | $8.10 \times 10^{-10}$, $1.42 \times 10^{-7}$, $4.99 \times 10^{-7}$ |

nium sulfate, and ammonium bicarbonate, are commonly determined by acid-base titration. Carbon, sulfur, phosphorus, silicon, and nitrogen in steel and in some raw materials can also be determined this way, and so can the raw materials, intermediates, and finished products of some organic synthesis and pharmaceutical industries. A few examples are given below.

### 4.7.1 Determination of Nitrogen in Organic Matter by the Kjeldahl Method

The Kjeldahl method was established by Johan Kjeldahl in 1883. Kjeldahl was then studying the chemistry of beer brewing and needed to determine the protein content of grain accurately. The traditional method was the Dumas method, which pyrolyzes the sample to nitrogen gas and calculates the nitrogen content from the gas volume; it involves many steps and is complicated and error-prone.

Kjeldahl found that organic matter is decomposed by concentrated sulfuric acid and that its nitrogen-containing constituents (proteins among them) are converted into ammonium sulfate. Adding alkali then decomposes the ammonium sulfate and liberates ammonia, which is distilled into a boric acid solution and finally titrated with hydrochloric acid; the nitrogen content follows from the amount of acid consumed. On this principle he built a new method for determin-

ing the nitrogen content of organic matter.

The method has three steps: digestion, distillation, and titration. Digestion is the critical step: the organic matter is decomposed by concentrated sulfuric acid, usually with a catalyst such as HgO or $CuSO_4$, and its nitrogen is converted into $(NH_4)_2SO_4$:

$$\text{(organic) N(aq)} + H_2SO_4(aq) \longrightarrow (NH_4)_2SO_4(aq)$$

Concentrated NaOH is then added to the digest until the solution is strongly alkaline, which decomposes the ammonium sulfate and liberates ammonia:

$$(NH_4)_2SO_4(aq) + 2\,NaOH(aq) = Na_2SO_4(aq) + 2\,NH_3(g) + 2\,H_2O(l)$$

The liberated $NH_3$ is carried over by steam into an excess of saturated $H_3BO_3$ solution, where

$$NH_3(g) + H_3BO_3(aq) = NH_4^+(aq) + H_2BO_3^-(aq)$$

The $H_2BO_3^-$ formed is a moderately strong base and may be titrated with a standard HCl solution:

$$H_2BO_3^-(aq) + HCl(aq) = Cl^-(aq) + H_3BO_3(aq)$$

The equations show that one mole of nitrogen finally consumes one mole of hydrochloric acid, so the molar ratio of nitrogen to acid is 1:1. Boric acid serves here as the medium that absorbs ammonia, with which it reacts to form $NH_4^+$ and $H_2BO_3^-$. Because boric acid is itself a very weak acid ($pK_a \approx 9.24$), the excess does not interfere with the titration of $H_2BO_3^-$ by hydrochloric acid, and this makes the boric acid absorption method an ideal way to capture ammonia.

Figure 4.20 shows the titration curve of $H_2BO_3^-$ with HCl; the stoichiometric point is at pH = 5.27, and in practice a mixed indicator is used. Two mixed

indicators are common: (1) two parts of methyl red in ethanol mixed with one part of methylene blue in ethanol just before use, and (2) one part of methyl red in ethanol mixed with five parts of bromocresol green in ethanol just before use.

$H_3BO_3$ also serves as the absorbing liquid; its concentration and volume may be increased and an excess ensured so that $NH_3$ is absorbed more efficiently. In the final titration a fixed volume of the absorbing liquid is taken, and an excess of $H_3BO_3$ in it does not appreciably affect the titration.

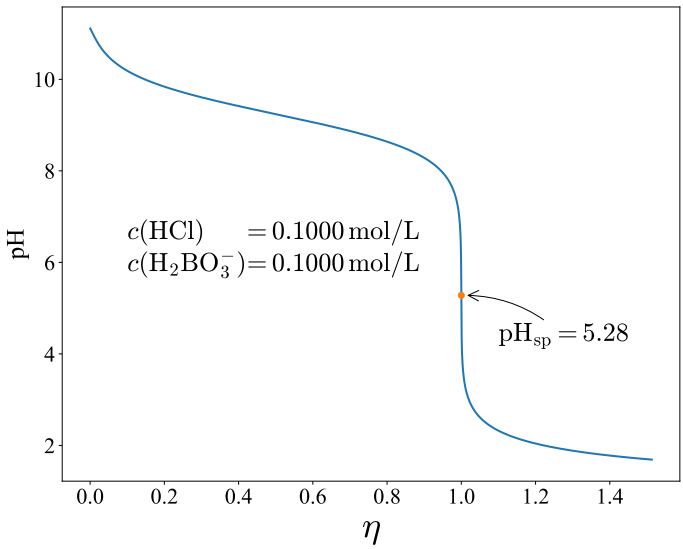



Figure 4.20: Titration curve of $H_2BO_3^-$ with HCl.

The current national standard for the method is GB/T 6432—2018, *Determination of crude protein in feeds —Kjeldahl method*. The procedure is as follows:

**Step 1:** Sample preparation. Draw a representative sample of feed according to GB/T 14699-2023 and reduce it by quartering. Prepare the test sample according to GB/T 20195-2024, grind it until it all passes a 0.42 mm test sieve, mix well, and store it in a closed container.

**Step 2:** Digestion. Run two determinations in parallel. Weigh 0.5 g $\sim$ 2 g of sample (containing 5 mg $\sim$ 80 mg of nitrogen, to 0.0001 g, denoted $m$) into a Kjeldahl flask or a digestion tube, add 6.4 g of mixed catalyst (or two Kjeldahl catalyst tablets), 12 mL of sulfuric acid, and two glass beads. Digest it by the heating program of GB/T 6432 until the solution is clear blue-green: for a Kjeldahl flask, raise the temperature from about 200°C to about 400°C and heat for at least 2 hours more; for a digestion tube, digest at 420°C for 1 hour. Remove and cool to room temperature.

**Step 3:** Transfer and dilution. When the digest has cooled, add 20 mL of water, transfer to a 100 mL volumetric flask, cool, dilute to the mark, and mix; this is the sample digest solution.

**Step 4:** Distillation of ammonia. Immerse the end of the condenser of a semimicro distillation apparatus in a conical flask containing 20 mL of boric acid absorbing solution 1 and two drops of mixed indicator. Pipette 10 mL $\sim$ 20 mL of the digest solution (denoted $V'$) into the reaction chamber, add 10 mL of sodium hydroxide solution, and distill. After 4 minutes lower the flask and distill for one more minute, until the distillate is neutral. Rinse the end of the condenser with water and let the washings run into the flask; then stop the distillation.

**Step 5:** Titration. Titrate the absorbing solution at once with 0.1 mol/L or 0.02 mol/L hydrochloric acid standard solution; the endpoint is the change from blue-green to gray-red (recorded as $V_2$).

**Step 6:** Blank test and calculation. Weigh 0.5 g of sucrose accurately in place of the sample and carry out a blank test in the same way (recorded as $V_1$); calculate the crude protein content by the formula.

The particle size in step 1 is critical: the sample must all pass the 0.42 mm sieve so that the organic nitrogen is completely converted into ammonium sulfate during digestion. Coarse particles give incomplete digestion and a systematic low result, while sieving, mixing, and sealing prevent the sample from taking up moisture or losing components.

The digestion conditions in step 2 decide the success of the method. The temperature follows the program of the standard (about 200°C to 400°C in a Kjeldahl flask, 420°C in a digestion tube); too high a temperature may decompose the ammonium sulfate and lose nitrogen, too low a one leaves the digestion incomplete. The end of digestion is marked by a clear blue-green solution, at which the organic carbon has been fully oxidized, but heating must continue for at least two more hours so that hard-to-digest components such as nitrogen heterocycles are fully converted. In the mixed catalyst, potassium sulfate (or sodium sulfate) raises the boiling point of the sulfuric acid and speeds up digestion, while copper sulfate pentahydrate catalyzes the oxidation. The glass beads prevent bumping during digestion.

Dilution in step 3 calls for care about the thermal contraction of the solution. After digestion the solution is cooled, diluted with water, and made up to the mark only when completely at room temperature, so that the volume is accurate. The digest solution should be distilled as soon as possible; on standing, ammonium salts may hydrolyze or microorganisms may change the form of the nitrogen.

The distillation in step 4 requires strict control of the amount of alkali and

the time. Adding 10 mL of sodium hydroxide solution (40 g of sodium hydroxide in 100 mL of water) ensures that the reaction chamber is strongly alkaline and that the ammonium salts decompose completely. Most of the ammonia escapes in the first 4 minutes; the "post-distillation" of the next minute drives out the rest. Judging the end of distillation by the neutrality of the distillate is more reliable than timing alone. The end of the condenser must stay below the surface of the absorbing liquid to prevent loss of ammonia.

The endpoint in step 5 must be observed keenly. The mixed indicator, methyl red with bromocresol green, is blue-green in the boric acid absorbing solution and turns gray-red as the hydrochloric acid is added. The color change corresponds to about pH = 5.1, within the titration jump of the boric acid–ammonium system.

The blank test in step 6 corrects for nitrogen impurities from the reagents and the environment. Sucrose, a pure organic substance, contains very little nitrogen; the blank must not consume more than 0.2 mL of 0.1 mol/L hydrochloric acid (or 0.3 mL of 0.02 mol/L acid), otherwise the purity of the reagents and the working environment should be checked.

In addition, the distillation step should be checked periodically: weigh 0.2 g of ammonium sulfate accurately in place of the sample; the nitrogen found should be (21.19 ± 0.2)%, which verifies the alkalization, distillation, and titration.

The crude protein content is calculated as

$$w = \frac{(V_2 - V_1) \times c(\mathrm{HCl}) \times \frac{14}{1000} \times 6.250}{m \times \frac{V'}{V}} \times 100$$

where 6.250 is the average factor converting nitrogen into crude protein.

Two parallel determinations are made for each sample and their arithmetic mean is reported to two decimal places. Under repeatability conditions the absolute difference between two independent results must not exceed the following

fractions of their mean: 1% when the crude protein content exceeds 25%, 2% between 10% and 25%, and 3% below 10%.

The result of the Kjeldahl method depends on the forms in which nitrogen occurs in the sample, and the two kinds of interference act in opposite directions. Non-protein nitrogen such as urea and ammoniacal nitrogen is also converted into ammonium during digestion and makes the "crude protein" higher than the true protein content; for such feeds (urea feeds, ammoniated straw) the protein nitrogen must be separated by methods such as trichloroacetic acid precipitation. Nitrate and nitrite nitrogen behave in the opposite way: under standard Kjeldahl conditions they are not quantitatively converted into ammonium but escape as NO and $NO_2$, giving low results; such samples require a modified Kjeldahl method (reducing nitrate to ammonium with salicylic acid and sodium thiosulfate before digestion) or the Dumas combustion method.

**Example 4.10** A 0.2500 g food sample is analyzed for nitrogen by the Kjeldahl method and titrated to the endpoint with 0.1000 mol/L HCl standard solution, of which 21.20 mL is consumed. Calculate the protein content of the food.

Solution: The nitrogen content of the sample is

$$w(\mathrm{N}) = \frac{0.1000\ \mathrm{mol/L} \times 21.20 \times 10^{-3}\ \mathrm{L} \times 14.01\ \mathrm{g/mol}}{0.2500\ \mathrm{g}} \times 100\%$$

$$= 11.88\%$$

The factor converting nitrogen into protein is 6.250, so

$$w(\text{protein}) = 6.250 \times 11.88\% = 74.25\%$$

The Kjeldahl method appeared just when the fertilizer industry and agriculture of the late nineteenth century needed it. Whether for assessing the quality of grain or for analyzing the fertility of soils and fertilizers, a fast and accurate method for the nitrogen content of organic matter was required; reliable and efficient, the method quickly became a standard in analytical chemistry and is still used today. Kjeldahl apparatus based on the same principle is now common in laboratories. Accurate as it is, however, the method is cumbersome: distillation and absorption, its core steps, are tedious, since distillation needs special apparatus and precise control of temperature and distillate flow to prevent loss of ammonia, while absorption requires an excess of saturated boric acid solution, which consumes reagents and reduces the economy of the experiment. These limitations have driven a continuing search for simpler and more efficient alternatives.

### 4.7.2 Determination of Nitrogen in Ammonium Salts

Nitrogen is one of the principal elements of living matter and a core constituent of proteins and nucleic acids. The Kjeldahl method converts the nitrogen of a nitrogen-containing substance into ammonium sulfate, the key species being $NH_4^+$. Once the amount of $NH_4^+$ is known, the nitrogen content of the organic matter and hence its protein content follow.

$NH_4^+$, however, is a very weak acid and cannot be titrated directly with a standard NaOH solution. Figure 4.21 shows the titration curve of $NH_4^+$ with NaOH: there is no pronounced jump at the stoichiometric point, so direct titration would give a large endpoint error. The reader may review Figures 4.17 and 4.18.

The formaldehyde method may have been suggested by Sørensen's "formaldehyde titration." It is a classical volumetric method for ammoniacal nitrogen based on acid-base titration: under carefully controlled neutral condi-

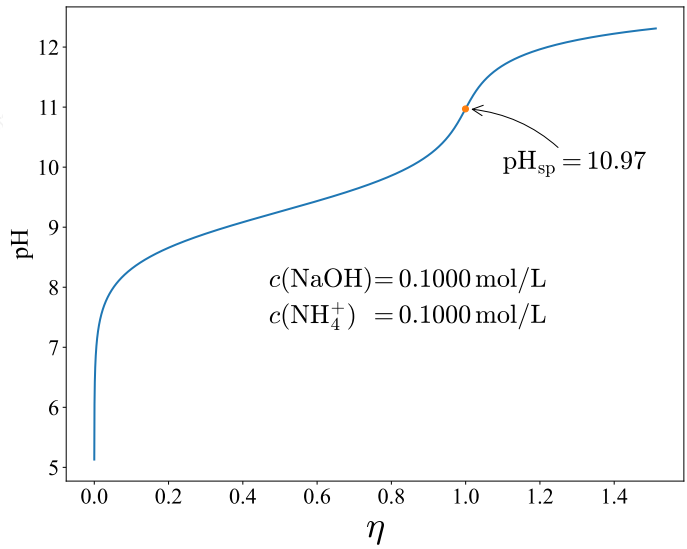



Figure 4.21: Titration curve of a $NH_4^+$ solution with NaOH.

tions the ammonium ion condenses quantitatively with excess formaldehyde to hexamethylenetetramine and liberates an equal amount of free acid, which is titrated with a standard sodium hydroxide solution, the ammoniacal nitrogen being calculated from the volume of alkali consumed. Compared with distillation, the formaldehyde method needs no distillation apparatus, is simple and fast, and shortens the analysis of one sample to 20 $\sim$ 30 min; it is particularly suited to the routine testing of large numbers of ammonium sulfate, ammonium chloride, and ammonium nitrate samples and of compound fertilizers free of urea and organic nitrogen. The current national standard, GB/T 3600-2000, *Determination of ammoniacal nitrogen content in fertilizers —Formaldehyde method*, specifies the preparation of reagents, the reaction conditions, the titration, and the calculation.

The reactions involved are

$$4NH_4^+(aq) + 6HCHO(aq) \rightleftharpoons (CH_2)_6N_4H^+(aq) + 3H^+(aq) + 6H_2O(l)$$

Since the dissociation constant of $(CH_2)_6N_4H^+$ is $K_a = 7.10 \times 10^{-6}$, it is markedly stronger than the ammonium ion ($K_a(NH_4^+) = 5.6 \times 10^{-10}$). Adding formaldehyde to an ammonium salt solution therefore changes the system from a single weak acid into a mixture of a strong acid (the liberated $H^+$) and a weak acid ($(CH_2)_6N_4H^+$).

Figure 4.22 shows the titration curve of a $NH_4^+$ solution treated with formaldehyde and titrated with NaOH. There is a sufficient jump at the stoichiometric point, covering pH 7 $\sim$ 11, so that phenolphthalein can be used for an accurate titration.

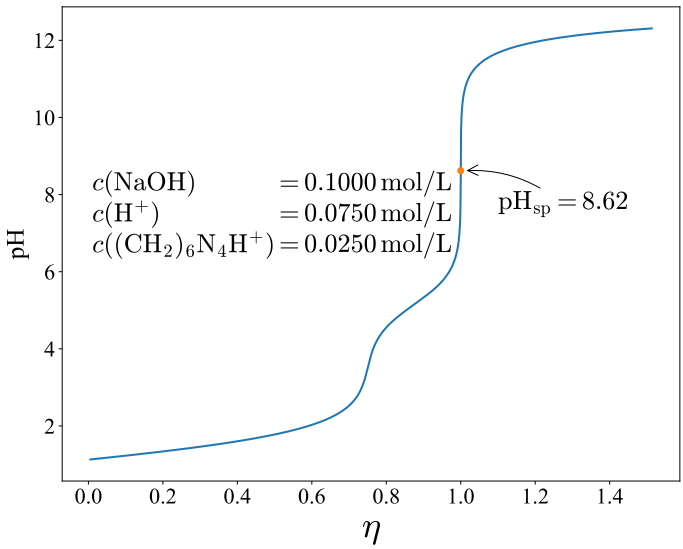



Figure 4.22: Titration curve of a $NH_4^+$ solution treated with formaldehyde and titrated with NaOH.

The procedure for the ammoniacal nitrogen of fertilizers is as follows:

**Step 1:** Dissolution. Weigh 0.5 g of sample (to 0.0001 g, denoted $m$) into a 250 mL conical flask, add about 50 mL of water, and swirl until the sample dissolves completely. If insoluble mat-

ter remains, filter through medium-speed paper and wash the residue several times with small portions of water, adding the washings to the filtrate.

**Step 2:** Neutralization. Add two or three drops of methyl red indicator to the solution. If it is red, add sodium hydroxide solution ($c(\mathrm{NaOH}) = 0.1\,\mathrm{mol/L}$) until it just turns orange; if it is yellow, add hydrochloric acid ($c(\mathrm{HCl}) = 0.1\,\mathrm{mol/L}$) until it just turns orange, so that the solution is nearly neutral.

**Step 3:** Reaction with formaldehyde. Add 15 mL of formaldehyde solution that has been neutralized beforehand to a faint pink with sodium hydroxide using phenolphthalein as indicator. Mix and let stand for 5 min so that the ammonium ion reacts completely.

**Step 4:** Titration. Add three or four drops of phenolphthalein indicator and titrate with standard sodium hydroxide solution ($c(\mathrm{NaOH}) = 0.5\,\mathrm{mol/L}$) to a faint pink that persists for 30 s (recorded as $V$).

**Step 5:** Blank test and calculation. Carry out a blank test in the same way (recorded as $V_0$) and calculate the ammoniacal nitrogen content by the formula.

The sample mass in step 1 is adjusted to the nitrogen content so that the titration consumes $10 \sim 30\,\mathrm{mL}$ of standard sodium hydroxide. The mass must match the capacity of the burette: too much sample generates more acid than the burette can deliver and dulls the jump, while too little increases the relative error. Fertilizers often contain a little insoluble filler or impurity, and filtering

it off avoids interference with the endpoint color, but the residue must not be washed too many times, lest ammoniacal nitrogen be lost.

The neutralization in step 2 is the key prerequisite of the method: the quantitative reaction between formaldehyde and the ammonium ion must take place in a neutral or weakly acidic medium. If the solution is too acidic, formaldehyde binds hydrogen ions or reacts otherwise and more alkali is needed in the titration, giving a high result; if it is alkaline, some ammonium ion may be converted into ammonia and escape, giving a low result. The transition range of methyl red is pH 4.4 (red) to 6.2 (yellow), and at the orange color the pH is about 5.0 $\sim$ 5.4, which keeps the ammonium ion stable and provides a suitable acidity for the formaldehyde reaction. Figure 4.23 shows the titration curve of the free sulfuric acid in the sample during neutralization.

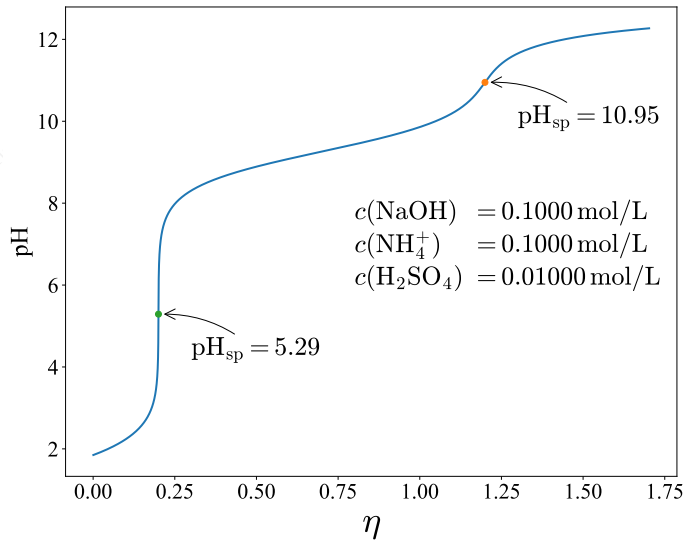



Figure 4.23: Titration curve of an ammonium solution containing free sulfuric acid, titrated with sodium hydroxide.

The formaldehyde must be neutralized beforehand in step 3 because commercial formaldehyde usually contains traces of formic acid and other acidic im-

purities; without neutralization these acids would consume titrant and give a systematic high result. The reaction between formaldehyde and the ammonium ion is fast at room temperature, and standing for 5 min makes it complete. The hexamethylenetetramine formed is a weak organic base ($pK_b \approx 8.8$) that shows no basicity within the transition range of phenolphthalein and therefore does not interfere with the titration of the free acid. By the stoichiometry, 4 mol of ammonium ion give 4 mol of hydrogen ion, that is, the molar ratio of nitrogen to hydrogen ion is 1:1, so an equal-mole calculation applies directly.

Phenolphthalein is used in step 4; its transition range is pH 8.2 ~ 10.0 (colorless to red). The endpoint is the first appearance of a faint pink, at which the free acid has just been neutralized and a slight excess of alkali gives the color. Because hexamethylenetetramine is very weakly basic it does not neutralize a strong acid appreciably, so the endpoint is sharp. If the sample contains a little colored impurity such as iron, the endpoint color may be dull, and a mixed indicator or potentiometric titration may be used instead.

The blank test in step 5 corrects for acidic impurities in the formaldehyde and the reagents. The ammoniacal nitrogen content is calculated as

$$w(\mathrm{N}) = \frac{c(\mathrm{NaOH}) \times (V - V_0) \times M(\mathrm{N})}{m} \times 100\%$$

where $c$(NaOH) is the concentration of the standard sodium hydroxide solution (mol/L); $V$ and $V_0$ are the volumes of that solution consumed by the sample and by the blank (mL); $M(\mathrm{N}) = 0.01401$ g/mmol is the millimolar mass of nitrogen; and $m$ is the sample mass (g).

The complex color changes during the titration and the judgment of the endpoint are explained clearly by the titration curve (Figure 4.22). The jump not only supports the choice of phenolphthalein but also reflects the conversion of the protonated species in solution. Figure 4.24 shows the concentration curves of

the key components, from which the hydrogen ion is titrated first and the weak acid $(CH_2)_6N_4H^+$ afterwards. Calculating the curve of a complex system from the general equation and using it to guide the actual procedure is therefore important: careful theoretical analysis is the foundation on which a complex titration is correctly designed and carried out.

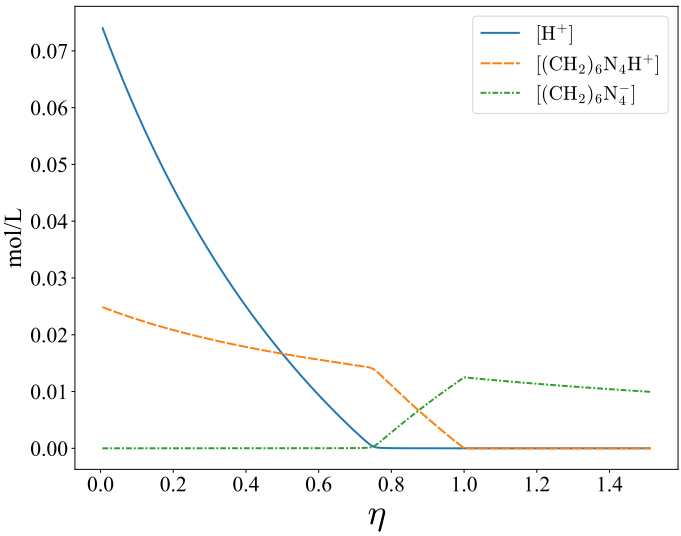



Figure 4.24: Concentration curves of the principal components during the titration of a formaldehyde-treated ammonium solution with sodium hydroxide.

The formaldehyde method has clear limits: it is not suitable for ammonium bicarbonate (the $HCO_3^-$ interferes with the reaction), for aqueous ammonia (loss by volatilization), or for fertilizers containing urea, cyanamides, or organic nitrogen compounds (which react with formaldehyde or produce extra acid and give high results). In practice the composition of the sample must therefore be known first, and the method should be used only after confirming that none of these interferences is present.

### 4.7.3 Determination of Sodium Hydroxide and Sodium Carbonate in Caustic Soda

Caustic soda is the common name of sodium hydroxide, now produced mainly by the ion-exchange membrane process: the NaOH solution formed is concentrated to solid NaOH. On an industrial scale, however, the packaged solid product is stored in large warehouses, where it inevitably absorbs carbon dioxide from the air and forms a little sodium carbonate, which affects the quality of the product.

Three methods are commonly used to determine sodium hydroxide and sodium carbonate in caustic soda: (1) potentiometric titration, (2) the double-indicator method, and (3) the barium chloride method. Potentiometric titration requires special instrumentation and is not discussed further here; only the last two methods are described.

#### The Double-Indicator Method

The double-indicator method titrates directly with a standard hydrochloric acid solution in two steps. In the first step phenolphthalein is used as the indicator and the solution is titrated with standard hydrochloric acid until the red color just disappears; in this step sodium carbonate is converted into sodium bicarbonate:

$$NaOH(aq) + HCl(aq) = NaCl(aq) + H_2O(l)$$

$$Na_2CO_3(aq) + HCl(aq) \rightleftharpoons NaHCO_3(aq) + NaCl(aq)$$

In the second step methyl orange is added and the titration with standard hydrochloric acid is continued to an orange-red endpoint; to make the endpoint easier to see, the titration may be paused near the end and the solution heated to

expel carbon dioxide. The reaction is

$$NaHCO_3(aq) + HCl(aq) = H_2CO_3(aq) + NaCl(aq)$$

Let the volume of hydrochloric acid consumed in the first step be $V_1$ and that in the second step $V_2$. The mass fractions of sodium carbonate and sodium hydroxide then follow from

$$w(Na_2CO_3) = \frac{c(HCl) \times V_2 \times M(Na_2CO_3)}{m_{烧碱试样}} \times 100\% \tag{4.133}$$

$$w(NaOH) = \frac{c(HCl) \times (V_1 - V_2) \times M(NaOH)}{m_{烧碱试样}} \times 100\% \tag{4.134}$$

By exploiting the color changes of phenolphthalein and methyl orange at different acidities, the double-indicator method determines sodium hydroxide and sodium carbonate in one titration, successively and in steps. Its core is the accurate location of the two jumps and the recording of the two volumes: the first volume $V_1$ corresponds to the acid needed to neutralize $OH^-$ to $H_2O$ and $CO_3^{2-}$ to $HCO_3^-$, while the second volume $V_2$ corresponds to the acid needed to convert all the $HCO_3^-$ (which comes entirely from $CO_3^{2-}$) into $H_2CO_3$. The calculation rests entirely on the stoichiometry of proton transfer during the titration.

The method is simple, fast, and needs no separation by precipitation. Its accuracy, however, depends strongly on judging the two endpoints correctly. The second endpoint (the methyl orange change) is less sharp because of the buffering action of carbonic acid, and it is the main source of error; the titration is therefore paused near the second endpoint and the solution boiled to remove carbon dioxide, which sharpens the endpoint.

### The Barium Chloride Method

The barium chloride method makes use of the ease with which carbonate forms a precipitate: barium chloride solution is added to the caustic soda solution

so that sodium carbonate is precipitated as barium carbonate, and the titration is then carried out in steps. The procedure is as follows. (1) Divide the prepared solution into two equal portions. Titrate the first portion with standard hydrochloric acid to the methyl orange endpoint; both sodium hydroxide and sodium carbonate are titrated, and the volume of acid consumed is recorded as $V_1$. (2) Add barium chloride solution to the second portion to precipitate the carbonate as barium carbonate, then titrate the sodium hydroxide with standard hydrochloric acid to the phenolphthalein endpoint, the volume consumed being recorded as $V_2$. Thus

$$w(\mathrm{NaOH}) = \frac{c(\mathrm{HCl}) \times V_2 \times M(\mathrm{NaOH})}{\frac{1}{2} \times m_{\text{烧碱试样质量}}} \times 100\% \tag{4.135}$$

$$w(\mathrm{Na_2CO_3}) = \frac{c(\mathrm{HCl}) \times (V_1 - V_2) \times M(\mathrm{Na_2CO_3})}{m_{\text{烧碱试样质量}}} \times 100\% \tag{4.136}$$

In the second portion, if the acidity is not properly controlled or the precipitate is left standing too long, the barium carbonate may dissolve. The dissolution equilibrium is

$$\mathrm{BaCO_3(s)} \rightleftharpoons \mathrm{Ba^{2+}(aq) + CO_3^{2-}(aq)}$$

If the pH falls, whether because the solution absorbs carbon dioxide or because the acidity is locally too high, $\mathrm{CO_3^{2-}}$ is converted into $\mathrm{HCO_3^-}$ or even $\mathrm{H_2CO_3}$ and the dissolution equilibrium shifts to the right. Some of the fixed carbonate then returns to solution and consumes extra hydrochloric acid during the phenolphthalein titration, so that $V_2$ is too large; the mass fraction of sodium hydroxide is then too high and that of sodium carbonate too low. The solution must therefore remain sufficiently alkaline after the barium chloride is added, and the precipitate must not stand for long, so that this error is kept small.

### 4.7.4 Determination of Silicon Dioxide in Silicates

Silicon is one of the most abundant elements in the earth's crust. In silicate rocks, cement, metallurgical slag, and refractories it usually occurs as silicon dioxide ($SiO_2$), and the accurate determination of that dioxide is a core index of the quality and process performance of silicate materials. Gravimetry (such as the poly(ethylene oxide) dehydration method) is the classical reference method, but it is lengthy, requires high-temperature ignition, and needs platinum ware, so it hardly meets the demand for rapid testing of large numbers of samples. The potassium fluorosilicate method converts the silicon into a potassium fluorosilicate precipitate that can be hydrolyzed quantitatively and then determines the hydrofluoric acid liberated by acid-base titration, a neat way of determining medium and high contents of silicon dioxide quickly and accurately.

The reactions involved are

$$SiO_2(s) + 6\,HF(aq) = H_2SiF_6(aq) + 2\,H_2O(l)$$

$$H_2SiF_6(aq) + 2\,K^+(aq) = K_2SiF_6(s) + 2\,H^+(aq)$$

$$K_2SiF_6(s) + 3\,H_2O(l) = 2\,KF(aq) + H_2SiO_3(s) + 4\,HF(aq)$$

$$HF(aq) + NaOH(aq) = NaF(aq) + H_2O(l)$$

After decomposition with hydrofluoric acid the silicon is present as fluorosilicic acid ($H_2SiF_6$). Enough potassium ion is added in strongly acidic medium to precipitate potassium fluorosilicate ($K_2SiF_6$) quantitatively. The precipitate is filtered and washed free of acid, then hydrolyzed with boiling water to liberate hydrofluoric acid (HF) quantitatively. The acid is titrated with a standard sodium hydroxide solution using phenolphthalein as indicator, and the silicon (or silicon dioxide) content follows from the volume of alkali consumed. By the stoichiometry, one mole of Si finally corresponds to four moles of HF, that is, four moles of

NaOH. The method thus converts the weakly acidic silicic acid, which can hardly be titrated directly, into the strong acid HF and thereby obtains a pronounced jump and reliable accuracy.

Compared with gravimetry the method needs no ignition, no constant weighing, and no platinum crucible; one sample takes $40 \sim 60$ min. It is particularly suited to the routine testing of large numbers of silicate materials such as cement, building materials, metallurgical slag, and refractories.

The procedure is as follows:

**Step 1:** Decomposition. Weigh $0.10 \sim 0.20$ g of sample (to 0.0001 g, denoted $m$) into a 300 mL plastic beaker, add 15 mL of potassium nitrate–nitric acid solution, swirl gently, and add 5 mL of hydrofluoric acid ($\rho = 1.15$ g/mL) drop by drop until the sample dissolves completely. Add 5 mL of urea solution (50 g/L) to expel nitrogen oxides, swirl until no more bubbles appear, and let stand for $1 \sim 2$ min.

**Step 2:** Precipitation. Add 5 mL of saturated potassium nitrate solution, stir, add a little paper pulp, and keep the beaker in a water bath below 25°C for 20 min so that the potassium fluorosilicate precipitates completely.

**Step 3:** Filtration and washing. Filter with suction through medium-speed quantitative paper in a Büchner funnel (or a plastic funnel), wash the beaker quickly with potassium nitrate solution (50 g/L) and transfer all the precipitate to the paper, then continue washing paper and precipitate. Use no more than 5 mL per washing and no more than 30 mL in total.

**Step 4:** Neutralization of residual acid. Put the precipitate, together

with the paper, into the original plastic beaker and add 15 mL of potassium nitrate–ethanol solution (50 g/L) to cover paper and precipitate. Add five drops of phenolphthalein (10 g/L) and stir thoroughly with a white or colorless plastic rod, breaking up the paper. Titrate the residual acid with standard sodium hydroxide solution ($c$(NaOH) $\approx$ 0.25 mol/L) to a stable rose color (this step is not recorded).

**Step 5:** Hydrolysis and titration. Add nearly 150 mL of boiling water (boiled beforehand to expel carbon dioxide) to the beaker and stir thoroughly so that the precipitate hydrolyzes rapidly. Titrate at once with standard sodium hydroxide solution, first quickly to near the endpoint, then add four more drops of phenolphthalein and continue slowly to a faint pink that persists for 30 s (recorded as $V$).

**Step 6:** Blank test and calculation. Carry out a blank test in the same way (recorded as $V_0$) and calculate the silicon dioxide content by the formula.

The sample mass in step 1 is adjusted to the silicon dioxide content so that the titration consumes 20 $\sim$ 40 mL of standard sodium hydroxide. Too much sample generates more acid than the burette can deliver and dulls the jump; too little increases the titration error. For hard-to-dissolve materials such as silicate rocks, the sample may first be fused with a sodium carbonate–sodium peroxide flux in a nickel crucible at 850°C $\sim$ 900°C; the melt is leached with hot water, transferred to a plastic beaker, acidified, and treated as above. Decomposition and precipitation must be carried out in plastic ware (polyethylene or polypropylene),

because hydrofluoric acid and potassium fluorosilicate attack glass strongly.

The temperature control in step 2 decides the success of the method: precipitation must be kept strictly below 25°C. At higher temperatures the precipitate is markedly more soluble and tends to form a colloid, which makes filtration difficult and gives low results. In hot summer weather an ice-water bath is necessary. A little paper pulp after precipitation makes the subsequent filtration easier.

The washing in step 3 must be controlled precisely. The precipitate is washed thoroughly with potassium nitrate solution to remove adsorbed free acid (mainly nitric and hydrofluoric acids). Insufficient washing leaves acid that consumes extra alkali in the later titration and gives high results, whereas excessive washing or too much wash liquid dissolves part of the precipitate and gives low results; the standard therefore limits each washing to 5 mL and the total to 30 mL.

The neutralization in step 4 is necessary for accuracy. Free acid left on the paper and the precipitate would otherwise interfere with the titration of the hydrofluoric acid from hydrolysis. The ethanol lowers the solubility of potassium fluorosilicate and prevents the precipitate from hydrolyzing during neutralization. The titration is stopped as soon as the rose color appears and its volume is not recorded; over-neutralization would also hydrolyze part of the precipitate and cause error.

In step 5 the water for hydrolysis must be neutral and boiling, and the titration must follow at once so that the hydrofluoric acid formed neither escapes nor absorbs carbon dioxide from the air. The solution should stay above 70°C so that the hydrolysis is complete and the endpoint sharp. Phenolphthalein is used, with a transition range of pH $8.2 \sim 10.0$ (colorless to red); after the free acid has been neutralized, the last half-drop of alkali turns the solution faintly pink. If a dark sample matrix obscures the endpoint, a bromothymol blue–phe-

nol red mixed indicator (yellow to purple-red) or potentiometric titration (with a pH meter, pH $\approx$ 8.2) may be used instead.

The blank test in step 6 corrects for acidic impurities in the reagents and the filter paper.

The silicon dioxide content is calculated as

$$w(\mathrm{SiO_2}) = \frac{c(\mathrm{NaOH}) \times (V - V_0) \times M(\mathrm{SiO_2})}{m} \times 100\%$$

where $c$(NaOH) is the concentration of the standard sodium hydroxide solution (mol/L); $V$ and $V_0$ are the volumes consumed by the sample and by the blank (mL); $M(\mathrm{SiO_2}) = 0.01502$ g/mmol is the millimolar mass of silicon dioxide (taking $\frac{1}{4}\mathrm{SiO_2}$ as the unit); and $m$ is the sample mass (g). If the result is to be reported as silicon, $M(\mathrm{SiO_2})$ is replaced by $M(\mathrm{Si}) = 0.00702$ g/mmol (taking $\frac{1}{4}\mathrm{Si}$ as the unit), or the silicon result is multiplied by 2.1393 ($M_{\mathrm{SiO_2}}/M_{\mathrm{Si}}$) to give silicon dioxide.

The hydrolysis of potassium fluorosilicate produces the strong acid HF quantitatively and the endpoint is therefore sharp. Sorting out the equilibria of the three steps, precipitation, hydrolysis, and titration, is the prerequisite for designing such a volumetric analysis correctly.

The method has clear limits: the precipitation temperature must be below 25°C, the titration above 70°C, and glassware must not be used. Large amounts of aluminum, titanium, and similar elements in the sample may form fluoride complexes in hydrofluoric acid, consuming the acid or contaminating the precipitate. For samples with complex matrices the applicability of the method should be verified with a reference material, and gravimetry (the reference method) used for confirmation when necessary. The composition and approximate content of the sample should therefore be known before the method is used, and the interferences and operating conditions must be shown to be acceptable.

### 4.7.5 Determination of the Saponification Value (Back Titration)

Some systems cannot be titrated directly by an acid-base titration because the reaction is too slow, the analyte is volatile, or no suitable indicator exists. Back titration is the classical answer: a known excess of a standard solution is added to the sample, the reaction is allowed to go to completion, and the excess reagent is then titrated with a second standard solution; the content of the analyte follows from the difference between the two volumes consumed. This section illustrates back titration with the determination of the saponification value of animal and vegetable fats and oils.

The saponification value is an important index in the analysis of fats and oils. It is defined as the number of milligrams of potassium hydroxide consumed in saponifying 1 g of fat or oil under specified conditions, expressed as mg KOH/g. The value is closely related to the mean relative molecular mass of the fatty acids of the oil: the shorter their average carbon chain, the more ester bonds a unit mass of the oil contains and the higher the saponification value. It is therefore an important parameter for judging the quality and purity of oils and is also commonly used to identify the kind of oil and to detect adulteration.

The current national standard is GB/T 5534—2024, *Animal and vegetable fats and oils —Determination of saponification value*. Its principle is as follows: an excess of potassium hydroxide–ethanol solution is added to the sample and heated under reflux in a water bath so that the oil is completely saponified; after saponification the remaining potassium hydroxide is titrated with a standard hydrochloric acid solution, phenolphthalein indicating the endpoint. A blank test is run at the same time, and the saponification value is calculated from the difference between the volumes of acid consumed by the blank and by the sample.

The saponification of an oil may be written in general form as

$$\mathrm{RCOOR'(l) + KOH(l) \longrightarrow RCOOK(l) + R'OH(l)}$$

where the KOH is added as an ethanol solution, and the oil, the potassium soap, and the glycerol are all in the liquid phase.

After saponification the remaining potassium hydroxide is neutralized by the hydrochloric acid:

$$\mathrm{KOH(aq) + HCl(aq) = KCl(aq) + H_2O(l)}$$

By the stoichiometry, the amount of potassium hydroxide consumed in saponification equals the difference between the amounts of hydrochloric acid consumed by the blank and by the sample, from which the saponification value is calculated.

The procedure is as follows:

**Step 1:** Weighing and saponification. Weigh the sample according to its expected saponification value and Table 4.5 (to 0.001 g, denoted $m$) into a 250 mL ground-glass conical flask. Pipette exactly 25.00 mL of potassium hydroxide – ethanol solution ($c(\mathrm{KOH}) \approx 0.5\,\mathrm{mol/L}$) into the flask and add a few boiling beads. Attach a reflux condenser, heat the flask in a water bath under reflux, and shake it from time to time so that the sample is completely saponified. The reflux time depends on the type of oil and is usually 60 min; for samples containing hard-to-saponify matter (such as waxes) it may be lengthened.

**Step 2:** Titration. When saponification is complete, remove the condenser, rinse its inner wall with a little water, and let the washings run into the flask. Add 0.5 ~ 1.0 mL of phenolph-

thalein indicator (10 g/L in 95% ethanol) to the hot solution and titrate while hot with standard hydrochloric acid solution ($c(\mathrm{HCl}) = 0.5\,\mathrm{mol/L}$) until the red color just disappears and does not return within 30 s (recorded as $V$).

**Step 3:** Blank test and calculation. Carry out a blank test in the same way (recorded as $V_0$) and calculate the saponification value by the formula.

Table 4.5: Reference sample masses

| Estimated saponification value (mg KOH/g) | Sample mass (g) |
|---|---|
| $150 \sim 200$ | 2.0 |
| $200 \sim 250$ | 1.0 |
| $250 \sim 300$ | 0.5 |
| $> 300$ | 0.3 |

The sample mass in step 1 should leave about half of the added potassium hydroxide unconsumed, so that saponification is complete and enough alkali remains for a reliable difference between the blank and the sample. Preparing the potassium hydroxide–ethanol solution calls for care: the hydroxide is dissolved in purified ethanol, the solution is kept in the dark, and the clear supernatant is used. Aldehydes in the ethanol must be removed beforehand, since under alkaline conditions they undergo aldol condensation and change the effective concentration of the base; worse, the products of these reactions may be colored near the phenolphthalein endpoint and interfere with its judgment. Inorganic acids or free fatty acids in the sample consume extra KOH during saponification and give a high result, while esters other than fatty acid esters (phosphoglycerides, for example) also take part in the reaction, so that the result reflects the total ester

value rather than the saponification value of the glycerides alone.

The titration in step 2 must be done before the saponified solution cools to room temperature; at too low a temperature the potassium soaps may solidify and occlude unreacted KOH, giving a low result. If the solution turns turbid during the titration (fatty acids separating), the saponifying agent has been over-neutralized and the sample must be weighed again. The endpoint is the change from pink to colorless; near the endpoint the flask should be shaken thoroughly to avoid local excess of acid. If a dark sample matrix obscures the endpoint, alkali blue 6B or thymolphthalein may be used instead; their transition ranges lie at higher pH and help to avoid the interference of the matrix color.

The blank test in step 3 must be carried out under exactly the same conditions as the sample, including reflux, standing time, and titration temperature. The blank must be determined anew with every new batch of potassium hydroxide–ethanol solution or hydrochloric acid. The blank usually consumes more acid than the sample, and the difference between the two should not be less than 5 mL; otherwise the sample mass should be increased or the amount of KOH added reduced.

The saponification value is calculated as

$$I_{\mathrm{s}} = \frac{(V_0 - V) \times c(\mathrm{HCl}) \times 56.11}{m}$$

where $I_{\mathrm{s}}$ is the saponification value (mg KOH/g); $c(\mathrm{HCl})$ is the concentration of the standard hydrochloric acid solution (mol/L); $V_0$ and $V$ are the volumes of that solution consumed by the blank and by the sample (mL); $m$ is the sample mass (g); and 56.11 is the molar mass of potassium hydroxide (g/mol), which, multiplied by $10^3$ and combined with the units of concentration (mol/L) and volume (mL), gives the saponification value directly in mg/g.

Two parallel determinations are made for each sample and their arithmetic

mean is taken as the result. Under repeatability conditions the absolute difference between two independent results must not exceed 2% of their arithmetic mean (or 3 mg KOH/g when the saponification value is below 50 mg KOH/g).

The saponification value reflects the total amount of ester bonds in the sample (including triglycerides, phospholipids, wax esters, and so on) and is therefore not identical with the content of neutral oil. For samples rich in free fatty acids the acid value must be taken into account to obtain an accurate ester value. Moreover, silicones, some mineral oils, and synthetic esters are not saponified, or only partly so, under the conditions of the determination and may make the result deviate from the true value. The origin and composition of the sample should be known beforehand so that the applicability of the method can be judged.

## 4.8 The Influence of Carbon Dioxide on Acid-Base Titrations*

Reagents in an acid-base titration may change in concentration through environmental factors, and carbon dioxide from the air deserves particular attention. A standard sodium hydroxide solution, once prepared and standardized, absorbs $CO_2$ during storage and forms some sodium carbonate:

$$2\mathrm{NaOH(aq)} + \mathrm{CO_2(g)} = \mathrm{Na_2CO_3(aq)} + \mathrm{H_2O(l)}$$

At first sight the process consumes NaOH and should weaken its capacity to titrate acids. Yet the sodium carbonate formed is itself basic and still reacts with acid, so that it appears to "compensate" for the loss of hydroxide, and one may even be tempted to conclude that $CO_2$ does not affect the result. The intuition is misleading: the real interference comes from the fact that $Na_2CO_3$ is a dibasic base and behaves differently under different indicators, which may bias the endpoint and cause a systematic error.

Consider a specific case. A sodium hydroxide solution that has been prepared and standardized to 0.1000 mol/L absorbs carbon dioxide from the air and forms sodium carbonate at 0.0100 mol/L; the concentration of the hydroxide itself falls to 0.0800 mol/L. Unless the solution is standardized again, we do not know how much carbonate has formed and still take $c(\mathrm{NaOH}) = 0.1000\,\mathrm{mol/L}$. What happens when this mixture is used to titrate 0.1000 mol/L hydrochloric acid?

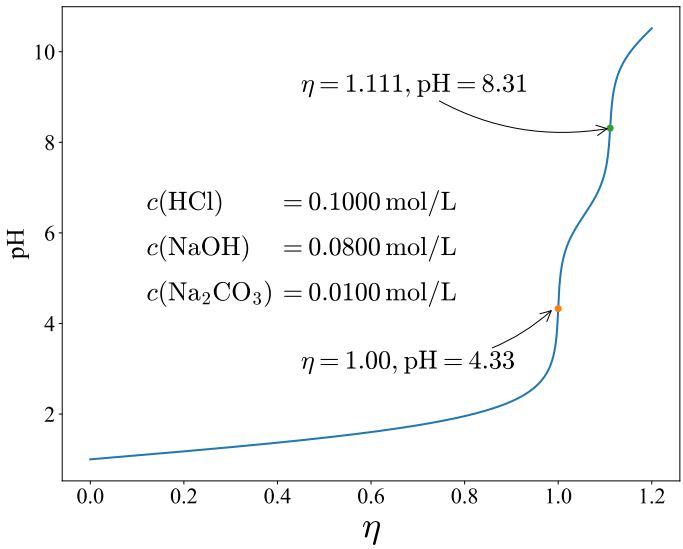



Figure 4.25: Titration curve of hydrochloric acid with a sodium hydroxide solution that has absorbed carbon dioxide.

Figure 4.25 shows the titration curve. Two jumps appear: the first corresponds to the true stoichiometric point at $\eta = 1.000$ and the second lies at $\eta = 1.111$. If methyl orange happens to be used, its color-change point falls within the first jump and the hydrochloric acid concentration calculated from $c(\mathrm{NaOH}) = 0.1000\,\mathrm{mol/L}$ is correct. If, however, phenolphthalein is used by habit, its color-change point lies at the second jump, $\eta = 1.111$, and the acid concentration is then calculated as $c(\mathrm{HCl})_{\mathrm{calc}} = c(\mathrm{NaOH}) \times 1.111$, clearly larger than the true value, a positive error.

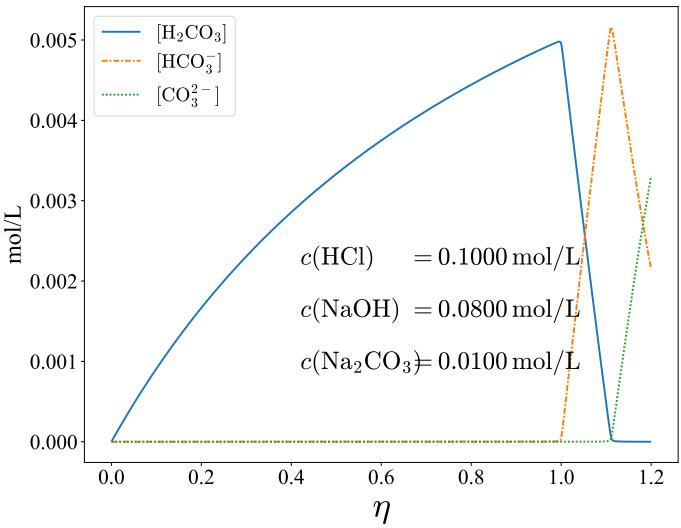



Figure 4.26: Concentration curves of the carbonate species during the titration of hydrochloric acid with a sodium hydroxide solution that has absorbed carbon dioxide.

How does the carbonate behave during the titration? Figure 4.26 shows the concentrations of the carbonate species. When the mixed solution of hydroxide and carbonate is added to the hydrochloric acid, the carbonate is in fact converted entirely into $H_2CO_3$, because the amount of acid present initially is far larger than that of the mixed base added. The carbonate does consume hydrochloric acid and does compensate for the loss of hydroxide.

As the titration proceeds the amount of acid decreases, while the carbonate entering the system with the mixed base is protonated completely to $H_2CO_3$ by the excess acid. The amount of $H_2CO_3$ therefore grows while $\eta < 1.000$ and reaches a maximum at $\eta = 1.000$: the acid has just been titrated completely, and the total carbonate concentration is $0.0100 \times \dfrac{1.000}{2.000} = 0.0050\,\text{mol/L}$, all present as $H_2CO_3$. Figure 4.26 shows that the concentration of $H_2CO_3$ is not zero at this point, and the mixed base added afterwards begins a new reaction with it. Since no hydrogen

ion is supplied to the system, the concentration of $H_2CO_3$ falls rapidly and at $\eta = 1.111$ it has just been converted completely into $HCO_3^-$. This is the root cause of the second jump.

In summary, although a sodium hydroxide solution that has absorbed carbon dioxide may still give an accurate result under particular conditions (titrating a strong acid with methyl orange, for instance), this is not the usual case and depends critically on a particular match between the system titrated and the indicator.

**Common Pitfall**

**$CO_2$ is not always harmless.** Some students suppose that "when NaOH absorbs $CO_2$ and becomes $Na_2CO_3$, a little alkali is lost and a few more drops of NaOH make it up." This ignores the fact that $Na_2CO_3$, as a dibasic base, behaves differently with different indicators: with methyl orange it consumes $2H^+$ (acting as an equivalent of NaOH), whereas with phenolphthalein it consumes only $1H^+$ (falling short). For the same bottle of "spoiled" NaOH, the systematic error when titrating a strong acid can reach about 11% if the wrong indicator is chosen. The conclusion: preventing the standard alkali from absorbing $CO_2$ is not a search for perfection but a safeguard of the reliability of the result.

To ensure that results are generally applicable and reliable, the fundamental measure is to prevent the standard sodium hydroxide solution from absorbing carbon dioxide: use carbonate-free reagents, store the solution well sealed, or protect it with a soda-lime tube.

## Chapter Summary

1. **The Brønsted-Lowry theory and the unified $K_a$ framework.** The Brønsted-Lowry theory centers on proton transfer, and an acid with its conjugate base forms a dialectical unity. $K_a$ is the only parameter needed: a large $K_a$ means a strong acid, a small one a weak acid, and $K_a \times K_b = K_w$ reduces the quantitative description of acids and bases to a single parameter.
2. **Distribution fraction $\delta$ and the distribution of species.** $\delta$ depends only on $K_a$ and $[H^+]$, not on the analytical concentration. At pH = p$K_a$, $\delta(HB) = \delta(B^-) = 0.5$, so the crossing point locates p$K_a$ directly; for a polyprotic acid the crossing point $\delta(H_2A) = \delta(A^{2-})$ lies at pH = (p$K_{a_1}$ + p$K_{a_2}$)/2. The distribution fraction is the mathematical basis of the titration equations and of the $F$ function.
3. **Calculating the hydrogen ion concentration.** The key is the proton balance equation, with $H^+$ coming from the acid and from the self-dissociation of water. The exact expression is usually an equation of high degree, and in practice the approximations are judged from the relative sizes of $c$ and $K_a$: the contribution of water may be ignored when $cK_a > 20K_w$, and the consumption of HB by dissociation when $c > 500K_a$. The pH of an amphoteric solution is approximately the arithmetic mean of two adjacent p$K_a$ values.
4. **Buffer solutions.** The Henderson-Hasselbalch equation

$$\mathrm{pH} \approx \mathrm{p}K_a + \lg(c_b/c_a)$$

gives the approximate pH of a buffer. The buffer index $\beta = \mathrm{d}c_b/\mathrm{dpH}$ quantifies the buffering power; $\beta_{max} = 0.575c_{total}$ occurs at pH = p$K_a$, and the useful buffer range is p$K_a \pm 1$. The first principle in choosing a buffer is to match p$K_a$ with the target pH.

5. **Titration equations and the general equation.** Starting from the simplest equation for a strong base titrating a strong acid,

$$\eta = \frac{c_{\mathrm{HCl}} - \Delta}{c_{\mathrm{NaOH}} + \Delta}$$

distribution fractions are introduced step by step to build the equations for weak acids (multiplied by $\delta$) and polyprotic acids (a linear combination of distribution fractions). The result is de Levie's general equation of acid-base titration,

$$\eta = -\frac{\sum F_{\mathrm{s}} c_{\mathrm{s}} + \Delta}{\sum F_{\mathrm{t}} c_{\mathrm{t}} + \Delta}$$

which brings any complex system under one framework. The $F$ function (proton dissociation/association function) quantifies the proton transfer capacity of each component as a linear combination of distribution fractions.

6. **Indicators and the endpoint error.** An indicator takes part in the titration as a weak acid or weak base and carries the inherent defect of being "referee and player" at once. The endpoint error

$$\epsilon_{\mathrm{ep}} = \frac{\eta_{\mathrm{ep}} - \eta_{\mathrm{sp}}}{\eta_{\mathrm{sp}}} \times 100\%$$

is calculated from the titration equation. $CO_2$ spoils NaOH into a mixture of NaOH and $Na_2CO_3$: with methyl orange the apparent concentration stays unchanged, whereas with phenolphthalein a marked positive error appears.

7. **Typical applications.** The Kjeldahl method (organic matter $\rightarrow NH_4^+ \rightarrow NH_3$, absorbed in $H_3BO_3$ and titrated with HCl), the formaldehyde method ($NH_4^+ + HCHO \rightarrow H^+$, phenolphthalein indicator), the double-indicator and barium chloride methods (separate determination of NaOH and $Na_2CO_3$ in caustic soda), the potassium fluorosilicate method ($SiO_2 \rightarrow K_2SiF_6 \rightarrow$ HF titrated with NaOH), and back titration (the saponification value: saponification with excess KOH and back titration with HCl).

## Exercises

★ 4.1 Write the conjugate base of each of the following acids: $H_3PO_4$, $H_2PO_4^-$, $NH_4^+$, $H_2O$, $HCO_3^-$, HCOOH.

★★ 4.2 For $H_3PO_4$, $pK_{a_1} = 2.12$, $pK_{a_2} = 7.20$, and $pK_{a_3} = 12.36$. At pH = 6.00, arrange the concentrations of $H_3PO_4$, $H_2PO_4^-$, $HPO_4^{2-}$, and $PO_4^{3-}$ in order of size.

★★ 4.3 Hydrochloric acid is added to a $Na_2CO_3$ solution. When the pH reaches 10.00, which components are present, and which is the main one?

★★★ 4.4 The distribution curves of the phosphoric acid species have five crossing points. (1) Calculate the pH of each crossing point. (2) Calculate the distribution fractions of the species at each crossing point. For $H_3PO_4$, $pK_{a_1} = 2.12$, $pK_{a_2} = 7.20$, and $pK_{a_3} = 12.36$.

★★★ 4.5 State the effect of each of the following on the result:

1. In standardizing a NaOH solution, the primary standard potassium hydrogen phthalate contains phthalic acid as an impurity;
2. A NaOH standard solution that has absorbed carbon dioxide is used to titrate $H_3PO_4$ to the first jump; what happens? And to the second jump?
3. A NaOH standard solution is known to have absorbed carbon dioxide, about 0.4% of the NaOH having become $Na_2CO_3$. When this solution is used to determine acetic acid, how does the endpoint error change relative to the case without absorption?

★★ 4.6 Starting from the proton association constant $K_b$, derive the distribution fractions of the two species in an aqueous ammonia solution and compare them with equations (4.19) and (4.20).

★★ 4.7 A solution may contain one of NaOH, $Na_2CO_3$, or $NaHCO_3$, or a mixture of two of them. With phenolphthalein as indicator, $V_1$ of standard HCl is consumed to the endpoint; methyl orange is then added and a further $V_2$ of HCl is consumed. Determine the composition of the solution from $V_1$ and $V_2$.

| | |
|---|---|
| $V_1 > V_2, V_2 > 0$ | |
| $V_2 > V_1, V_1 > 0$ | |
| $V_1 = V_2$ | |
| $V_1 = 0, V_2 > 0$ | |
| $V_1 > 0, V_2 = 0$ | |

★★ 4.8 Calculate the pH of a 0.05 mol/L sulfuric acid solution.

★ 4.9 Calculate the pH of a 0.10 mol/L NaAc solution. For HAc, $pK_a = 4.74$.

★★ 4.10 Calculate the pH of a 0.10 mol/L $H_2C_2O_4$ solution. For the acid, $K_{a_1} = 5.9 \times 10^{-2}$ and $K_{a_2} = 6.4 \times 10^{-5}$. What relative error is introduced by treating $H_2C_2O_4$ as a monoprotic acid and neglecting the second dissociation?

★★ 4.11 Calculate the pH of a 0.10 mol/L $Na_2HPO_4$ solution. For $H_3PO_4$, $pK_{a_1} = 2.12$, $pK_{a_2} = 7.20$, and $pK_{a_3} = 12.36$.

★★ 4.12 How many jumps appear on the titration curve when each of the following solutions is titrated with NaOH or HCl? Explain briefly.

(1) $H_2SO_4$+$H_3PO_4$ (2) HCl+$H_3BO_3$ (3) HF + HAc

(4) NaOH + $Na_3PO_4$ (5) $Na_2CO_3$+$Na_2HPO_4$ (6) $Na_2HPO_4$+$NaH_2PO_4$

★★★ 4.13 For the diprotic acid $H_2A$, $K_{a_1} = 1.0 \times 10^{-3}$ and $K_{a_2} = 1.0 \times 10^{-7}$. A 0.1000 mol/L solution of $H_2A$ is titrated with 0.1000 mol/L NaOH. Can two separate jumps be observed on the titration curve? How does the conclusion change if $K_{a_2}$ rises to $1.0 \times 10^{-5}$? Explain with the criterion $cK_a \geq 10^{-8}$ for stepwise

titration.

★ 4.14 A buffer of pH $\approx$ 3 is to be prepared. Which of the following acids and their conjugate bases should be chosen?

(1) dichloroacetic acid (2) formic acid (3) chloroacetic acid (4) acetic acid

★★ 4.15 How many grams of solid NaAc must be added to 100.0 mL of 0.100 mol/L HCl to raise its pH from 1.00 to 4.44? (Ignore the change in volume.)

★ 4.16 An $NH_3 - NH_4Cl$ buffer solution is prepared with $c(NH_3) = 0.200$ mol/L and $c(NH_4Cl) = 0.300$ mol/L. Calculate its pH.

★★ 4.17 A 0.6850 g mixture of $Na_2CO_3$ and $NaHCO_3$ is dissolved in water. With methyl orange as indicator, 50.00 mL of 0.2000 mol/L HCl is consumed to the endpoint. How many milliliters of the same acid would be consumed with phenolphthalein as indicator?

★★ 4.18 A 0.1000 mol/L NaOH solution titrates a 0.1000 mol/L HAc solution to pH = 7.00. Calculate the endpoint error.

★★ 4.19 A 0.1000 mol/L HCl solution titrates 0.1000 mol/L $Na_2CO_3$ to the first stoichiometric point (forming $NaHCO_3$). Calculate the pH at that point and the endpoint errors when cresol red ($pK_a = 8.3$) and phenolphthalein ($pK_a = 9.1$) are used as indicators. For $H_2CO_3$, $pK_{a_1} = 6.38$ and $pK_{a_2} = 10.25$.

★★★ 4.20 A 0.1000 mol/L NaOH solution titrates a mixture that is 0.1000 mol/L in hydroxylammonium chloride ($NH_3^+OH \cdot Cl^-$) and 0.1000 mol/L in $NH_4Cl$. For the acids, $K_a(NH_3OH^+) = 10^{-5.96}$ and $K_a(NH_4^+) = 10^{-9.26}$. (1) What is the pH at the first stoichiometric point? (2) What percentage of the $NH_4Cl$ has reacted at that point?

★★ 4.21 A 0.1000 mol/L HCl solution titrates a 0.1000 mol/L $NH_3$ solution. Calculate the endpoint errors with phenolphthalein ($pK_a = 9.1$) and with methyl orange ($pK_a = 3.4$) as indicators.

★★★ 4.22 A sample may contain one or two of NaOH, $Na_2CO_3$, and $NaHCO_3$, together with inert impurities that do not react with acid. A 0.5000 g portion is titrated with 0.1000 mol/L HCl; with phenolphthalein as indicator 15.00 mL of acid is consumed, and with methyl orange as indicator a further 22.00 mL is consumed. Determine the composition of the sample and the mass fraction of each component.

★★★ 4.23 A NaOH standard solution of concentration $c$(NaOH) has been contaminated with NaAc at concentration $c$(NaAc). Derive the titration equation, in terms of the volume ratio, for titrating HCl of concentration $c$(HCl) with this solution.

★★ 4.24 Use the program pH_at_eta.m to study the various approximations based on equation (4.47) and the relative errors they introduce. Present the results as a graph or a table.

★★★ 4.25 An industrial wastewater contains NaOH and $Na_3PO_4$. A 25.00 mL sample is titrated with 0.1000 mol/L HCl standard solution. (1) With phenolphthalein as indicator, to what species are NaOH and $Na_3PO_4$ titrated, and how is the volume of HCl consumed related to the amounts of the two components? (2) With methyl orange as indicator, the $HPO_4^{2-}$ formed in the first stage is titrated further to $H_2PO_4^-$; write the relation between the volume of acid consumed in this stage and the amount of $Na_3PO_4$. (3) If the phenolphthalein endpoint consumes 18.50 mL and the methyl orange endpoint a further 8.00 mL of HCl, calculate the molar concentrations of NaOH and $Na_3PO_4$ in the sample. For $H_3PO_4$,

$pK_{a_1} = 2.12$, $pK_{a_2} = 7.20$, and $pK_{a_3} = 12.36$.

★★ 4.26 A 100.0 mL solution contains 0.1000 mol/L HCl, 0.05000 mol/L HAc, and 0.03000 mol/L $NaH_2PO_4$. How many grams of solid NaAc must be added to bring the pH to 5.00? (Ignore the change in volume.) For HAc, $pK_a(HAc) = 4.74$; $M(NaAc) = 82.03$ g/mol; for $H_3PO_4$, $pK_{a_1} = 2.12$, $pK_{a_2} = 7.20$, and $pK_{a_3} = 12.36$.

★★ 4.27 A 0.10 mol/L NaOH solution titrates $H_2SO_4$ of the same concentration with methyl orange as indicator, the endpoint being at pH = 4.40. Calculate the endpoint error. For $HSO_4^-$, $pK_a = 1.99$.

★★ 4.28 A 0.10 mol/L NaOH standard solution titrates a 0.050 mol/L $H_2SO_4$ solution. Calculate the pH at volume ratios $\eta$ of 0.00 and 1.0. For $H_2SO_4$, $pK_{a_1} = -10.0$ and $pK_{a_2} = 1.99$.

★★ 4.29 Protonated ethylenediamine (written $H_2EN^{2+}$) has $pK_{a_1} = 6.85$ and $pK_{a_2} = 9.93$. How many grams of solid NaOH must be added to prepare 1.000 L of an ethylenediamine buffer of total concentration 0.15 mol/L and pH = 6.65? For NaOH, $M(NaOH) = 40.00$ g/mol.

★★ 4.30 To prepare 500.0 mL of a phosphate buffer of pH = 7.20 and total concentration 0.1000 mol/L, what volumes of 0.1000 mol/L $H_3PO_4$ solution and 0.1000 mol/L $Na_2HPO_4$ solution should be taken? For $H_3PO_4$, the three $pK_a$ values are 2.12, 7.20, and 12.36.

★★ 4.31 A 0.2000 mol/L NaOH solution titrates a mixture that is 0.2000 mol/L in HAc and 0.4000 mol/L in $H_3BO_3$ to pH = 6.20. Calculate the endpoint error. For HAc, $K_a(HAc) = 1.8 \times 10^{-5}$, and for $H_3BO_3$, $K_a(H_3BO_3) = 5.8 \times 10^{-10}$.

★★ 4.32 A buffer solution is prepared from a weak base BOH and its salt BCl so that pH = 10.00. When 60. mL of 1.0 mol/L hydrochloric acid is added to 140. mL

of this buffer, the pH falls to 9.00. Calculate the concentrations of BOH and BCl in the original buffer. For BOH, $K_b(\mathrm{BOH}) = 5.0 \times 10^{-5}$.

★★★ 4.33 A phosphate buffer of pH = 7.50 is to be prepared as 1.00 L from 0.500 mol/L $H_3PO_4$ and 0.500 mol/L NaOH. To check its buffering power, 5.00 mL of 0.100 mol/L HCl is added to 50.0 mL of the buffer and the pH is found to fall to 7.10. Calculate the volumes of phosphoric acid and sodium hydroxide solution needed. For $H_3PO_4$, the three p$K_a$ values are 2.12, 7.20, and 12.36.

★★ 4.34 A 0.8150 g sample of a pure monoprotic weak acid HB is dissolved in water to give a 0.1000 mol/L solution. With phenolphthalein as indicator, 24.60 mL of 0.1100 mol/L NaOH is consumed to the endpoint; when 11.00 mL of the NaOH has been added, the pH of the solution is 4.80. Calculate the p$K_a$ of HB.

★★ 4.35 The mass fractions of NaOH and $Na_2CO_3$ in an industrial caustic soda are to be determined. A 2.546 g sample is dissolved and made up to 250.0 mL. A 25.00 mL portion is titrated to the methyl orange endpoint with 24.86 mL of standard HCl; another 25.00 mL portion is treated with excess $BaCl_2$ and titrated with the same acid to the phenolphthalein endpoint, which consumes 23.74 mL. Neutralizing 0.4852 g of borax ($Na_2B_4O_7 \cdot 10\,H_2O$) requires 24.37 mL of the same acid. Calculate the mass fractions of NaOH and $Na_2CO_3$ in the sample. For $M(Na_2B_4O_7 \cdot 10\,H_2O) = 381.4$ g/mol, $M(\mathrm{NaOH}) = 40.00$ g/mol, and $M(Na_2CO_3) = 106.0$ g/mol.

★★ 4.36 A 25.00 mL portion of a mixed phosphate solution requires 10.00 mL of 0.5000 mol/L standard HCl to reach the phenolphthalein endpoint, and 50.00 mL of the same acid to reach the methyl orange endpoint. Calculate the pH of the original solution. For $H_3PO_4$, the three p$K_a$ values are 2.12, 7.20, and 12.36.

★★ 4.37 A buffer solution is prepared from a weak acid HB and its conjugate base,

with [HB] = 0.250 mol/L. When 200. mg of solid NaOH is added to 100. mL of this buffer (the volume change being negligible), the pH of the resulting solution is 5.60. What was the pH of the original buffer? For $M(\mathrm{NaOH}) = 40.00\,\mathrm{g/mol}$ and $K_\mathrm{a}(\mathrm{HB}) = 5.0 \times 10^{-6}$.

★★ 4.38 Glycine exists in aqueous solution as the zwitterion $\mathrm{NH_3^+CH_2COO^-}$. To prepare 100. mL of a glycine buffer of total concentration 0.100 mol/L and pH = 2.00, how many grams of glycine are needed, and how many milliliters of 1.00 mol/L strong acid or strong base must be added? For the glycinate, $K_{\mathrm{a}_1} = 4.5 \times 10^{-3}$ and $K_{\mathrm{a}_2} = 2.5 \times 10^{-10}$; $M(\mathrm{NH_2CH_2COOH}) = 75.07\,\mathrm{g/mol}$.

★★ 4.39 Equal volumes of a solution 0.20 mol/L in $\mathrm{NH_4Cl}$ and 0.20 mol/L in $\mathrm{NH_3}$ and a solution 0.020 mol/L in HAc and 0.020 mol/L in NaAc are mixed. Calculate the pH of the mixture.

★★ 4.40 A 0.10 mol/L HCl solution titrates a 0.10 mol/L NaOH solution that also contains 0.10 mol/L NaAc. Calculate the endpoint error at pH = 7.00. For HAc, $K_\mathrm{a}(\mathrm{HAc}) = 10^{-4.74}$.

★★ 4.41 Pure dry NaOH and $\mathrm{NaHCO_3}$ are mixed in a mass ratio of 2 : 1. A portion of the mixture is dissolved in water and titrated with the same standard hydrochloric acid. Calculate the ratio of the volume of acid needed with phenolphthalein as indicator to that needed with methyl orange. For $M(\mathrm{NaOH}) = 40.00\,\mathrm{g/mol}$ and $M(\mathrm{NaHCO_3}) = 84.01\,\mathrm{g/mol}$.

★★★ 4.42 A volume $V_0$ of a solution of the weak acid salt NaHB of analytical concentration $c$(NaHB) is titrated with HCl of analytical concentration $c$(HCl). The acid dissociation constants of the diprotic acid are $K_{\mathrm{a}_1}$ and $K_{\mathrm{a}_2}$. Derive the titration equation in terms of the volume ratio $\eta$.

★★★ 4.43 A mixed acid solution is 0.1000 mol/L in sulfuric acid and 0.1000 mol/L in phosphoric acid. Calculate its pH. For $H_2SO_4$, $pK_a(H_2SO_4) = -10.0$ and $pK_a(HSO_4^-) = 1.99$; for $H_3PO_4$ the three $pK_a$ values are 2.12, 7.20, and 12.36.

★★★ 4.44 A 20.00 mL mixed solution of disodium malonate (written $Na_2M$) and sodium hydrogen malonate (NaHM) is titrated with 0.01000 mol/L HCl. When 1.00 mL of the acid has been added the pH is 5.70, and when 10.00 mL has been added it is 4.28. How many milliliters of the acid in total are needed to reach the stoichiometric point of malonic acid ($H_2M$)? For malonic acid, $pK_{a_1} = 2.86$ and $pK_{a_2} = 5.70$.

## Answer

4.1 $H_2PO_4^-$, $HPO_4^{2-}$, $NH_3$, $OH^-$, $CO_3^{2-}$, $HCOO^-$

4.2

$$[H_2PO_4^-] > [HPO_4^{2-}] > [H_3PO_4] > [PO_4^{3-}]$$

4.3 The main component is $HCO_3^-$.

4.4

(1) 2.12, 4.66, 7.20, 9.78, 12.36.

(2) 0.000, 0.000, 0.003, 0.500.

4.5 (omitted)

4.6 (omitted)

4.7 (omitted)

4.8 pH = 1.24

4.9 pH = 8.87

4.10 pH = 1.28 (first dissociation only); including the second dissociation gives 1.2770, a relative error of about 0.1%

4.11 pH = 9.74

4.12 (omitted)

4.13 (1) $K_{a_1}/K_{a_2} = 10^4$, exactly at the critical condition; two jumps are observable but not pronounced. (2) $K_{a_1}/K_{a_2} = 10^2$, only one overall jump.

4.14 Chloroacetic acid, because its $pK_a = 2.86 \approx 3$.

4.15 1.2 g.

4.16 9.07

4.17 12.50 mL

4.18 −0.6%

4.19 The first stoichiometric point has pH = 8.32; cresol red gives −1.1% and phenolphthalein −6.6%, so cresol red is the better choice.

4.20 (1) pH = 7.61. (2) 2.2% of the $NH_4Cl$ has reacted.

4.21 Phenolphthalein: −41%; methyl orange: 0.80%

4.22 $Na_2CO_3$ : 31.80%, $NaHCO_3$ : 11.76%

4.23

$$\eta = \frac{c(\mathrm{HCl}) - \Delta}{c(\mathrm{NaOH}) + \delta(\mathrm{HAc})c(\mathrm{NaAc}) + \Delta}$$

4.24 (Programming exercise) When $c > 500K_a c^{\ominus}$ and $cK_a > 20K_w c^{\ominus}$ are not satisfied, the relative error of the approximate expressions grows rapidly.

4.25 $c(\mathrm{NaOH}) = 0.04200\ \mathrm{mol/L}$; $c(Na_3PO_4) = 0.03200\ \mathrm{mol/L}$

4.26 $c(\mathrm{NaAc}) = 0.3731\ \mathrm{mol/L}$, so 3.06 g of solid NaAc must be added

4.27 $\epsilon_{ep} = -0.3\%$

4.28 pH = 1.24 at $\eta = 0.00$; pH = 7.27 at $\eta = 1.0$

4.29 About 2.3 g of solid NaOH must be added

4.30 125.0 mL of $H_3PO_4$ solution and 375.0 mL of $Na_2HPO_4$ solution

4.31 $\epsilon_{ep} = -3.2\%$

4.32 [BCl] = 0.29 mol/L, [BOH] = 0.57 mol/L

4.33 89.4 mL of $H_3PO_4$ solution and 149 mL of NaOH solution

4.34 $pK_a = 4.89$

4.35 $w(NaOH) = 38.94\%$, $w(Na_2CO_3) = 2.43\%$

4.36 pH ≈ 11.9

4.37 The original buffer had pH = 5.45

4.38 0.751 g of glycine and 6.91 mL of 1.00 mol/L hydrochloric acid

4.39 pH = 9.17

4.40 $\epsilon_{ep} = +0.55\%$

4.41 $V_{\text{phenolphthalein}}/V_{\text{methyl orange}} = 0.81$

4.42 $$\eta = \frac{\left[\delta(H_2B) - \delta(B^{2-})\right]c(NaHB) + \Delta}{c(HCl) - \Delta}$$

4.43 pH = 0.94

4.44 28.00 mL of hydrochloric acid in total

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